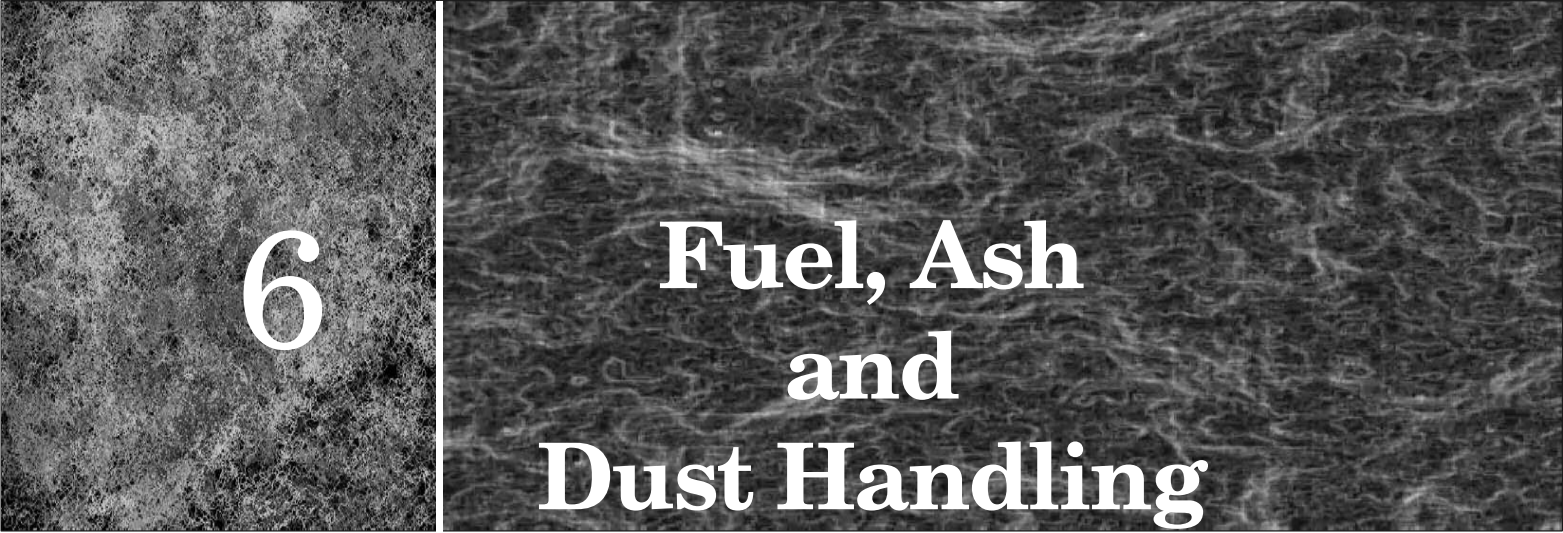




6

Fuel, Ash and Dust Handling

6.1 INTRODUCTION

Due to continuously increasing demand for power at lower cost, it has become necessary to set up higher capacity thermal power stations. Rise in capacity of these plants pose problems in coal supply system from the coal mines to the power station and then during its internal handling. An annual consumption of a 2,000 MW station require approximately 20,000 to 25,000 tons daily. The transportation, loading, unloading, storage and processing of huge amount of coal from mines to the plant and also within the plant are the major problems.

When designing a power station careful thought should be given to the coal handling system including the design and layout of the individual units. A thermal plant may use coal, oil or gas as the fuel, but since main fuel used in coal due to shortage and high cost of petroleum products, we are giving more attention to the handling of coal in this chapter.

Layout of a power plant is the most important task for obtaining least total cost of coal handling, as the orderly flow minimises expenses on its handling and productivity. For a good layout, it is most essential to have minimum movement, bottle neck, congestion, back-tracking and maximum space utilisation.

A good handling system should have following objectives:

1. Minimum handling
2. Facilitate in power generation process
3. Flexibility in arrangement
4. Maximum utilisation of equipment
5. Maximum utilisation of space
6. Efficient utilisation of manpower
7. Care for employee safety and convenience
8. Reduced internal transport
9. Easy to supervise
10. Minimum production delays and back-tracking

Coal handling is a prime consideration in designing new plants by proper application of modern handling devices/equipment. Coal handling involves movement of coal mechanically in most efficient manner. It must arrive at its place at right time and ensure that each location continually receives the right quantity.

Planning for coal handling should follow the systems approach so as to achieve the following:

1. Increase the production effectiveness by having right quantity of coal, at right places at the right time, by avoiding delays and following the orderly flow. This helps in improving the productivity
2. Reduce accident rates
3. Effective utilisation of space.

Following *principles* should be adopted while designing coal handling system:

1. *Planning Principle*: All handling activities should be planned.
2. *Systems Principle*: Handling activities be integrated and coordinated.
3. *Simplification Principle*: Reduce, combine or eliminate unnecessary movements and / or equipment by applying motion economy principles.
4. *Material Flow Principle*: Material flow pattern must be determined by operation sequence and pattern of equipment arrangement. Avoid overcrowding, back-tracking and obstacles, and follow a direct path.
5. *Gravity Principle*: Utilize gravity where possible.
6. *Unit size Principle*: Increase size, quantity, weight of the load handled. Since larger the load, lesser will be the cost per unit handled.
7. *Space utilisation Principle*: Optimum utilisation of space.
8. *Safety Principle*: Safe handling methods and equipment for better working conditions and to avoid unsafe conditions.
9. *Mechanisation Principle*: For increasing the efficiency use right type, size and standard equipment.

6.1.1 Ash Handling

After coal, ash handling is another major task. In the case of coal fired boilers about 10 to 15% of the total weight of the coal fired is collected in the form of ash. A thermal power plant of 2000 MW capacity produces about 2000 to 3000 tons of ash per day. This will require a large ash handling system and also requires a large area of land for dumping.

6.2 COAL HANDLING SYSTEMS

Coal can be handled manually or mechanically. Manual handling is not possible for modern steam power plants. All the modern plants are employing mechanical handling. Large plants use large size equipment, while small plants use smaller size equipment for mechanical handling. Advantages of mechanical handling as compared to manual handling are:

1. It is economical.
2. Operation is easy and smooth.
3. It is reliable
4. Lesser manpower required.

5. Improved working conditions. Minimum labour is required on unhealthy working conditions.

6.3 ACTIVITIES OF COAL HANDLING

General arrangements of activities/processes involved in mechanical system of coal handling for small and large plants located away from coal mines are shown in flow chart given below (Fig. 6.1). It is not necessary that this flow chart may be followed as such in all the plants. However, some changes, where required, can take place depending upon the circumstances, including distance of mine from plant location, type of equipment used in handling, type of boilers and fuel burning system, space available at plant site etc. Some of the intermediate steps may be eliminated or rearranged or bifurcated depending upon the type of plant, its capacity and other factors as mentioned above.

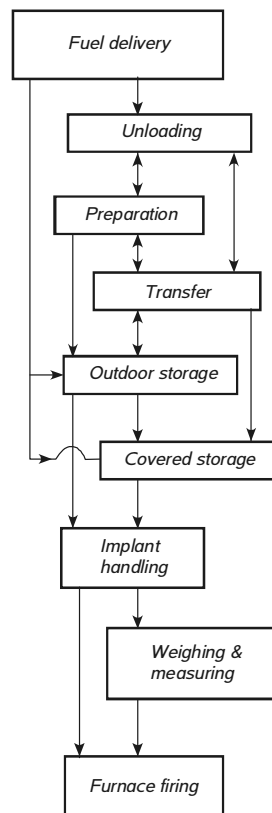


Fig. 6.1. Flow chart of basic steps in coal handling system of a power plant.

The activities or basic steps of coal handling are generally done in two phases, namely, outplant handling and inplant handling. The outplant handling involves the handling of coal after mining is done at mines to the supply near the plant. The inplant handling involves the handling of coal from storage near the plant till it is fed to the burners.

6.4 OUTPLANT HANDLING

Flow diagram of outplant handling activities include the following basic steps:

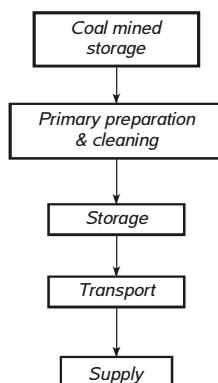


Fig. 6.2. Flow diagram of outplant handling of coal

Table 6.1: Equipments for various handling activities

S.N.	Coal handling activities	Equipment used
1.	Coal Benefication (Primary preparation and cleaning)	coal sizing, screens, coal drying equipment, washing, and dewatering.
2.	Storage	Bulldozer, drag scraper, cranes with grab buckets, and conveyor system.
3.	Transport	Rail and ships with loading system, tipping trucks, dumpers, ropeways.
4.	Unloading	Unloading tower and bridges, car shakers, rotary car dumpers, cranes with buckets, lift trucks with scoop, self unloading boats.
5.	Coal preparation	Crushers, sizers (screens), driers, magnetic separators.
6.	Transfer	Belt conveyor, screw conveyor, bucket elevator, skip hoists, grab bucket elevators, flight conveyors
7.	Covered (Live) storage	Bins, bunkers, indicator gates, valves.
8.	Inplant handling	Same as for transfer, and chutes for transfer from bunkers to firing units.
9.	Weighing	Coal meters, weigh lorry hoppers, weigh platform, weighing devices built in feeders and inbelt conveyors.
10.	Coal pulverizing	Crushher, magnetic separator, drier, pulveriser, bin, feeder. Pulveriser may be: Ball mill, ball and race mill, impact or hammer mill, or bowl mill.

6.4.1. Primary Preparation and Cleaning

Run-off-Mine Coal (ROM Coal) is the raw coal extracted from mines before cleaning and preparation. ROM coal is subjected to *Coal Cleaning and Preparation* process before despatch to the consumers premises. The coal cleaning and primary preparation of coal prior to use is

known as *coal beneficiation*. The purpose of this is to make coal acceptable to the consumers. It is better to remove unwanted material from coal near the coal fields itself, rather than transporting them along with coal to a long distance, i.e., upto to the power plant.

The main objectives of primary preparation (or beneficiation) are:

1. Differentiating between coal and impurities like hard lumps, stones etc.
2. Removal of impurities, dirt and mud.
3. Removal of some sulphur from coal.
4. Sorting out and screening different sizes of coal pieces. Segregating high grade, medium grade and low grade coals.

Thus the functions of primary preparation includes sizing, removal of ash and sulphur bearing minerals, cleaning and blending of different coals to achieve desired physical and chemical properties.

Following activities are generally performed for primary preparation and cleaning.

1. **Sorting out.** Dull and hard lumps are sorted out from dark and softer coal, dirt and other non-coal lumps are removed manually from picking up belt before loading the coal on wagons.
2. **Removal of dirt.** The coal is passed through primary screens to separate the large coal above 8 cm in size from the small coal. The large coal is then cleaned by hand and the coal below 8 cm in size is cleaned by washing or dry cleaning plants. In the dry screening plants, during screening, the fine dirt associated with coal is automatically removed by vibrations imparted to the screen. The process of dry cleaning can be used when the coal to be cleaned has less than 3% free moisture.
3. **Washing and cleaning.** This process is optional and is used when sulphur, sand and washable impurities like dirt, pyrite etc., are required to be removed from the coal and the specified purity as required is to be obtained.
4. **Removal of sulphur.** The most objectionable impurity in coal is sulphur. When coal is burned, poisonous oxides of sulphur are released into environment. High sulphur coals are washed and chemically treated for removal of sulphur. Organically bound sulphur, however, cannot be removed by physical sorting, water soluble sulphate are removed by washing the coal. Pyritic sulphur is also removed by washing.
5. **Removal of metallic particles.** These are removed with the help of magnetic separators, generally these are provided with the belt conveyer pulley, where these particles cling to the belt and then dropped into a reject chute.
6. **Dewatering.** Excessive surface moisture on the coal reduces the heating value of coal, increases transportation costs and creates handling problems. Therefore, dewatering is necessary after washing to produce clean coal. Vibrating screens are used to dewater the coal. Centrifugal driers can also be used to dewater the coarse and medium size coal.

6.4.2. Advantages of Primary Preparation and Cleaning of Coal

1. **Reduced transportation.** As this eliminates about 15 to 20% waste, which results a significant long term saving especially where the hauls are long and volume handled are large.

2. **Improve coal quality.** It eliminates non-combustibles and produce fuel of uniform size. This creates high quality fuel which costs less to transport, handle, crush and pulverise. Washed and cleaned coal reduces the outages caused by excessive wear of crushers and pulverisers as pyrites, rock, tramp iron and other abrasive constituents are removed.
3. **Reduces pulveriser operation and maintenance costs.** These costs are reduced because of lesser hard mineral matter to pulverise.
4. **Improves boiler performance.** The boiler efficiency increases due to improved quality of coal being fed.
5. **Less ash to handle.** Cleaner coal reduces the ash in the coal, thereby, reducing ash handling and ash disposal costs. Cleaner coal also mean less heat loss in bottom ash and gives improved heat efficiency.
6. **Removal of sulphur.** Physical cleaning can not remove all sulphur associated with coal because a significant amount of sulphur is chemically bound with the coal. However, unbound sulphur can be removed (up to 50%) by simple washing.

Higher surface moisture of a washed coal may partially negate the benefits gained out of ash and sulphur reduction, therefore, before going to adopt cleaning by washing, a cost-benefit analysis is necessary.

6.4.3 Storage of Coal

Coal may be stored at the coal fields, at intermediate location near the plant and at final locations near the boiler furnace. The store locations and amount of coal to be stored depends upon the availability of space for storage, transportation facilities, the amount of coal that will weather away, nearness to coal mines of the power station, losses on storage and deterioration of coal, interest on capital, cost of insurance, handling costs, local conditions of wind, sunlight and other climatic conditions.

Considering above factors, the storage locations, store space and capacity at each location is decided.

The coal storage has following basic purposes:

1. It is insurance against complete shutdown of a plant occurring from failure of normal operating supplies to arrive; and
2. It permits some choice of the date of purchase of coal, allowing the purchaser to take advantage of seasonal market conditions.

Thus, storage of coal is primarily a matter of protection against the coal strikes, failure of transportation system, and general coal shortages.

Here, we have discussed the storage neat the coal fields, the aspects related to other storage locations shall be discussed later on in the subsequent paragraphs.

The storage methods vary depending upon temperature, rains, wind conditions at site. Tropical areas need special storage precautions to avoid spontaneous fires due to self-heating of exposed coals. Type of coal also dictates the choice of handling and storage methods.

During the storage, the coal undergoes oxidation and deterioration due to chemical actions. This effect is known as “weathering” of coal. The weathering depends upon the nature of

coal, method of storage, total bulk of storage and exposed area, exposure to atmospheric air, sunlight, temperature etc.

Stored coal weathers and may ignite spontaneously. Weathering releases some of the volatile matter to reduce the heating value. Oxidation of coal may under certain conditions heat the coal to the point of ignition. Ventilating the pile can some times remove the heat and avoid burning.

Coal piles are generally built up in about 25 cm layers and thoroughly compacted by running the tractors over them. This excludes the air to a large extent and reduces oxidation and therefore spontaneous heating. For long storage periods, sealing the outside surface with asphalt or fine coal makes the pile practically air tight. Another method of removing the heat of oxidation is, allow the air to move through the layers evenly so that the heat of reaction is carried away and the temperature of the coal is maintained below the combustion temperature (70°C).

The coal can be stored in following methods:

1. *Stacking the coal in heaps*—The coal is piled, on the ground in open area, upto 10-12 m height. The pile top should be given a slope in the direction in which the rain may be drained off. The coal should be stored at a site located on solid ground, well drained, free of standing water preferably on high ground not subject to flooding. The sealing of stored pile is desirable in order to avoid the oxidation of coal. Asphalt, coal dust and bituminous coating are the materials commonly used for the sealing purposes.
2. Stacking the coal under cover.
3. Placing the coal in bunkers.
4. Storage under water. The possibility of slow oxidation and spontaneous combustion can be completely eliminated by storing the coal under water.

Storage piles can be of several alternative shapes out of which conical and wedge shaped are common. In tropical climate the limit of single storage bulk (pile) are, generally kept as lignite coal 50 tons, bituminous coal 200 tons.

Following precautions should be taken during coal storage for avoiding spontaneous combustion:

1. Exposed surface area of the bulk storage should be minimum.
2. Temperature is monitored and is not allowed to exceed safe limits.
3. Temperature is reduced by water spraying.
4. Coal is packed layer by layer and moving roller over each layer.
5. Top surface and sides should be covered with fine coal so that internal coal is protected against oxidation.
6. Coal should not be stored near heat source.
7. Underground or underwater storage is preferred for huge quantity over long time.

6.4.4 Transportation of Coal

The coal is supplied from the coal mines to the power station. As already mentioned that an annual consumption of coal for a 2000 MW power station is approximately 5 million tons or 20,000 tons per day over 250 days. Therefore, the transportation of coal is a major coal handling

problem. The method of transportation of coal to power station depends primarily on the location of the station.

The economics of coal transportation is compared with economics of electrical energy production at mine mouth power station and electrical power transmission over long distances.

Following are the methods of transportation of coal:

- | | |
|-------------------------|---------------------------------------|
| 1. Continuous transport | — Buckets on wireropes—Conveyor belts |
| 2. Road transport | — Tipping trucks |
| 3. Rail transport | — Rail wagons |
| 4. River transport | — Barges |
| 5. Ocean transport | — Barges—Cargo ships |
| 6. Pipeline transport | — Pipeline coal slurry transport |
| | — Pipeline gaseous coal transport |

Recently, the coal slurry transportation and gasified coal transportation technologies are being considered as economic alternatives, to transportation of solid coal. Coal slurry pipeline carries coal powder suspended in water. At sending end, the slurry is prepared and pumped. At receiving end, the slurry is filtered and the coal is reclaimed.

If the plant is situated on the bank of a river or near the sea shore, it is often economical to transport coal in barges from where the coal is then unloaded mechanically by cranes or grab buckets. The coal from the coal field to the sea port is usually transported by rail or conveyor.

Dedicated rail routes are most common. The number of wagons per train and amount of coal per train varies between 50 to 30 wagons and 3000 to 8000 tons of coal approximately.

6.5. DELIVERY AND STORAGE AT PLANT

6.5.1. Coal Delivery

1. By Ship. When the coal is transported by ship, the unloaded coal from the ship is either sent to storage yard or directly to the conveyor system which carries coal directly to the inplant storage near the boiler house.

2. By Rail. The coal supply to the Indian power plants is mainly by rail as the river transportation is not available. While transporting by rail, a railway siding line is taken to the power station and coal is either delivered to the storage yard or close to the point of consumption. The capacity of the railway siding should be equal to three days requirement of the plant.

Presently, in rail transportation, **Marry-go-round** system is being extensively used. The system uses 'hopper wagons' which are provided with bottom door by opening of which the coal can be discharged into a ground hopper while the train moves at a slow steady speed of about 1.0 kmph round a loop surrounding the coal store. These trains may be shuttling between the coal loading point of the mines and the unloading pointing at the power station. Coal discharges upto about 2000 tons per hour are possible with this system.

This system is suitable for round the clock operation. For this system, wagons can be loaded automatically by a sophisticated electronically controlled, integrated loading and weighing system capable of feeding 60 tons of coal into each wagon in one minute with the train moving at about 1.0 to 1.2 kmph.

3. By Ropeways. This method is very efficient for transporting the coal from the mine to power station, particularly when the distance between the mine and power station is less than 10 km. This system can supply the coal continuously with little manpower.

4. By Road. The transportation of coal by road is used only for small capacity plants located near the mines and where the traffic problems are not existing. The vehicles used for this purpose are self tipping usually by hydraulic cylinders actuated by the engine itself.

5. By Pipeline. Since the liquid fuel prices are increasing day by day, and at many locations, the existing railway facilities are not sufficient to cope up with the increasing demands of thermal power stations, the transportation of coal slurry by pipeline is considered most speedy method. This method shows great promise for future development.

In this system, the coal is washed and crushed to a suitable size (upto 14 mesh), and then mixed with water to form a homogeneous slurry. This slurry is then pumped at a pressure between 50 bar to 300 bar depending upon the distance and the type of terrain. The booster stations are provided when the pipe line is considerably long. The number of booster stations and their locations are decided as per the topography of the land along the pipeline.

At the delivery terminal the coal is dewatered and dried before use.

The coal transport through pipeline has following **advantages**.

1. This system is capable of transporting very large quantities continuously unaffected by the climate.
2. It is reliable and safe, because of absence of moving machines.
3. It can also be used in difficult terrain where use of other modes of transport is very difficult.
4. Manpower requirement is very low and maintenance charges are also low.
5. Loss or theft of coal during transport is eliminated.
6. It has least environmental hazards like, noise, dust problem and traffic congestion.
7. It requires very less areas as the site dumping and storage requirement is eliminated.

However, this system has following **disadvantages**:

1. This system requires large quantity of water to make slurry.
2. Preparation of coal at the pumping terminal, and dewatering and recovery of coal at the delivery terminal requires special arrangements.
3. This results in some loss of useful heat of coal due to added surface moisture of about 10%.

6.5.2 Coal Unloading

For unloading coal in all modern power stations, mechanical unloading devices are required as number of wagons arrived everyday is large. Following equipments are generally used for **unloading from railway wagons**:

- | | |
|---------------------------------------|-------------------------|
| (i) Unloading bridges or tower cranes | (ii) Grab buckets |
| (iii) Car shakers | (iv) Rotary car dumpers |
| (v) Coal accelerators etc. | |

When the coal is **transported by sea**, the unloading devices used are:

- (i) Unloading bridges or tower cranes
- (ii) Self unloading boats
- (iii) Coal accelerators
- (iv) Portable conveyors.

A travelling gantry crane on unloading bridges is used for unloading large coal carrying vessels. Some modern boats are equipped with their own unloading conveyors and long swinging belt conveyor boom which can be adjusted to carry the coal from the vessel to the desired storage point on the shore.

When the coal is **transported by road**, the use of tipping trucks with scoops is always recommended. As already mentioned, the method of transport of coal by road is employed only for small plants.

The type of coal unloading equipment used in thermal power plant depends upon the type of out-plant handling mode, *i.e.*, by road, rail or ship.

6.5.3. Storage at Plant

Coal reserves for large plants are usually stored outdoors next to the plant. Bulldozers are generally used as outside storage equipment and this enables coal to be pushed to which ever part of the stockyard is most convenient. It has the great advantage of flexibility of application.

The function / purpose of storage have already been discussed in earlier sections. The size of the storage near the plant depends upon the storage in the mining. In case the storage in the mining area is not sufficient, the storage near the plant should be large enough to meet 3 to 4 months requirement of the plant.

6.6 IN-PLANT HANDLING

6.6.1 Preparation

Though primary preparation and cleaning, as already discussed, should be done in coal fields but in case it is not done, it should be performed in the plant with final preparation. The cleaning and primary preparation is shown in Fig. 6.3.

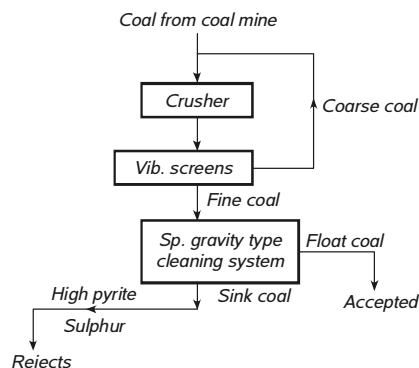


Fig. 6.3. Flow chart of primary preparation and cleaning

As seen in the figure crushing may be done before and/or after screening. The equipment used for crushing includes: rotary rushers, double roll crushers, hammer mills etc. Fine coals are washed in specific gravity type cleaning system wherein sulphur, sand and washable impurities like dirt, pyrite sulphur is removed from coal and specified purity is obtained in the form of floating coal, which is accepted.

The coal preparation plant can either be located near the coal receiving point or at the point of actual use *i.e.* just before indoor live storage (bunker). The coal preparation plant includes the following equipment:

- (a) Crushers
- (b) Sizers
- (c) Dryers and
- (d) Magnetic separators.

The coal crushers are necessary to prepare the coal of required size before feeding to the combustion chamber. The part of the coal obtained from mines which does not require sizing is by passed. The capacity of coal crushers (available upto 600 tons per hour) must be sufficient enough to meet the peak load requirements. The crushed coal is passed over the sizer (generally vibrating screen) which removes over sized coal and feedback to the crusher.

The sized coal is then further passed on to the drier to remove the free excess moisture from the coal. In the drier, hot flue gases are passed through the coal storage in closed spaces for removing the moisture from the coal. Before supplying the coal after preparation to the storage hopper, the iron scrap and metallic particles are removed with the help of magnetic separators. These iron particles though may not harm in mechanical stokers type of combustion chambers, but may choke the burners and may also increase the wear of the handling equipment. The flow diagram of the coal preparation plant is shown in Fig. 6.4.

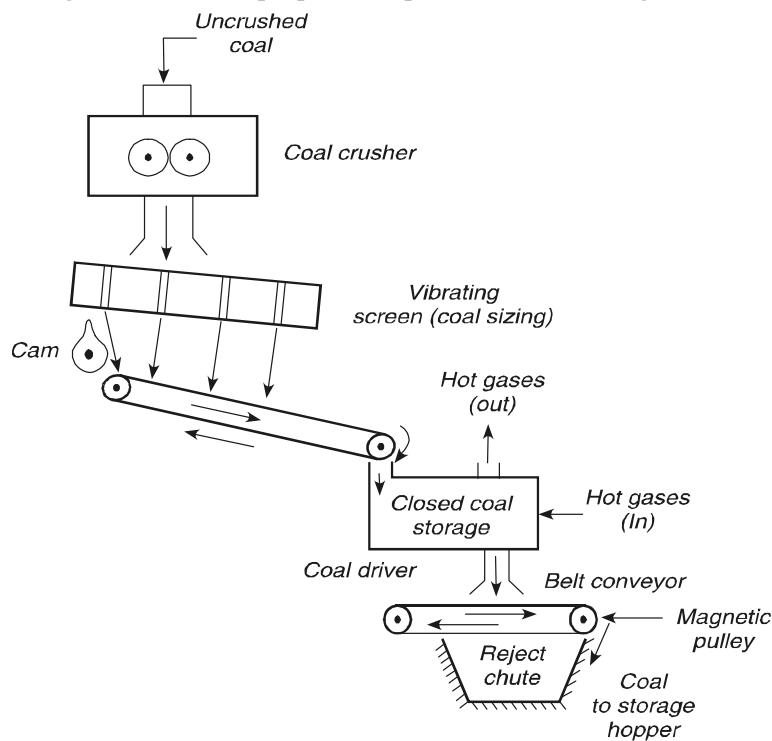


Fig. 6.4. Flow diagram of coal preparation plant.

6.6.2 Coal Transfer

After the coal is prepared, it is transferred to live storage (bunkers), for which one or more of the following equipment in suitable combination are used depending upon the local conditions:

1. Belt conveyors.
2. Screw conveyors.
3. Bucket elevators.
4. Grab bucket elevators.
5. Skip hoists.
6. Flight conveyors.

1. Belt conveyor. Fig. 6.5 shows a belt conveyor. It consists of an endless belt moving over a pair of end drums (rollers). At some distance a supporting roller is provided at the centre. The belt is made up of rubber or canvas. Belt conveyor is suitable for the transfer of coal over long distance. It is used in medium and large power plants. The initial cost of the system is not high and power consumption is also low. The inclination at which coal can be successfully elevated by belt conveyor is about 20° . Average speed of belt conveyors varies between 200—300 r.p.m. This conveyor is preferred over other types.

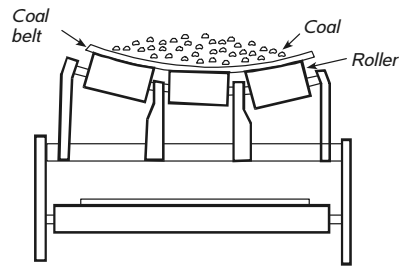


Fig. 6.5. Belt conveyor.

Advantages of belt conveyor

- (i) Its operation is smooth and clean.
- (ii) It requires less power as compared to other types of systems.
- (iii) Large quantities of coal can be discharged quickly and continuously.
- (iv) Material can be transported on moderate inclines.

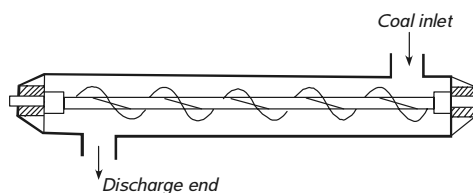


Fig. 6.6. Screw conveyor.

2. Screw conveyor. It consists of an endless helicoid screw fitted to a shaft (Fig. 6.6). The screw while rotating in a trough transfers the coal from feeding end to the discharge end.

This system is suitable, where coal is to be transferred over shorter distance and space limitations exist. The initial cost of the system is low. It suffers from the drawbacks that the power consumption is high and there is considerable wear of screw. Rotation of screw varies between 75—125 r.p.m.

3. Bucket elevator. It consists of buckets fixed to a chain (Fig. 6.7). The chain moves over two wheels. The coal is carried by the buckets from bottom and discharged at the top.

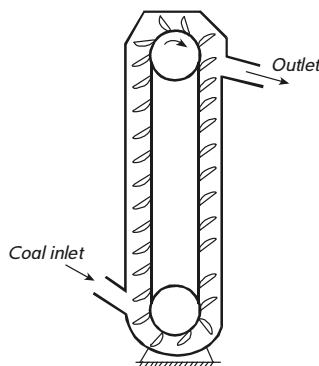


Fig. 6.7. Bucket elevator.

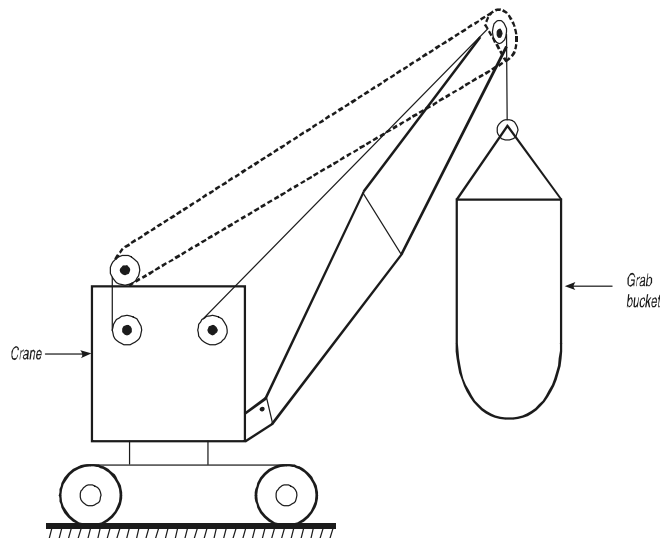


Fig. 6.8. Grab bucket elevator.

4. **Grab bucket elevator.** It lifts and transfers coal on a single rail or track from one point to the other. The coal lifted by grab buckets is transferred to overhead bunker or storage. This system requires less power for operation and requires minimum maintenance.

The grab bucket conveyor can be used with crane or tower as shown in Fig.6.8. Although the initial cost of the this system is high but operating cost is less.

5. **Skip hoist.** It consists of a vertical or inclined hoistway a bucket or a car guided by a frame and a cable for hoisting the bucket. The bucket is held in up right position. It is simple and compact method of elevating coal or ash. Fig. 6.9 shows a skip hoist.

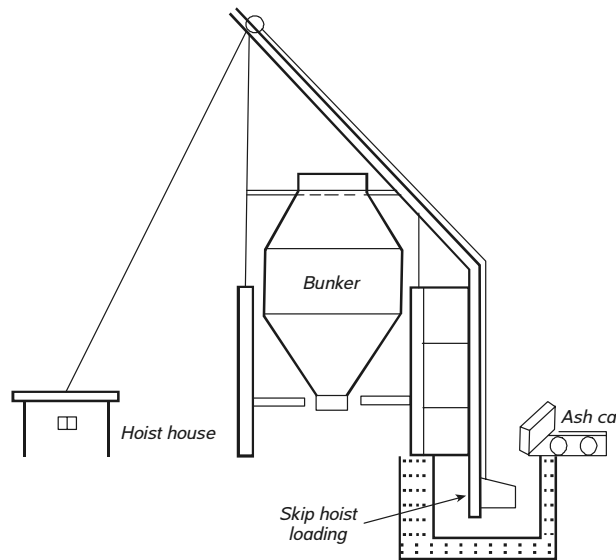


Fig. 6.9. Skip hoist.

6. **Flight conveyor.** It consists of one or two strands of chain to which steel scraper or flights are attached which scrap the coal through a trough having identical shape. This coal is discharged in the bottom of trough. It is low in first cost but has large energy consumption. There is considerable wear.

Skip hoist and bucket elevators lift the coal vertically while Belts and flight conveyors move the coal horizontally or on inclines.

Fig. 6.10 shows a flight conveyor. Flight conveyors possess the following advantages:

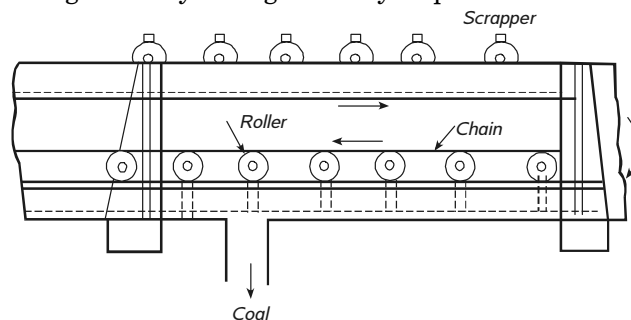


Fig. 6.10. Flight Conveyor.

- (i) They can be used to transfer coal as well as ash.
- (ii) The speed of conveyor can be regulated easily.
- (iii) They have a rugged construction.
- (iv) They need little operational care.

Disadvantages. Various disadvantages of flight conveyors are as follows :

- (i) There is more wear due to dragging action.
- (ii) Power consumption is more.
- (iii) Maintenance cost is high.
- (iv) Due to abrasive nature of material handled the speed of conveyors is low (10 to 30 m/min).

6.6.3 Live Storage or Covered Storage (or Boiler-room Storage)

This is the storage from which coal is withdrawn and supplied to combustion equipment with little or no rehandling. This storage, also known as bunker, consists of about 24 to 30 hrs of coal requirements of the plant and usually is in covered area in the plant near the boiler furnace. The live storage can be provided with bunkers and coal bins.

Bunkers are usually made of steel or reinforced concrete. From bunkers the coal is transferred to the boiler grates (mechanical stokers). A cylindrical bunker is shown in Fig. 6.11.

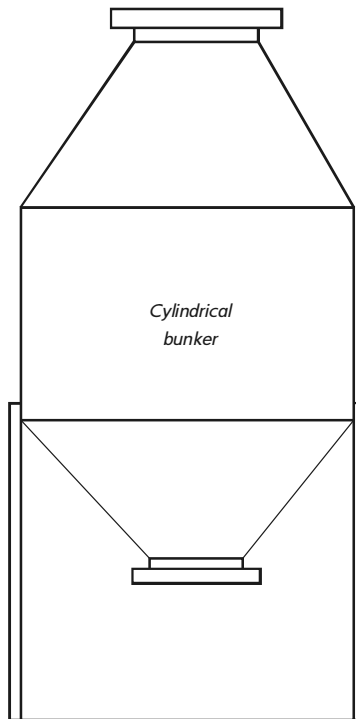


Fig. 6.11. A typical cylindrical bunker.

6.6.4 In-Plant Handling of Coal

An inplant handling system feeds the coal from covered storage (boiler room storage) to the firing equipment. In case of mechanical stoker firing, only chutes are required to feed the coal from storage bunker to the firing units. Gates and valves are provided in this system to control the flow according to the load of the plant.

The pulverized fuel firing system requires a number of equipments for inplant handling such as chutes, pulverized mills, feeders, weighing equipment, etc. The chutes are made of corrosion and abrasion resistant materials.

Inplant handling also uses the same equipment as used during transfer of coal, such as belt conveyer, screw conveyors, bucket elevators etc. Weigh lorries, hoppers and automatic scales are used to record the quantity of coal delivered to the furnace.

Coal weighing methods. Weigh lorries, hoppers and automatic scales are used to weigh the quantity of coal. The commonly used methods to weigh the coal are as follows:

- (i) Mechanical
- (ii) Pneumatic
- (iii) Electronic.

The mechanical method works on a suitable lever system mounted on knife edges and bearings connected to a resistance in the form of a spring of pendulum. The pneumatic weighters use a pneumatic transmitter weight head and the corresponding air pressure determined by the load applied. The electronic weighing machines make use of load cells that produce voltage signals proportional to the load applied.

As the cost of fuel used is a major operating cost of the power plant, it is necessary to exercise proper control over total quantity of coal used and to have an idea whether the coal supplied to the individual boiler is properly burned or not as per the load on the plant. The coal may be weighed at the unloading point and at individual stoker or pulverizer hoppers.

6.7 COAL CRUSHING AND PULVERIZING

We have already discussed pulverized coal firing system, and pulverized coal burners, in the previous chapter. Here in this section, we shall now discuss as to how the coal is pulverized.

After the preparation, the coal is taken to the crusher for being broken into the required feed size.

Pulverizing process of coal has three functions: (a) feeding, (b) drying, and (c) grinding. The feeding system controls the fuel feed rate according to the boiler demand and the required primary air rate for drying, and then transporting the pulverized coal and primary air stream to the burner. Dryers are an integral part of the pulverizing equipment. Adjustment of air temperature and quantity to suit the moisture content of coal and also to provide the requisite proportion in air fuel ratio is an essential function of the pulverizer. The grinding function is performed by impact, attrition, crushing, or combination of these. The main requirement of the grinding is that, it should not only be fine but also of a consistent degree of fineness.

For large scale generation of energy the efficient method of burning coal is confined still to pulverised coal combustion. The pulverised coal is obtained by grinding the raw coal in pulverising mills. The various pulverising mills used are as follows :

- (i) Ball mill,
- (ii) Hammer mill

(iii) Ball and race mill

(iv) Bowl mill.

The coal pulverising mills reduce coal to powder form by three actions as follows :

(i) Impact

(ii) Attrition (abrasion)

(iii) Crushing.

Most of the mills use above mentioned all the three actions in varying degrees. In impact type mills hammers break the coal into smaller pieces whereas in attrition type the coal pieces which rub against each other or metal surfaces to disintegrate. In crushing type mills coal caught between metal rolling surfaces gets broken into pieces. The crushing mills use steels balls in a container. These balls act as crushing elements.

1. Ball Mill. A line diagram of ball mill using two classifiers is shown in Fig. 6.12. It consists of a slowly rotating drum which is partly filled with steel balls. Raw coal from feeders is supplied to the classifiers from where it moves to the drum by means of a screw conveyor. As the drum rotates the coal gets pulverised due to the combined impact between coal and steel balls. Hot air is introduced into the drum. The powdered coal is picked up by the air and the coal air mixture enters the classifiers, where sharp changes in the direction of the mixture throw out the oversized coal particles. The over-sized particles are returned to the drum. The coal air mixture from the classifier moves to the exhauster fan and then it is supplied to the burners.

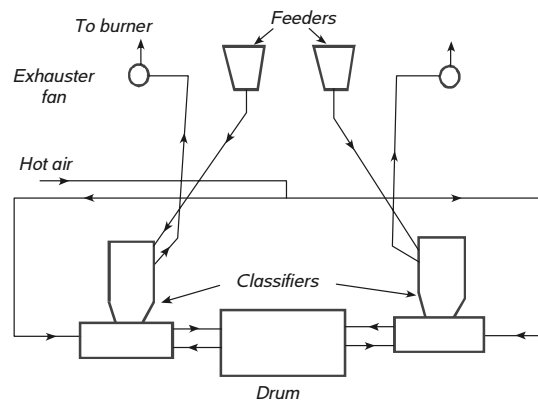


Fig. 6.12. Ball mill, line diagram.

2. Ball and Race mill. Fig. 6.13 shows a ball and race mill. In this mill the coal passes between the rotating elements again and again until it has been pulverised to desired degree of fineness. The coal is crushed between two moving surfaces, namely, ball and races. The upper stationary race and lower rotating race driven by a worm and gear hold the ball between them. The raw coal supplied falls on the inner side of the races. The moving balls and races catch coal between them to crush it to a powder. The necessary force needed for crushing is applied with the help of springs. The hot air supplied picks up the coal dust as it flows between the balls and races, and then enters the classifier. Where oversized coal particles are returned for further grinding, whereas the coal particles of required size are discharged from the top of classifier.

In this mill coal is pulverised by a combination of crushing, impact and attrition between the grinding surfaces. The advantages of this mill are as follows:

(i) Low capital cost,

(ii) Lower power consumption,

(iii) Lower space required,

(iv) Lower weight.

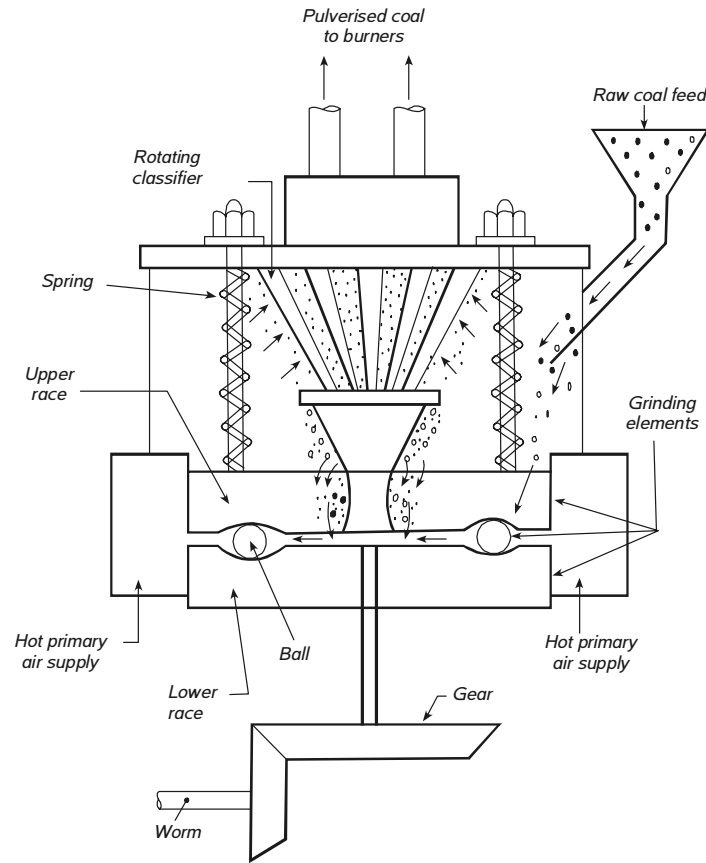


Fig. 6.13. Ball and race mill.

However, in this mill there is greater wear as compared to other pulverisers.

The use of pulverised coal has now become the standard method of firing in the large boilers. The pulverised coal burns with some advantages that result in economic and flexible operation of steam boilers.

Preparation of pulverised fuel with an intermediate bunker is shown in Fig. 6.14. The fuel moves to the automatic balance and then to the feeder and ball mill through which hot air is blown. It dries the pulverised coal and carries it from the mill to the separator.

The air fed to the ball mill is heated in the air heater. In the separator dust (fine pulverised coal) is separated from large coal particles which are returned to the ball mill for regrinding. The dust moves to the cyclone. Most of the dust (about 90%) from cyclone moves to bunker. The remaining dust is mixed with air and fed to the burner.

Coal is generally ground in low speed ball tube mill. It is filled to 20-35% of its volume. With steel balls having diameter varying from 30-60 mm. The steel balls crush and ground the lumps of coal. The average speed of rotating of tube or drum is about 18-20 r.p.m. (Fig. 6.15).

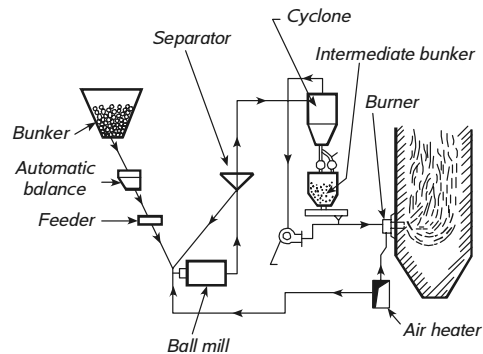


Fig. 6.14. Flow diagram of pulverized fuel.

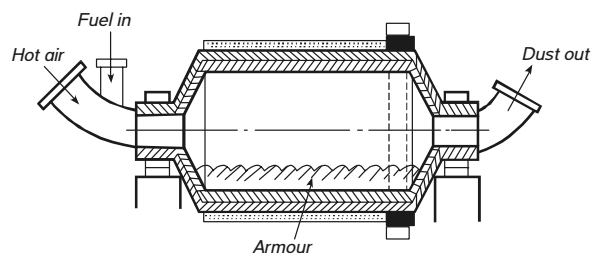


Fig. 6.15. Ball tube mill

3. **Shaft Mill.** Fig 6.16 shows fuel pulverisation with a shaft mill. The fuel from bunker is moved to feeder via automatic balance. Then from duct fuel goes to mill where it is crushed by heaters secured on the spindle of the mill rotor.

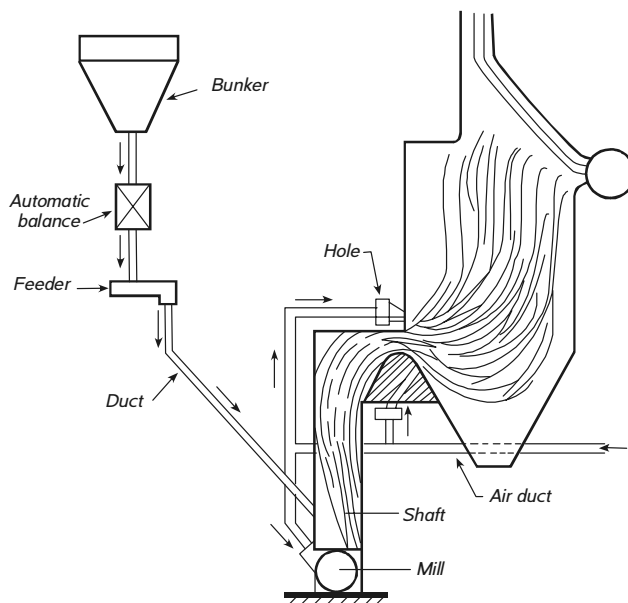


Fig. 6.16. Fuel pulverization with a shaft mill.

The pulverised fuel is dried up and then blown into shaft by hot air. Secondary air is delivered into the furnace through holes to burn the fuel completely.

6.8 ASH HANDLING

A large quantity of ash produced in steam power plants using coal. Ash produced is about 10 to 20% of the total coal burnt in the furnace. Handling of ash is a problem because ash coming out of the furnace is too hot, it is dusty and irritating to handle and is accompanied by some poisonous gases. It is desirable to quench the ash before handling due to following reasons:

1. Quenching reduces the temperature of ash
2. It reduces the corrosive action of ash
3. Ash forms clinkers by fusing in large lumps and by quenching clinkers will disintegrate, and
4. Quenching reduces the dust accompanying the ash.

Handling of ash includes its removal from the furnace, loading on the conveyors and delivered to the fill from where it can be disposed off. A station using low grade fuel is generally required to deal with large quantities of ash.

If station is located near the sea, it has the advantage of disposing ash in sea. In land stations ash is usually disposed off for use in brick making, building and road construction, industrial processing, filling up the disused quarries or low lying areas etc.

The ashes should be discharged and dumped at a sufficient distance from the power plant. Ash handling is a major problem due to the following **difficulties in the handling and disposal**:

- (i) It is hot when comes out of the boiler furnaces
- (ii) It is abrasive and wears out the conveyors, containers and other handling equipment.
- (iii) It produces dust nuisance.
- (iv) It produces poisonous gases.
- (v) It has corrosive action.
- (vi) It forms clinkers by fusing together in lumps.
- (vii) It is dusty, hence irritating and annoying to handle.

6.8.1 Principle requirements of an ash handling system

1. The ash handling plant should be able to handle large clinkers, boiler refuse, soot dust, etc., with minimum efforts and attention.
2. The plant should be able to deal with hot and wet ashes, and operate with minimum noise and within desirable / mandatory limits of dust nuisance.
3. Minimum operation and maintenance charges.
4. Minimum effect of abrasion.
5. Should be able to handle the maximum quantity of ash produced / likely to be produced.
6. It should have provision of meeting the additional capacity with minimum changes.

6.9 ASH HANDLING EQUIPMENT

Mechanical means are required for the disposal of ash. The handling equipment should perform the following functions: 1. Capital investment, operating and maintenance charges of the equipment should be low. 2. It should be able to handle large quantities of ash. 3. Clinders, soot, dust etc. create troubles, the equipment should be able to handle them smoothly. 4. The equipment used should remove the ash from the furnace, load it to the conveying system to deliver the ash to a dumping site or storage and finally it should have means to dispose of the stored ash. 5. The ash equipment should be corrosion and wear resistant.

Fig. 6.17 shows a general layout of ash handling and dust collection system. The commonly used ash handling systems are as follows :

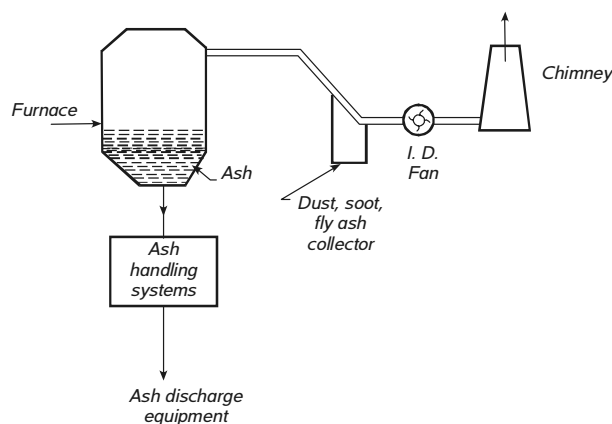


Fig. 6.17. Layout of ash handling and dust collection system.

- | | |
|-------------------------|-----------------------|
| (i) Hydraulic system | (ii) Pneumatic system |
| (iii) Mechanical system | (iv) Steam jet system |

6.9.1. Hydraulic system.

In this system, ash from the furnace grate falls into a system of water possessing high velocity and is carried to the sumps. It is generally used in large power plants. Hydraulic system is of two types, namely low pressure hydraulic system used for continuous removal of ash and high pressure system which is used for intermittent ash disposal. Fig. 6.18 shows hydraulic system.

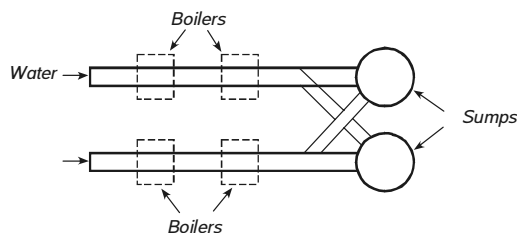


Fig. 6.18. Ash handling hydraulic system

In this method water at sufficient pressure is used to take away the ash to sump. Where water and ash are separated. The ash is then transferred to the dump site in wagons, rail cars or

trucks. The loading of ash may be through a belt conveyor, grab buckets. If there is an ash basement with ash hopper the ash can fall, directly in ash car or conveying system.

Water Jetting. Water jetting of ash is shown in Fig. 6.19. In this method a low pressure jet of water coming out of the quenching nozzle is used to cool the ash. The ash falls into a trough and is then removed.

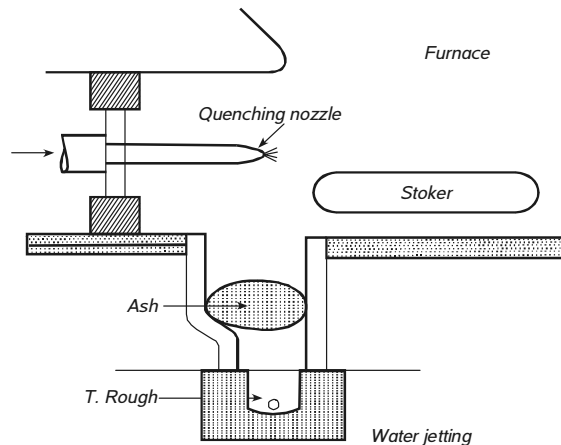


Fig. 6.19. Water jetting in ash handling.

Ash sluice ways and ash sump system. This system shown diagrammatically in Fig. 6.20 uses high pressure (H.P.) pump to supply high pressure (H.P.) water jets which carry ash from the furnace bottom through ash sluices (channels) constructed in basement floor to ash sump fitted with screen. The screen divides the ash sump into compartments for coarse and fine ash. The fine ash passes through the screen and moves into the dust sump (D.S.). Dust slurry pump (D.S. pump) carries the dust through dust pump (D.P.), suction pipe and dust delivery (D.D.) pipe to the disposal site. Overhead crane having grab bucket is used to remove coarse ash. A.F.N. represents ash feeding nozzle and S.B.N. represents subway booster nozzle and D.A. means draining apron.

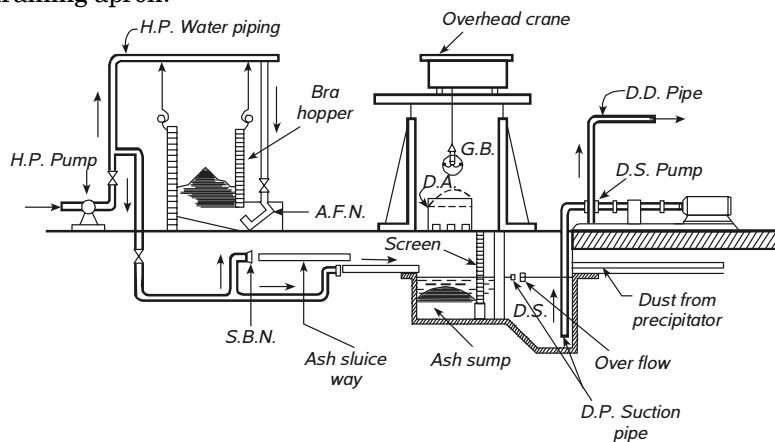


Fig. 6.20. Ash sump system.

The hydraulic system of ash handling is clean, dustless and healthy, because it is totally enclosed. Its ash carrying capacity is considerably large, therefore, it is more suitable for large

thermal power plants. This system is gaining more popularity in modern thermal power stations because of these advantages.

6.9.2. Pneumatic System

In this system (Fig.6.21) ash from the boiler furnace outlet falls into a crusher where larger ash particles are crushed to small sizes. The ash is then carried by a high velocity air or steam to the point of delivery. Air leaving the ash separator is passed through filter to remove dust, etc., so that the exhauster handles clean air which will protect the blades of the exhauster. A good installation of pneumatic system should avoid sharp bends and the lifts should be provided through a gradual sloping of the conveyor pipe.

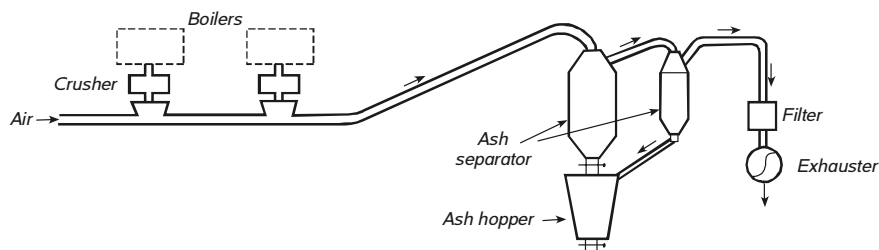


Fig. 6.21. Pneumatic system of ash handling.

This system has been specifically developed for the handling of both abrasive and very fine, dusty materials such as ashes, flyash, soot and fine coal. A pneumatic system has following elements:

- (i) Conveyor pipeline with feed inlets.
- (ii) Air tight receiver combined with an efficient separator.
- (iii) Air tight discharger gates for the receiver and separator.
- (iv) Air washer or filter.
- (v) Air exhauster.
- (vi) Storage bin with discharge gate and chute.

A pneumatic system has following *advantages*:

- (i) In this all dust nuisance is eliminated because the materials are handled in a totally enclosed conveyor conduit.
- (ii) The materials are conveyed in a dry condition, and therefore it is discharged freely by gravity.
- (iii) The entire conveyor layout is quite flexible and can thus made to fit varying physical plant conditions.
- (iv) It is easy to install and is economical.

This system has following *disadvantages*:

1. Large amount of wear in conveyor pipes.
2. High labour and maintenance charges.
3. It is comparatively dirty and noisy.

6.9.3. Mechanical ash handling system

Commonly used mechanical handling system include wheel borrows, bucket elevator, belt conveyors, bucket conveyor, skip hoist, rail road cars, motor trucks, barges etc.

- (a) **Wheel barrows.** The ash is collected directly through the ash outlet doors from the boiler into the containers / barrows moving on rails towards the duming site. This system is suitable for small stations.
- (b) **Bucket elevator.** This is useful equipment when vertical lift is involved in movement of ash. The bucket is attached to an endless chain and collecting the ashes under the boiler, elevating and dumping them in storage. This system is also limited to small power plants.
- (c) **Belt conveyors.** The ash must be thoroughly quenched before it is fed on to the conveyor belt. The ash is allowed to fall through a water seal over the belt conveyor and then is carried continuously towards the dumping site or overhead bunkers. The ash is carried to the dumping site from the ash bunker with the help of trucks. The control valve is opened and close manually to load the truck. This system is shown in Fig. 6.22.

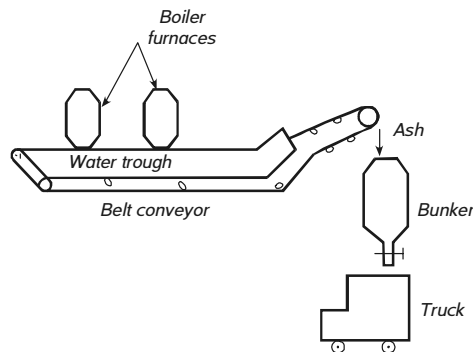


Fig. 6.22. Ash handling by belt conveyor.

The ash may also be conveyed to the dump site in *wagons* or *rail cars*. The loading may be through a belt conveyor, bucket elevator or direct by gravity from the boiler hoppers. This system has high initial cost.

6.9.4. Steam Jet System

This system utilizes jets of high pressure steam, blowing in the direction of travel through a conveying pipe in which the ash from the boiler ash hoppers is fed. The application of this system is for small and medium size plants. In this system ash can be removed economically through a horizontal distance of 200 meters and through a vertical distance of about 30 meters. This equipment can be installed in a limited space and also in awkward position. Moreover, the operation is noisy and the pipes carrying ash have greater wear and therefore, the pipe is required to be lined with nickel alloy.

6.10 USES OF ASH AND DUST

1. Ash is widely used in the production of cement. Fly ash is pozzolanic and develops self-hardening characteristics. Concrete made of ash has shown improved work ability and strength greater than all cement based concrete. The percentage of fly ash and/or bottom ash to be used in the production of concrete depends on quality of coal, combustion characteristics, as these effects the chemical and physical quality of ash.

2. Due to better alkaline values, ash is used for treating acidic soils. Its use for agriculture purposes have also been recently established. It has been found that ash supplies essential nutrients such as sulphur, calcium, zinc, etc. It has also been found that if the ash issued in limited quantity in soil, it increases the yield of corn, turnip, and white clover.
3. When about 10% of normal cement is added to pulverized fuel ash, the product acts as a hardening materials and can be used for factory floors, side roads, foot paths, road base and various other construction activities.
4. Pulverized fuel ash can be used as load bearing fill material.
5. It can also be used for building the blocks or bricks which can be obtained by adding it to clay.
6. It is used for filling up the disused quarries or abandoned mines.

6.11 DUST COLLECTION AND DISPOSAL

In coal fed furnaces the products of combustion contain particles of solid matter floating in suspension. This may be smoke or dust. The production of smoke indicates that combustion conditions are faulty and amount of smoke produced can be reduced by improving the furnace design.

In spreader stokers and pulverised coal fired furnaces the coal is burnt in suspension and due to this dust in the form of fly ash is produced. The size of dust particles is designated in microns ($1\mu = 0.001 \text{ mm}$). Dust particles are mainly ash particles called by ash intermixed with some quantity of carbon ash material called cinders. Gas borne particles larger than 1μ in diameter are called dust and when such particles becomes greater in size than 100μ they are called cinders.

Smoke is produced due to the incomplete combustion of fuels, smoke particles are less than 10μ in size. The disposal of smoke to the atmosphere is not desirable due to the following reasons:

1. A smoky atmosphere is less healthful than smoke free air.
2. Smoke is produced due to incomplete combustion of coal. This will create a big economic loss due to loss of heating value of coal.
3. In a smoky atmosphere lower standards of cleanliness are prevalent. Buildings, clothings, furniture, etc., becomes dirty due to smoke. Smoke corrodes the metals and darkens the paints.

To avoid smoke nuisance the coal should be completely burnt in the furnace. The presence of dense smoke indicates poor furnace conditions and a loss in efficiency and capacity of boiler plant. A small amount of smoke leaving chimney shows good furnace conditions whereas smokeless chimney does not necessarily mean a better efficiency in the boiler room.

To avoid the atmosphere pollution the fly ash must be removed from the gaseous products of combustion before they leaves the chimney.

The removal of dust and cinders from the flue gas is usually effected by commercial dust collectors which are installed between the boiler outlet and chimney, usually in the chimney side of air preheater.

The principal characteristics of an ash collector is the degree of collection.

η = Degree of collection

$$= \frac{G_1 - G_2}{G_1} = \frac{C_1 - C_2}{C_1}$$

where G_1 = Quantity of ash entering an ash collector per unit time (kg/s)
 G_2 = Quantity of uncollected ash passing through the collector per unit time (kg/s)
 C_1 = Concentration of ash in the gases at the inlet to the ash collector (kg/m³)
 C_2 = Ash concentration at the exist (kg/m³).

Depending on the type of fuel and the power of boiler the ash collection in industrial boilers and thermal power stations can be effected by mechanical ash collectors, fly ash scrubbers and electrostatic precipitators.

For fly ash scrubbers of large importance is the content of free lime (CaO) in the ash. With a high concentration of CaO the ash can be cemented and impair the operation of a scrubber.

The efficiency of operation of gas cleaning devices depends largely on the physico-chemical properties of the collected ash and of the entering waste gases.

Following are the principal characteristics of the fly ash:

- (i) Density
- (ii) Dispersity (Particle size)
- (iii) Electric resistance (For electrostatic precipitators)
- (iv) Coalescence of ash particles.

Due to increasing boiler size and low sulphur high ash content coal the problem of collecting fly ash is becoming increasingly complex. Fly ash can range from very fine to very coarse size depending on the source. Particles colour varies from light tan to grey to black. Tan colour indicates presence of iron oxide while dark shades indicate presence of unburnt carbon. Fly ash particles size varies between 1 micron (1 μ) to 300 μ fly ash concentration in flue gases depends upon mainly the following factors:

- (i) Coal composition
- (ii) Boiler design and capacity.

Percentage of ash in coal directly contributes to fly ash emission while boiler design and operation determine the percentage retained in the furnace as bottom ash and fly ash carried away by flue gas. Fly ash concentration widely varies around 20-90 gm/m³ depending on coal and boiler design. Fly ash particle size distribution depends primarily on the type of boiler such as pulverised coal fired boiler typically produces coarser particles than cyclone type boilers. Electrostatic precipitator (ESP) is quite commonly used for removal of fly ash from flue gases.

6.11.1 Types of Dust collectors

Dust collectors can be classified into two broad categories:

1. Mechanical Dust Collectors.
2. Electrical Dust Collectors.

Different types of dust collectors for different firing system are as under:

- (a) Gas and oil firing:
 - No dust collector s necessary.
- (b) Stoker firing:
 - Cinder catchers, cinder vane fans.
 - Centrifugal collectors (cyclonic separator).

(c) Pulverized fuel firing:

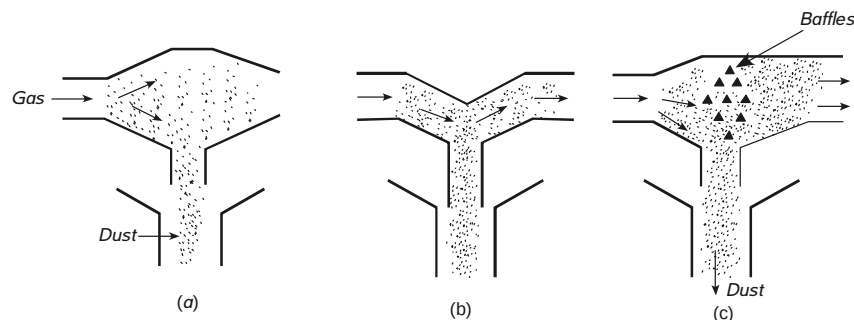
— Centrifugal collectors, wet scrubbers, electrostatic collectors.

1. Mechanical Dust Collectors. Mechanical dust collectors are sub-divided into : (a) dry type, and (b) wet types. In wet type collectors, also known as scrubbers, water sprays are used to wash dust from the flue gases.

(A) Dry type Mechanical Dust Collectors

Gravitational Separators. The basic principles of these mechanical dust collectors are shown in Fig. 6.23.

As shown in Fig. 6.23.(a) by increasing the cross-sectional area of duct through which dust laden gases are passing, the velocity of gases is reduced and causes heavier dust particles to fall down. Changing the direction of flow [Fig. 6.23.(b)] of flue gases causes the heavier particles to settle out. Sometime baffles are provided as shown in Fig. 6.23 (c) to separate the heavier particles.



(a) Sudden velocity decrease by enlarging section, (b) Abrupt change of flow and enlarging section, (c) Impingement upon small baffles.

Fig. 6.23. Principles used in dust separation.

1. Cinder Catcher. Cinder catcher (Fig. 6.24) is used to remove dust and cinders from the gas. In this catcher the dust laden gas is made to strike a series of vertical baffles that change in direction and reduce its velocity. The separated dust and cinders fall to the hopper for removal. Cinder catchers are ordinarily used with stoker firing.

2. Cinder Vane Fans. It is essentially an induced draft fan with an arrangement to withdraw part of the whirling gas from the casing through small tangential slotted openings. Fig. 6.25. shows dust laden gas from a stoker fired furnace entering the scroll outer side. While the gas sharply changes direction to enter the rotor, the centrifugal force carries the dust particles along the edge of the scroll.

3. Cyclonic (Centrifugal) Separators. In this type of dust collector, a high velocity gas stream carrying the dust particles enters at high velocity and tangential to the conical shell as shown in Fig. 6.26. This produces a whirling motion of the gas within the chamber and throws heavier particles to the sides and fall out of gas stream and are collected at the

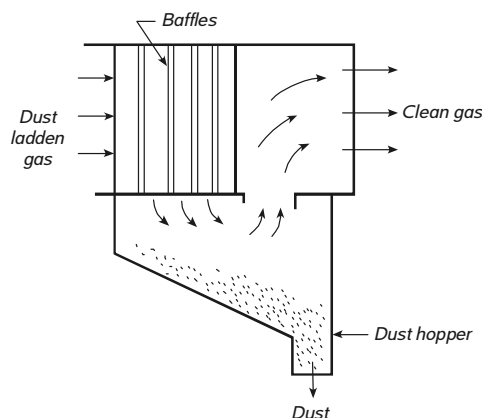


Fig. 6.24. Cinder Catcher.

bottom of the collector. These cyclone collectors can be used as single unit or multi unit in parallel.

The performance of these collectors depend upon flue gas volume, particulate loading, inlet velocity, temperature, diameter to height ratio of cyclone and dust characteristics.

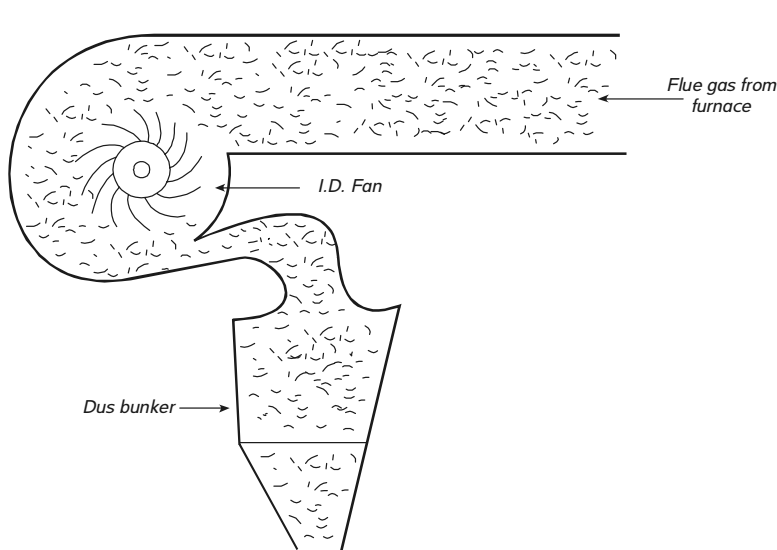


Fig. 6.25. Principle of cinder vane fan

4. Bag House Dust Collectors. Generally, low sulphur coal (less than 1%) and temperature under 300°C favour a bag-house collector (i.e. fabric filters), while higher sulphur content and higher temperatures favour an electrostatic precipitators. A well designed and maintained bag-house collector (particle size above $1\ \mu$) will collect 99.9% of dust and the efficiency is independent of the amount of dust in the flue gas.

Fabric filters are made of porous material which retains particulate matter while the carrier gas flows out through the voids. A fabric filter element is usually made in the form of a long, hollow cylindrical tube that provides a large surface area per unit of gas volumetric flow rate (sq. m/cum gas flow). The reciprocal of this parameter is called the air-to-cloth ratio. A fabric filter system usually contains a large number of fabric filter elements arranged in parallel rows, called a *bag-house*. A power plant bag house might contain several thousand vertical hollow cylinders, each ranging in diameter from 125 to 300 mm and upto 12.5 m in height.

(B) Wet type Mechanical Dust Collectors (Scrubber)

Fig. 6.27 shows a fly ash centrifugal scrubber. It is simi-

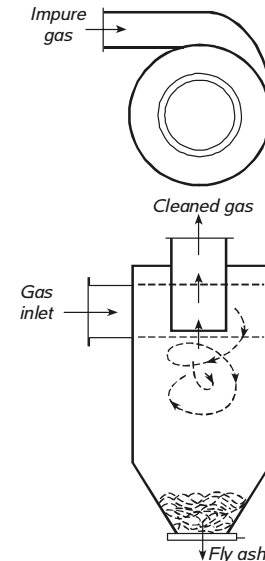


Fig. 6.26. Cyclonic separators

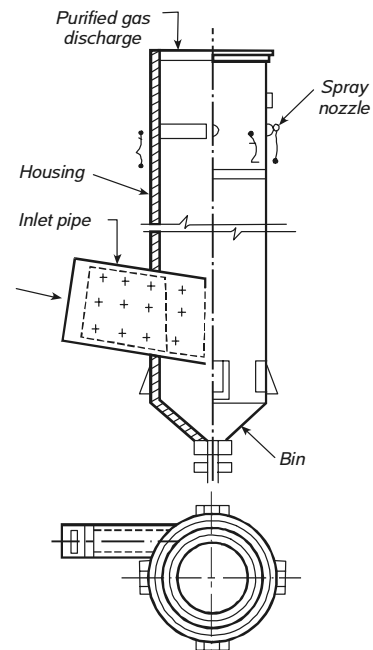


Fig. 6.27. Wet type dust collector

lar to a mechanical ash collector but has a flowing water film on its inner walls. Due to this film, the collected ash is removed more rapidly from the apparatus to the bin and there is less possibility for secondary capture of collected dust particles by the gas flow. The degree of ash collection in scrubbers varies from 0.82 to 0.90. The dust laden gas enters through the inlet pipe.

6.11.2 Electrostatic Precipitators (ESP)

It has two sets of electrodes, insulated from each other, that maintain an electrostatic field between them at high voltage (Fig. 6.28). The flue gases are made to pass between these two sets of electrodes. The electric field ionises the dust particles that pass through it attracting them to the electrode of opposite charge. The other electrode is maintained at a negative potential of 30,000 to 60,000 volts. The dust particles are removed from the collecting electrode by rapping the electrode periodically. The electrostatic precipitator is costly but has low maintenance cost and is frequently employed with pulverised coal fired power stations for its effectiveness on very fine ash particles and is superior to that of any other type.

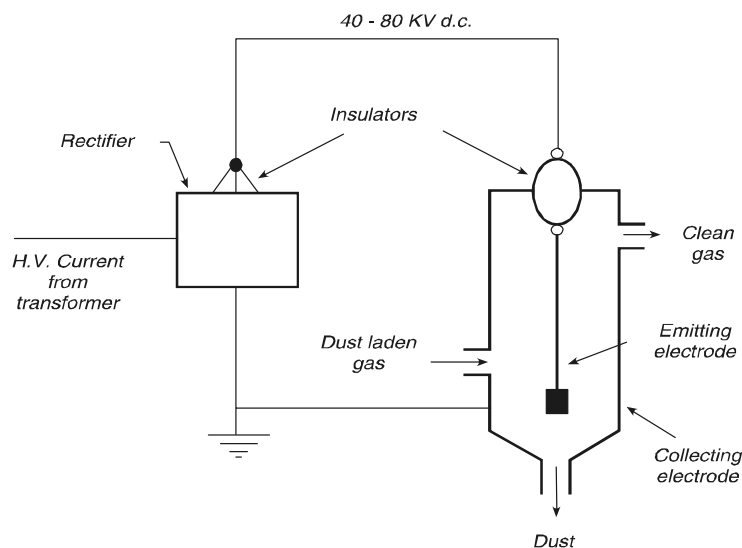


Fig. 6.28. Basic elements of ESP

An electrostatic precipitator has following basic components:

1. Source of high voltage
2. Ionizing and collecting electrodes
3. Dust removal mechanism
4. Shell to house the elements.

Gas Conditioning. Gas conditioning is the injection of small quantities of SO_3 into the flue gas. This reduces electrical resistivity of fly ash and thus makes the dust more amenable to collection in the electrical precipitator. Thus, it increases the collection efficiency of the precipitator.

6.11.3 Installation of Dust Collectors

Dust collectors are installed between the boiler outlet and the chimney; generally on the chimney side of the air-heater. In this case, the heat utilised for heating the air in the air-heater

is more as there is no loss of heat in collectors, as heaters are prior to the dust collectors. But some prefer keeping the dust collector before the air-heater because it maintains the air-heater more clean and reduces the cleaning charges. However, the general practice is to keep the dust collectors after the air-heater and use soot blowers to keep the heater surface clean.

6.11.4 Combined Operation of Dust Collectors

In order to take advantages of different types of dust collectors, these are used in various combinations. Some of the commonly used combinations are:

- (a) Combined cyclone and bag house collectors.
- (b) Combined dry mechanical and electrical collectors.
- (c) Combined wet mechanical and electrical collectors.

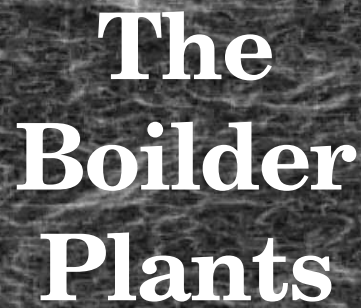
QUESTIONS

1. What are the advantages of mechanical handling of coal over manual handling? State the elements involved in the handling system.
2. Describe a modern method of coal handling in a 200 MW capacity power station.
3. Describe a coal handling system for any power station which you have visited.
4. What precautions should be observed in arranging storage of coal?
5. What are live and dead storage of coal in a power station? Why each of these is necessary in a large central power station?
6. Name the different types of coal conveyors. Describe the construction and operation of belt conveyor and screw conveyor.
7. Describe a grab bucket elevator.
8. What is coal beneficiation? Enumerate its major advantages of coal beneficiation.
9. What are the principal requirements of an ash handling system?
10. Describe a modern ash handling system in a coal operated power plant having 200 MW capacity.
11. Describe a pneumatic ash handling system for a thermal power station.
12. Name the various methods of ash handling. Describe the belt conveyor system of ash handling. Why it is essential to quench the ash before handling?
13. Write short notes on:
 - (a) Screw conveyor
 - (b) Bucket elevator
 - (c) Skip hoist
 - (d) Grab bucket conveyor
14. Describe the electrostatic precipitator (ESP) for dust collection in a steam power station.
15. What is a fabric filter? What do you mean by the term 'air-to-cloth ratio'?
16. (a) What are the different types of fabric filters?
(b) What is a bag house?
17. Explain the operation of an electrostatic precipitator.
18. What are the factors to be considered for an ash disposal system?
19. What are the uses of fly-ash?
20. Describe various mechanical devices used for coal handling in a large steam power plant including its transportation to storage, reclamation and inplant handling?

21. What do you understand by 'Outplant-handling of coal'? What are the different methods of outside coal handling.
22. What is necessity of coal storage? Discuss different methods used for coal storage at plant.
23. What is 'inplant coal handling'? Draw a neat line diagram of 'inplant coal handling' and indicate the names of equipment used at different stages.
24. Explain the different types of coal conveyors.
25. Why ash and dust handling is more difficult than coal handling?
26. What are the different ash handling systems? Discuss their relative merits and demerits?
27. How dust collection system differs from ash collection? Why it is more serious in case of pulverized coal-fired boilers?
28. Explain with neat diagram the working of different types of wet type mechanical dust collectors.
29. (a) What do you understand by the term 'performance of a dust collector'?
(b) Explain as to why the combined operation of different types of dust collectors is preferred.



7



The Boilder Plants

7.1 INTRODUCTION

The function of a boiler (or steam generator) is to convert water into steam at the required rate of evaporation and at the desired temperature and pressure to suit the turbine which it serves. In the boiler, the chemical energy of a fuel is liberated, safely, efficiently and economically. The basic components of a boiler are: fuel burning equipment, furnace and boiler surface. When superheated steam is required, superheater surface may be added. Feed water heating (economiser) surface, air heater surface, and steam resuperheating or simply known as (re-heating) surface may also be added as thermodynamic need and economic requirement.

In a boiler, furnace is a confined space where fuel burns. The furnace directs the energy released by the fire (combustion of fuel) to the heat transfer surface of the boiler, water walls, convection tubes, superheater, reheater, economiser and air preheater.

The fuel burning systems; namely, stoker firing and pulverized fuel firing have already been discussed. The remaining components of the boilers shall be discussed in this chapter.

Steam produced in the boiler is mainly required for power generation, process heating and space heating purposes. The capacity of the boilers used for power generation is considerably large compared with other boilers. Due to the requirement of high efficiency, the steam for power generation is produced at high pressures and in very large quantities.

Boiler is an apparatus used to produce steam. Thermal energy released by combustion of fuel is transferred to water which vaporises and gets converted into steam at the desired pressure and temperature. The steam produced is used for:

- (i) producing mechanical work by expanding it in steam engine or steam turbine
- (ii) heating the residential and industrial buildings
- (iii) performing certain processes in the sugar mills, chemical and textile industries.

Boiler is a closed vessel in which water is converted into steam by the application of heat. Usually boilers are coal or oil fired. A boiler should fulfill the following requirements:

- (i) **Safety.** The boiler should be safe under operating conditions.
- (ii) **Accessibility.** The various parts of the boiler should be accessible to the requirements.

- (iii) **Capacity.** The boiler should be capable of supplying steam according to the requirements.
- (iv) **Efficiency.** To permit efficient operation, the boiler should be able to absorb a maximum amount of heat produced due to burning of fuel in the furnace.
- (v) It should be simple in construction and its maintenance cost should be low.
- (vi) Its initial cost should be low.
- (vii) The boiler should have no joints exposed to flames.
- (viii) The boiler should be capable of quick starting and loading.

The performance of a boiler may be measured in terms of its evaporative capacity also called power of a boiler. It is defined as the amount of water evaporated or steam produced in kg per hour. It may also be expressed in kg per kg of fuel burnt or kg/hr/m² of heating surface.

7.2 CLASSIFICATION OF BOILERS

Boiler can be classified as under:

1. According to Flow of Water and Hot Gases

- (a) Water tube boilers
- (b) Fire tube boilers.

In water tube boilers, water circulates through the tubes and hot products of combustion flow over these tubes. In fire tube boiler the hot products of combustion pass through the tubes which are surrounded by water. Fire tube boilers have low initial cost, and are more compact. But they are more likely to explosion, water volume is large and due to poor circulation they cannot meet quickly the change in steam demand. For the same output the outer shell of fire tube boilers is much larger than the shell of water-tube boiler. Water tube boilers require less weight of metal for a given size, are less liable to explosion, produce higher pressure, are accessible and can response quickly to change in steam demand. Tubes and drums of water-tube boilers are smaller than that of fire-tube boilers and due to smaller size of drum higher pressure can be used easily. Water-tube boilers require lesser floor space. The efficiency of water-tube boilers is more.

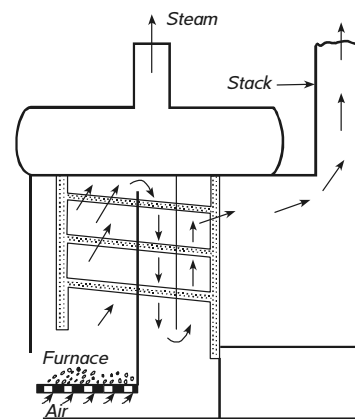


Fig. 7.1. Water tube boiler-principle.

The basic principle of water-tube boiler is shown Fig. 7.1 and that of fire-tube boiler is shown in Fig. 7.2. Examples of water tube boilers are Babcock and Wilcox and La Mont Boilers whereas Cornish, Locomotive and Cochran Boilers are fire tubes boilers.

Water tube boilers are classified as follow:

- 1. Horizontal straight tube boilers
 - (a) Longitudinal drum
 - (b) Cross-drum
- 2. Bent tube boilers
 - (a) Two drum
 - (b) Three drum
 - (c) Low head three drum
 - (d) Four drum
- 3. Cyclone fired boilers

Advantages: Various advantages of water tube boilers are as follows:

- (i) High pressure of the order of 200 kg/cm^2 can be obtained.
- (ii) Heating surface is large. Therefore steam can be generated easily.
- (iii) Large heating surface can be obtained by use of large number of tubes.
- (iv) Because of high movement of water in the tubes the rate of heat transfer becomes large resulting into a greater efficiency.
- (v) Generation of steam is much quicker due to small ratio of water content to steam content. This also helps in reaching the steaming temperature in short time.
- (vi) Heating surfaces are more effective as the hot gases travel at right angles to the direction of water flow.
- (vii) The combustion efficiency is higher because complete combustion of fuel is possible as the combustion space is much larger.
- (viii) Damage due to bursting of water tube is less serious.

Disadvantages

- (i) It is less suitable for impure and sedimentary water, as a small deposit of scale may cause the overheating and bursting of tube. Therefore, use of pure feed water is essential.
- (ii) There require more attention, hence the maintenance costs are higher.
- (iii) Failure in feed water supply even for a short period may overheat the boiler.

Fire tube boilers are classified as follows:

1. External furnace:
 - (i) Horizontal return tubular (ii) Short fire box (iii) Compact
2. Internal furnace:
 - (i) Horizontal tubular
 - (a) Short fire box (b) Locomotive
 - (c) Compact (d) Scotch
 - (ii) Vertical tubular
 - (a) Straight vertical shell, vertical tube
 - (b) Cochran (vertical shell) horizontal tube

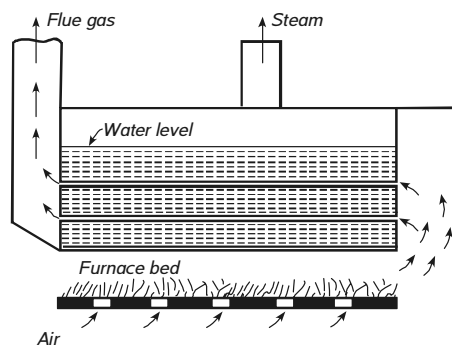


Fig. 7.2. Principle of fire tube boilers.

Advantages

Various advantages of fire tube boilers are as follow:

- (i) Low cost
- (ii) Fluctuations of steam demand can be met easily
- (iii) It is compact in size
- (iv) Failure in feed water supply for some time does not cause damage to boiler as it contains large quantity of water.

Disadvantages

The disadvantages of fire tube boiler are as follow:

- (i) The steam obtained is generally wet.
- (ii) Larger steam pressures can not be obtained. These boilers can produce a maximum pressure of about 17 kg/cm².
- (iii) More time is needed to generate steam. The output of the boiler is also limited.
- (iv) Fire tube boilers have large ratio of water to steam and therefore makes the boiler slow in reaching the operating pressure.
- (v) Explosion of the fire tube boilers becomes very serious because of its large water capacity.
- (vi) The transportation of fire tube boiler is very inconvenient because of large size of the shell.
- (vii) Slow rate of steam generation renders the fire tube boilers unsuitable for use in steam power plants.

2. Classification According to the axis of shell

- (a) Vertical boilers,
- (b) Horizontal boilers

In vertical boilers, the axis of the shell is vertical, whereas it is horizontal in case of horizontal boilers.

3. Classification According to the location of furnace

- (a) External fired boilers,
- (b) Internal fired boilers

In externally fired boilers, the furnace is placed outside the boiler shell. The advantage of this type is that the fire space can be enlarged easily. Water tube boilers are always externally fired. In internally fired boilers, the furnace is placed inside the boiler shell. Most of the fire tube boilers are of this type.

4. According to the method of water circulation

- (a) Natural circulation boilers,
- (b) Forced circulation boilers.

In natural circulation boilers, water is circulated by natural convection currents which are set up due to the temperature difference. Most of the low capacity boilers are of this type. In forced circulation boilers, water is circulated with the help of a pump driven by motor. The forced circulation is used in high pressure and high capacity boilers.

5. According to the Use Application of Boiler

- (a) Stationary Boilers,
- (b) Mobile Boilers

Most of the Industrial boilers and boilers used for power generation are of stationary type. The boilers used to run locomotives and ships are mobile boilers as they continuously move from one place to other.

6. According to Steam Pressure

- (a) Low pressure boilers
- (b) Medium pressure boilers
- (c) High pressure boilers

7.3 SELECTION OF BOILER

Selection of boiler depends upon the following factors:

1. Type of load to be supplied by the boiler.
2. Amount and type of fuel to be burnt.
3. Space available for boiler installation.
4. Cost of fuel.
5. Controls necessary to enable the load to be picked up quickly.
6. Desirability of using the heat recovery equipment such as air preheater, economiser etc.
7. Cost of boiler. The cost of boiler depends upon the following factors:
 - (a) Operating pressure
 - (b) Operating temperature
 - (c) Type of firing
 - (d) Efficiency of boiler
 - (e) Amount of steam to be generated.

7.4 COMMON TYPES OF BOILERS

7.4.1. Babcock and Wilcox Boiler.

Fig. 7.3. shows a Babcock and Wilcox boiler. It is a water tube boiler. It consists of a drum, super heater, headers, tubes etc. The headers are inter-connected by means of tubes which form the real heating element of the boiler. The fuel is burnt on the grate and hot gases of combustion pass over the tubes and heat the water circulating through the tubes. The baffles provide a Zig-zag path for the flue gases. The boiler is fitted with various mountings and accessories. The hot water and steam rise up through the uptake header into the boiler shell where steam separates from water and collects in the steam space. The cold water flows down into the tubes through the downtake header. Superheater is used to superheat the steam. The superheated steam can be taken out to the steam stop valve. Any sediment or mud in water gets collected in mud box and is blown off from time to time through the blow off valve.

The outstanding features of this boiler are :

1. The evaporative capacity of this boiler is high (20,000 to 40,000 kg/hr) and the operating pressure lies between 11.5 to 17.5 bar.
2. Draught loss is minimum compared with other boilers.
3. The defective tubes can be replaced easily.
4. The entire boiler rests over an iron structure independently of bricks work, so that the boiler may expand or contract freely.

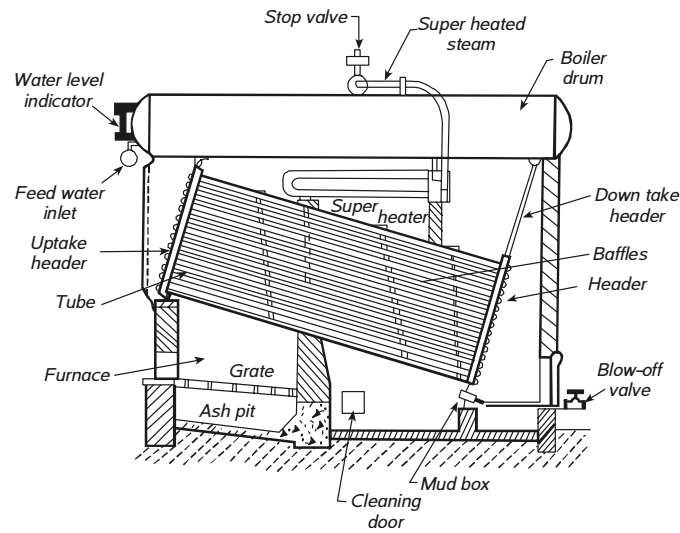


Fig. 7.3. Babcock and Wilcox boiler.

7.4.2. Cochran Boiler

A brief outline of this boiler is shown in Fig. 7.4. The coal burns on the grate inside the fire box. The flue gases after passing through the combustion chamber make their way through the fire tubes. The gases heat the water and convert it into steam which collects in steam space, and can be taken out through steam stop valve. The flue gases escape to the atmosphere after passing through smoke box and stack. Man-hole is provided to clean the boiler when necessary. A fire brick lining is provided to prevent the shell from being damaged. It is a vertical type boiler, requires lesser space and is quite compact in design. In such boilers the ratio of heating surface to grate area ranges between 10 to about 25 and steam can be produced upto a pressure of 10 kg/cm². This boiler is made in size upto about 2.743 metres diameter and 5.791 metres height and the maximum evaporative capacity from and at 100°C is 4730 kg per hour.

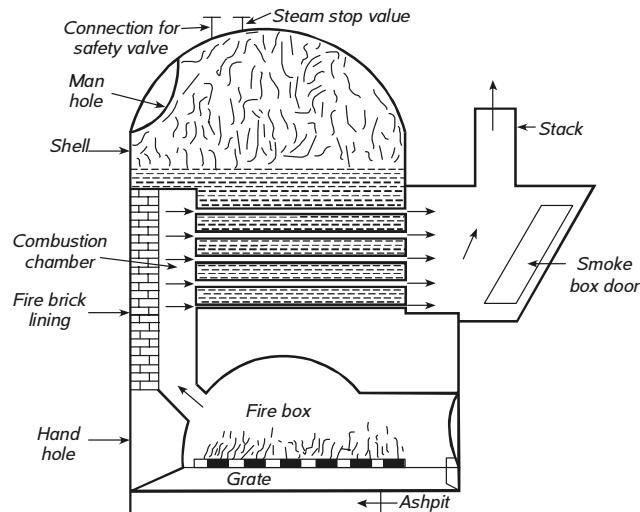


Fig. 7.4. Cochran Boiler.

The outstanding features of this boiler are:

1. It is very compact and requires minimum floor area.
2. Any type of fuel can be used with this boiler.
3. It is well suited for small capacity requirements.
4. It gives about 70% thermal efficiency with coal firing and about 75% with oil firing.

This boiler is provided with all required mountings such as : pressure gauge, water level indicator, safety valve, fusible plug, blow-off cock, steam stop valve, feed check valve.

7.4.3. Lancashire Boiler

Fig. 7.5. shows a brief outline of this boiler. The coal is fed through the fire hole on the grate (*G*) where its combustion takes place. F.B. represents fire-box. The flue gases move through the entire length of the furnace tube (*T*) upto the back end then move downward and flow along the bottom flue and finally leave to the atmosphere through a chimney. The surface tubes are nearly one metre in diameter, the diameter being larger at the front end and smaller at the rear end.

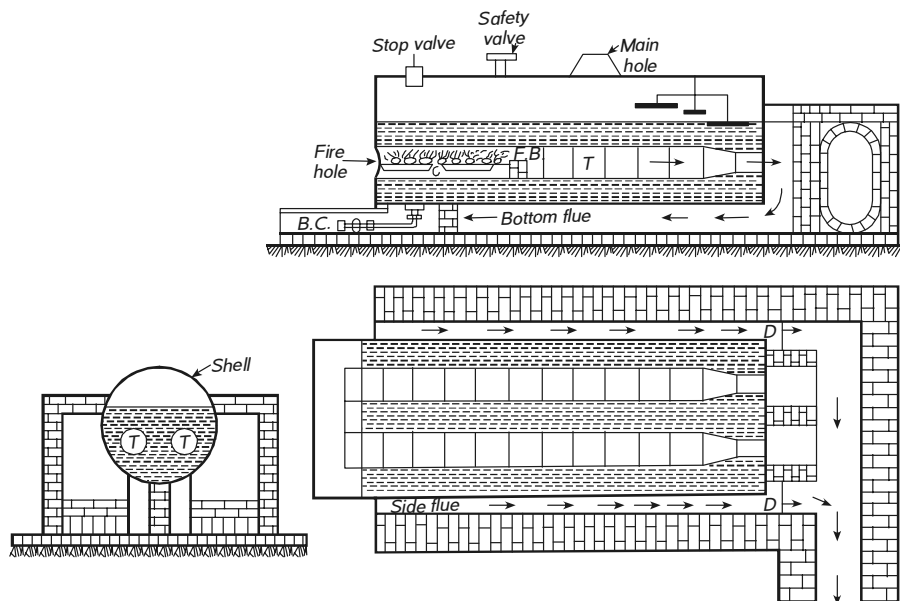


Fig. 7.5. Lancashire Boiler.

The two furnace tubes are surrounded by water on all sides. The boiler shell is filled with substantial quantity and is supported over the brickwork. The steam produced collects in the space above the water level. Blow Cock (B.C.) is provided to empty the boiler so that repairing, inspection or cleaning can be carried out when desired. Such boilers are generally constructed for working pressure upto 17.5 kg/cm^2 and evaporation capacity of 8300 kg/hour .

It is stationary fire tube, internally fired, horizontal, natural circulation boiler. It is widely used boiler because of its good steaming quality and its ability to burn coal of inferior quality. This boiler is fitted with usual mountings like, pressure gauge, water level indicator, steam

stop valve, safety valve, low water and high steam safety valve, manhole, blow-off cock, fusible plug and feed check valve.

Outstanding features of this boiler are:

1. Its heating surface area per unit volume of the boiler is considerably large.
2. Its maintenance is easy.
3. It is suitable where a large reserve of hot water is needed. Load fluctuations can be easily met by this boiler due to the large reserve capacity.
4. Superheater and economiser can easily be incorporated into the system, therefore the overall efficiency of the boiler can be considerably large (80-85%).

7.4.4 Scotch Marine Boiler

It is a fire tube boiler. It is compact, more efficient and has the ability to use any type of water. These boilers may be :

- (i) single ended, (ii) double ended.

Single ended boiler as shown in Fig. 7.6. may have one to four furnaces. A single ended boiler has length up to 3.5 metres whereas double ended boiler may be up to 6.5 metre long.

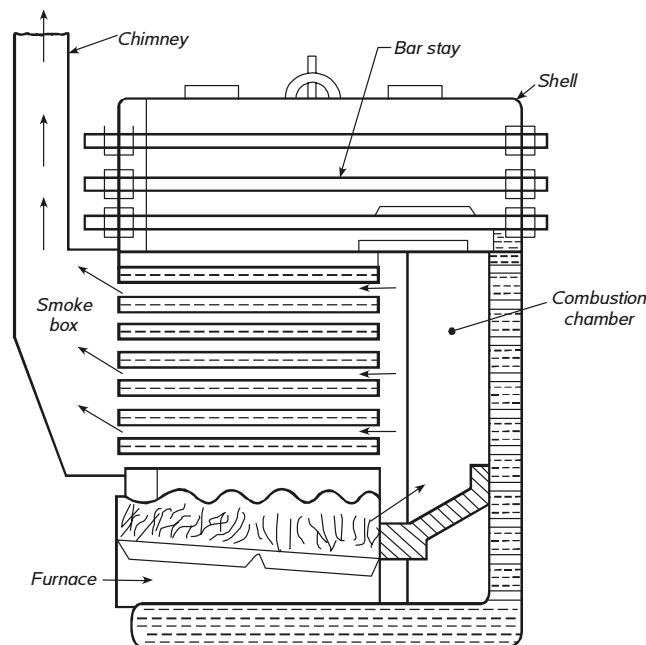


Fig. 7.6. Scotch marine boiler

The furnace tubes, smoke tubes and the combustion chamber, all being surrounded by water, give a very large heating surface area in proportion to the cubical size of boiler. The water circulates around the smoke tubes. The level of water is maintained a little above the combustion chamber. The flue gases, from the combustion chamber, are forwarded by draught through the smoke tubes, and finally up the chimney.

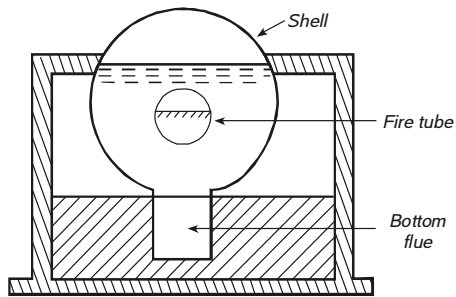


Fig. 7.7. Cornish boiler

7.4.5 Cornish Boiler

Fig. 7.7. shows this boiler. Its construction is similar to Lancashire boiler. It is a fire tube boiler. It has only one flue tube as compared to Lancashire boiler which has two flue tubes. The diameter of flue tube may be about 0.6 times that of shell. The diameter of boiler may be 1 to 2 m and length 5 to 7.5 metre. As compared to Lancashire boiler the capacity of cornish boiler is less.

7.5 WATER TUBE BOILERS

As already discussed, the fire-tube boilers are used to produce steam for industrial purposes, and are not used in power plants. These are still often used to produce saturated steam at the upper limits of about 18 bar pressure and 20,000 kg/hr steaming capacity.

The water tube boilers are universally used in the large capacities (usually pressure is above 17 kg/cm² and steam generation more than 7000 kg/hr). These boilers, in which water flows through the tubes and flue gases flow outside them, puts the pressure in the tubes and the relatively small diameter drums, which are capable of withstanding extreme pressures of the modern steam generator. Positive fluid circulation assures good heat transfer and enables building large capacity units. Tubes and drums are smaller in size as compared to fire-tube boilers, this enables high steam pressure feasible. Tubes, external to the drums, can be placed for best heat absorption.

Straight-tube Boilers. The water tube boilers may be straight tube or bent tube boilers. Babcock and wilcox boiler is a straight tube boiler. The straight tubes can be made, installed and replaced easily. Depending on capacity, one or more than one drum can be installed in parallel. Longitudinal drum boilers were limited to low pressures (12-23 bar) and steam capacities 0.6 to 10 kg/sec, having surfaces of 93 to 930 m². Cross-drum boilers can accommodate many more tubes than longitudinal drum boilers and built with heating surfaces of 93-2200 m², pressures 12-100 bar and steam capacities upto 63 kg/sec.

Bent-tube Boilers. Bent-tube boilers offer greater accessibility for inspection, cleaning and maintenance, and ability to operate at higher steaming rates and to deliver drier steam. In a bent-tube boiler, the tubes were so bent that they enter and leave the drums radially. These may have two, three or four drums. Lowest drum serves as mud drum and provided with blow-offs, and the uppermost last drum contains the driest steam, To increase the steaming and combustion rates, water walls are provided. More recent designs used water cooled furnace walls by lining the interior of the walls with tubes carrying the same boiler water, where partial boiling of water occurs.

Heat transfer. In a water-tube boiler, feed water is heated in three main kinds of heat exchangers *viz.*, economizer, evaporator, and superheater. Feedwater enters the economiser where it is heated by the outgoing flue gases till it is saturated liquid at that pressure (1-2) and then it is fed to the steam drum of the boiler. Saturated water falls through the downcomer into the bottom header and moves up through the riser where water is partially boiled back into the drum (2-3). Heat exchange circuit is shown in Fig. 7.8(a). Saturated steam from the drum goes to the superheaters for being heated to the desired temperature (3-4). The line 1-4 represents the entire heating process in T-S (temperature-entropy) diagram shown in Fig. 7.8(b).

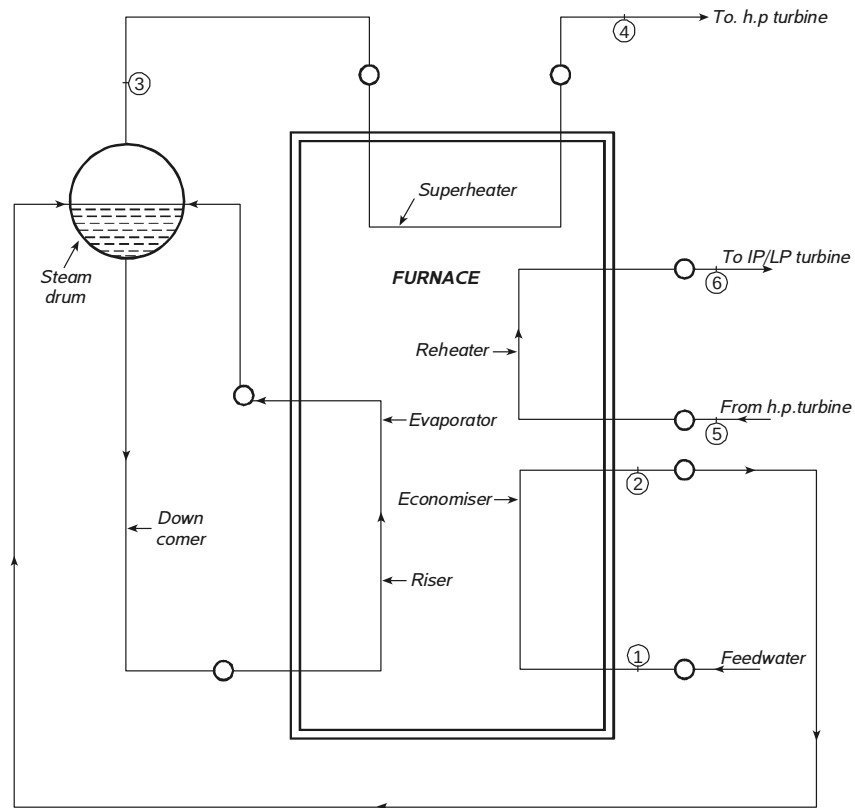


Fig. 7.8(a) Circuit of heat transfer in a water-tube boiler.

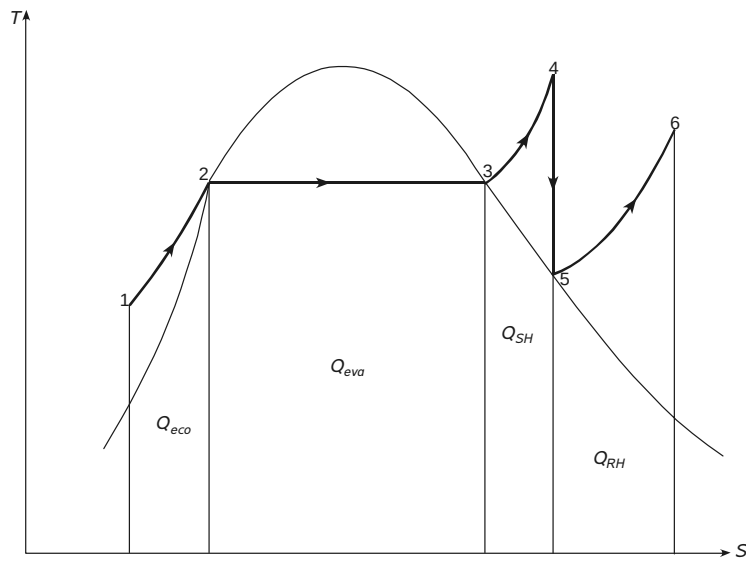


Fig. 7.8(b) Heat transfer in heat exchangers on T-S diagram.

For each kg of steam formed, the heat absorbed in the economiser (in the liquid phase), in the evaporator (liquid to vapour), and in the superheater (in the vapour or gas phase) are shown by Q_{eco} , Q_{eva} , Q_{SH} respectively in the Fig. 7.8(b). If a reheater is used, Q_{RH} indicates the heat absorbed in the reheater.

Heat transfer in different heat exchangers is given by:

$$Q_{eco} = h_2 - h_1$$

$$Q_{eva} = h_3 - h_2$$

$$Q_{SH} = h_4 - h_3$$

Circulation. Flow of water and steam within the boiler circuit is called circulation. Adequate circulation is necessary to carry away the heat from the furnace. If the circulation is caused by density difference, the boiler is said to have *natural circulation*. When the circulation is caused by a pump, it is known as *forced or controlled circulation*. Nearly saturated water falls by gravity from the drum through the downcomer into the bottom header and then flows up along the riser, where it partially boils, back into the steam drum, as shown in Fig. 7.8(a). The density of steam-water mixture in the riser is less than that of saturated water in the downcomer, and as a result of this density difference a circulation current is set up within the downcomer-riser circuit. The feedwater enters the drum from the economiser and saturated steam is taken out of the drum to the superheater.

The smaller the diameter of the risers, the larger the surface area exposed to hot gas for heat transfer. Therefore, the risers are of smaller diameter and larger in number compared to downcomers.

Steam Drum. The risers discharge a mixture of water, steam, foam and sludge into the steam drum. Any moisture carried with steam to the superheater tubes contains dissolved salts. In the superheater, the water is evaporated and salts are deposited on the inside surface of the tubes to form a scale, which is difficult to remove. This scale reduces the rate of heat absorption, and ultimately leading to the failure of the superheater tubes by overheating and rupture. Therefore, utmost care should be taken so that no damage is done to it by the excessive carryover of moisture with steam. The water should therefore be adequately treated prior to feeding it to the boiler as make up.

The main function of the steam drum are as following:

1. To store water and steam sufficiently to meet varying load requirement.
2. To aid in circulation.
3. To provide enough surface area for liquid-vapour disengagement.
4. To separate vapour or steam from water-steam mixture, discharged by the risers.

Steam-water Separation. Mechanical separators, also known as *drum internals*, like baffles, screens and cyclones are housed inside the drum for separation of steam-water mixture. Baffle plates, change or reverse the steam flow direction to assist gravity separation, act as primary separators. Screens made of wire mesh act as secondary separators where the individual wires attract and intercept the fine droplets. The accumulating water drops then fall by gravity back to the main water body. A boiler drum is said to be “priming”, if there is too much moisture carryover because of a high water level or a high steaming rate. Erratic feedwater control and rapid changes in steaming rate can induce priming. Therefore, a certain water level should be maintained to give adequate steam space and reduce priming.

Foaming is a condition resulting from the formation of bubbles on the water surface, and is responsible for the delivery of impure steam. Concentrated solid precipitates at drum bottom are removed through blowdown, which may be either intermittent or continuous. Blowing down of boiler water is the removal of some of the concentrated saline water and its replacement with relatively dilute feedwater. The amount of blow down is governed by the allowable solids concentration and the purity of feed water.

7.6 HIGH PRESSURE BOILERS

Special features of these boilers are: generating tubes, water walls, steam generating drums, forced circulation, small tubes, lack of drums, pressurised combustion, multiple tube circuits, extreme compactness and low weight. It is a common practice to use high pressure and temperature steam in power plants increase the efficiency of the plant and to reduce the cost of electricity production. The modern high pressure boilers used for power generation are of steam capacities 30 to 650 tons/hr and above with a pressure upto 160 bar and maximum steam temperature of about 550°C.

By using high pressure boilers, low grade fuels can be burnt easily, a saving in costly construction material can be achieved and a higher efficiency is possible. High pressure boilers are water-tube boilers, use pulverised coal firing and in such boilers furnace walls are water cooled and water tubes form the walls of the furnace. They use slag type furnace.

7.6.1 Unique Features of High Pressure Boilers

The high pressure boilers possess special features which make them more advantageous. The unique features are as follows:

1. Method of water circulation. In the boilers water circulation is maintained with the help of pump which forces the water through the boiler plant. However, sub-critical boilers may use natural circulation.

Circulation by a pump :

- (i) ensure positive circulation
- (ii) increases evaporative capacity of the boiler and
- (iii) requires less number of drums.

2. Type of tubing. These boilers generally use several sets of tubings through which water passes. This arrangement helps:

- (i) to reduce the pressure loss
- (ii) to achieve better control over the quality of the steam.

3. Improved method of heating. In high pressure boilers following improved methods of heating may be used to increase the heat transfer rate.

- (i) To save heat by evaporation of water above critical pressure of the steam.
- (ii) To heat water by mixing the superheated steam. It gives high heat transfer coefficient.
- (iii) To use high water velocity inside the tubes and to increase gas velocity above sonic velocity. This will increase over all heat transfer coefficient.

Advantages

The various features (advantages) of such boilers which make them useful are as follows:

- (i) In high pressure boiler pumps are used to maintain forced circulation of water through the tubes of the boiler. This ensures positive circulation of water and increases evaporative capacity of the boiler and less number of steam drums will be required.
- (ii) The heat of combustion is utilised more efficiently by the use of small diameter tube in large number and in multiple circuit.
- (iii) Pressurised combustion is used which increases rate of firing of fuel thus increasing the rate of heat release.
- (iv) Due to compactness less floor space is required.
- (v) It is economical to use high pressure and high temperature steam produced by such boilers.
- (vi) All the parts are uniformly heated, therefore, the danger of over heating is reduced.
- (vii) The efficiency of the plant is increased to about 40%.
- (viii) The tendency of scale formation is eliminated due to high velocity of water through the tubes.

7.6.2 Types of High Pressure Boilers

The various types of high pressure boilers commonly used are as follows:

- (i) La Mont Boiler
- (ii) Loeffler Boiler
- (iii) Benson Boiler
- (iv) Velox Boiler

(i) **La Mont Boiler** Fig. 7.9. shows the schematic arrangement of the boiler. Feed water from hot well is supplied to the separating drum (boiler) through the economiser. Water circulating pump maintains the forced circulation of water through the tubes in evaporation section. The steam produced flows to the drum. Dry steam from the top of drum is then passed through a convection superheater and the superheated steam flows to the turbine. This type of boiler

has a working pressure of 170 kg/cm^2 .

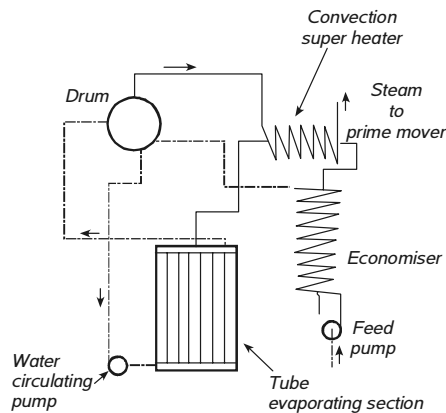


Fig. 7.9. La Mont Boiler

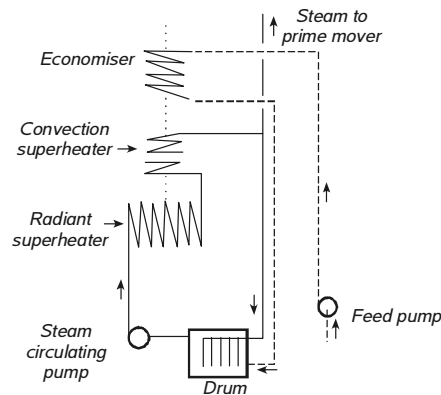


Fig. 7.10. Loeffler boiler

(ii) **Loeffler Boiler** In this boiler (Fig. 7.10) the drum is placed away from the furnace. Feed water passing through the economiser moves to the drum through which superheated

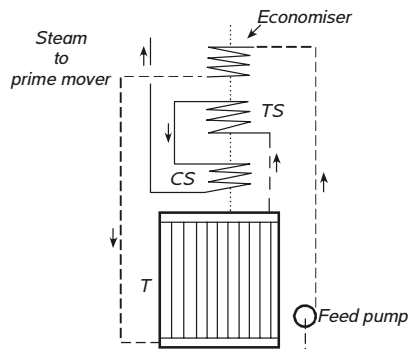


Fig. 7.11. Benson boiler

steam is also passing. The superheated steam converts the water into saturated steam. A steam circulating pump is used to circulate steam through radiant and convective superheaters, where steam is heated to the required temperature. A part of this steam flows to the prime mover and the remaining steam flows to the drum. Such boilers have a working pressure of 135 kg/cm^2 .

(iii) **Benson Boiler.** The schematic arrangement of the boiler is shown in Fig. 7.11. This boiler does not use any drum. The feed water after circulating through the economiser tubes flows through the radiant parallel tube section (T) to evaporate partly. The steam water mixture produced then moves to the transit section (TS) where this

mixture is converted into steam. The steam is now passed through the convection superheater (CS). Such boilers have a working pressure of 230 kg/cm^2 .

(iv) **Velox Boiler.** In this boiler (Fig. 7.12) pressurised combustion is used. The axial flow compressor (AC) driven by gas turbine (GT) raises the incoming air from atmospheric pressure to furnace pressure. The combustion gases after heating the water and steam flow through the gas turbine to the atmosphere. The feed water after passing through the economiser is pumped by a water circulating pump (WC) to the tube evaporating section (T). Steam separated in steam separating section flows to the superheater.

This boiler can generate a pressure of about 84 kg/cm^2 .

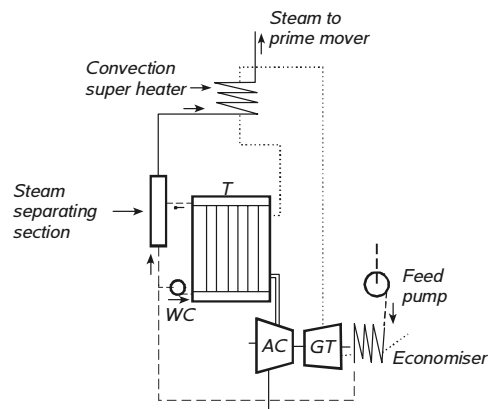


Fig. 7.12. Velox boiler

7.6.3 Modern Trends in Generating Steam

Some of the special features of generating steam using high pressure boilers are as follows:

- (i) Use of forced circulation of water inside the boiler.
- (ii) Use of small tubes in large number in order to increase the surface area of heat receiving surface.
- (iii) Treating the feed water thoroughly before use.
- (iv) Use of multiple tubes circuits.
- (v) Use of pressurised combustion.
- (vi) Use of superheaters.

The process of steam generation can be divided into three stages namely:

- (i) heating of water upto boiling point
- (ii) evaporation of boiling water and its conversion into dry saturated steam,
- (iii) transformation of dry saturated steam into superheated steam.

7.7 SLAG REMOVAL

Ash is removed in dry state in case of furnaces burning coals of high fusion temperature and in the liquid form if the coal used is of low fusion temperature. Ash when in the form of liquid slag is collected on the furnace by the various methods shown. Slag collected on the furnace floor is tapped off through a slag spout in the side wall (Fig. 7.13) or it may be removed at the rear of furnace (Fig. 7.14). Two-stage furnace arrangement (Fig. 7.15) and (Fig. 7.16) is also used to remove the slag. In primary furnace nearly complete combustion of fuel takes place and temperature is quite high and cooling takes place in secondary furnace.

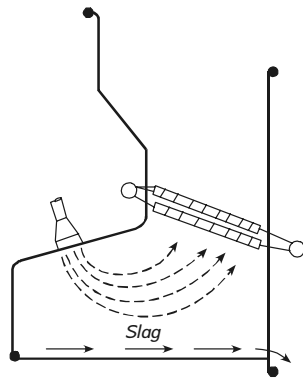


Fig. 7.13.

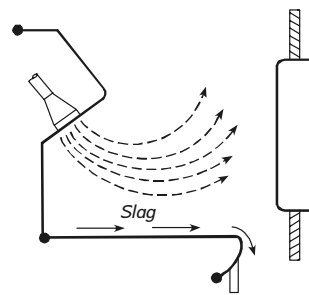


Fig. 7.14.

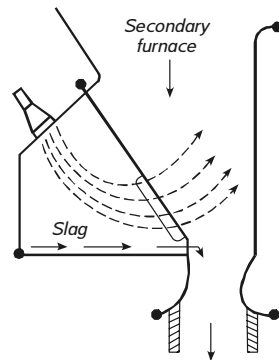


Fig. 7.15.

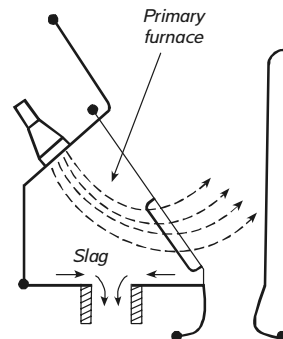


Fig. 7.16.

7.8 BOILER MOUNTINGS

Different fittings and devices necessary for the operation and safety of a boiler are known as boiler mountings. The safety valve, water level indicator and the fusible plug are the devices used for the safety of the boiler. The pressure gauge, feed check valve, blow off cock and steam stop valve fall under the category of fittings and are essential for the operation of the boiler. The functions of various boiler mountings are given here under:

1. **Pressure gauge.** A pressure gauge is used to measure the pressure of the steam inside the boiler.

2. **Water level indicator.** This indicates the water level inside the boiler. The water in the boiler should not fall below a particular level otherwise the boiler will be overheated and the tubes may burn out. Usually two water level indicators are fitted in front of the boiler.
3. **Safety valve.** The function of safety valve is to prevent an increase of steam pressure in the boiler above the safe design pressure. When the pressure increases above design pressure, the valve opens and discharges the steam to the atmosphere, and when the pressure falls just below design pressure, the valve closes automatically. A spring loaded safety valve is commonly used.
4. **Fusible plug.** When the temperature of the boiler shell increases above a particular level, the fusible plug, which is mounted over the grate, melts and forms an opening. The high pressure steam pushes the remaining water through this hole on the grate and extinguish the fire. Thus, it prevents the overheating of the boiler shell and tubes when the water level in the boiler falls below a predetermined level.
Under normal water level conditions in the boiler, this plug is covered with water which keeps the temperature of the fusible metal below its melting point. When the water level in the boiler falls and the plug is uncovered, the fusible metal quickly melts and the plug drops out.
5. **Blow-off cock.** Since the water supplied to the boiler always contains some impurities like mud, sand and salt, therefore on heating, these are deposited, at the bottom of the boiler. If these are not removed, they are accumulated at the bottom of the boiler and reduce its capacity and also heat transfer rates. In addition, the salt content goes on increasing due to evaporation of water. When the blow-off cock, located at the bottom of the boiler is opened during the running of the boiler, the high pressure steam pushes the water and the collected material at the bottom is blown out. The blow-off cock is operated at regular interval of 5-6 hours of working for few minutes to keep the boiler clean.
6. **Steam stop valve.** This valve is fitted to the highest point of the boiler shell. The function of the stop valve is to regulate the flow of steam from the boiler to the prime mover as per requirement and shutoff the steam flow when not required.
7. **Feed check valve.** The function of the feed check valve is to allow the supply of water to the boiler at a higher pressure than boiler pressure continuously and to prevent the back flow of water from the boiler when the pump pressure is less than boiler pressure or when pump fails.
8. **Anti-priming pipe.** The water particles are thrown up into the steam when the boiler is generating steam rapidly. These water particles are carried away with the steam. This is known as **priming**. As wet steam is not desirable when it is used for power generation in prime mover, it is necessary to remove these water particles from the steam. Therefore, an anti-priming pipe, which is a simple device to prevent the carrying of water particles with the steam, is used. An anti-priming pipe, which is perforated by providing slots or holes, is fitted into the boiler shell in the steam space and the steam has to pass through this before coming out through the steam stop valve. When the steam enters into the anti-priming pipe through the perforated holes, the water particles are agglomerated and fall back into the boiler.
9. **Man-hole.** Man-holes, generally elliptical in shape, are provided in the boiler shell so that the people can get inside the shell for inspection and maintenance.

7.9 BOILER WATER WALLS

Modern power stations have boilers with water walls which consists of a number of water or evaporation tubes (risers) spaced all around the walls of the furnace and carries away the heat released in the furnace by burning of fuel. The combustion space of a furnace is shielded wholly or partially by small diameter tubes placed side by side. The arrangement is mostly suitable for pulverised fuel firing system.

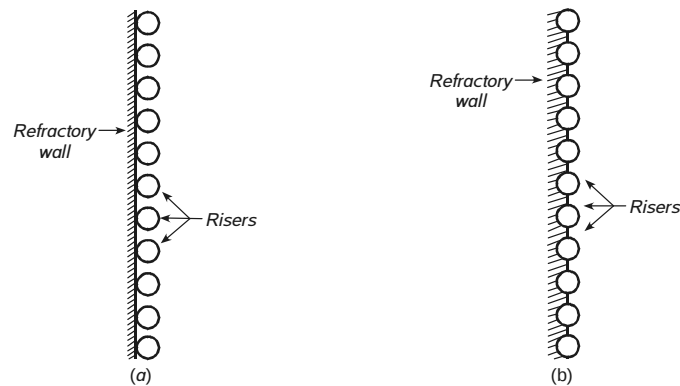


Fig. 7.17. Water-walls : (a) tangent tube touching the refractory wall, (b) tubes embedded in the refractory.

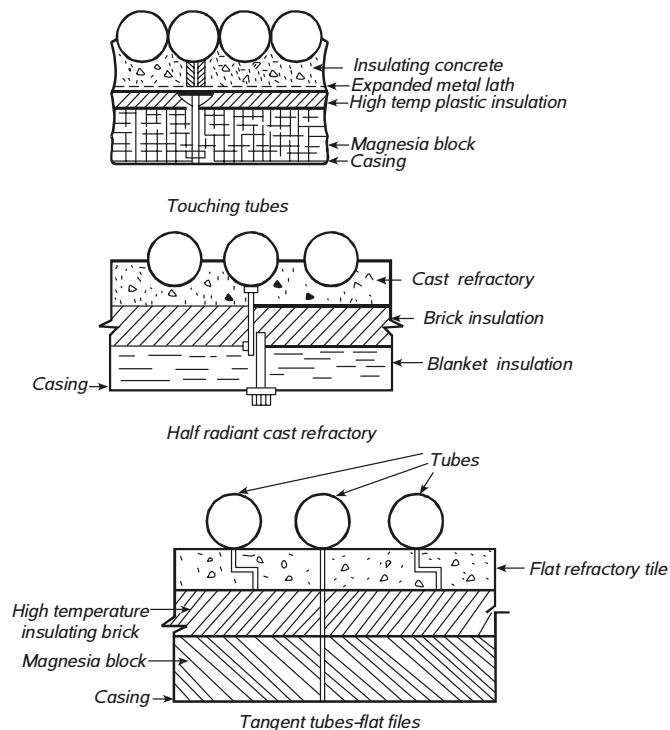


Fig. 7.17 (c).

The provision of water walls is **advantageous** due to the following reasons:

1. These walls provide a protection to the furnace against high temperature.
2. They avoid the corrosion of the refractory material and insulation.
3. Reduces furnace volume due to high heat transfer capacity.
4. The evaporation capacity of the boiler is increased.
5. Very high heat transfer rate.

The tubes are attached with the refractory materials on the inside or partially embedded into it. Fig. 7.17 shows the : (a) tangent tubes touching the refractory wall, and (b) tubes embedded in the refractory.

7.10 CIRCULATION OF WATER IN BOILERS

Reliable circulation of water in the boiler is essential for proper boiler design. Circulation means continuous flow of water in a closed circuit designed to create continuous motion of the steam and water in the boiler and intensively heat from the heating surface. Heat transfer rate is improved and boiler walls are protected from chemical destruction by continuous motion of water as it washes of the bubbles of steam from the heating surface. Circulation of water in boiler is carried out in two ways:

(i) Natural circulation, (ii) Forced circulation.

Natural circulation of water is based on the different densities of water and steam and water mixture. Fig. 7.18 shows the simplest natural circulation arrangement consisting of a drum, a header and two pipes. As heat is supplied, the steam starts forming in left hand pipe. The steam and water mixture rises and separates into steam and water, steam accumulating in the drum above water. The amount of steam produced in the right hand pipe is small and also formation of steam is less intensive. Due to this reason the density of steam and water mixture in right hand pipe will be greater than in left hand pipe and therefore a continuous flow of water from right hand pipe to left hand pipe, will take place.

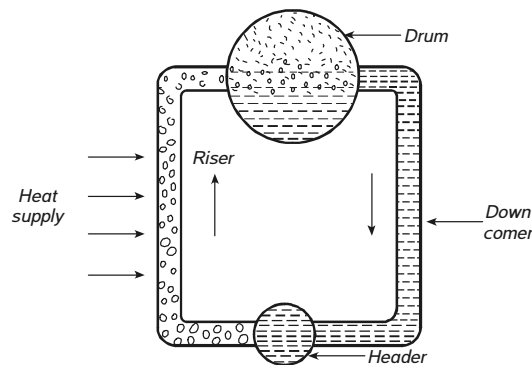


Fig. 7.18. Natural circulation.

In **Forced circulation** of water, pumps are used to maintain the continuous flow of water in the boiler. The relation between the amount of water passing through a circulation circuit in a definite time and the amount of steam generated in the circuit during the same time is called circulation ratio. The value of circulation ratio varies from 8 to 50 in natural circulation

water tube boilers. Boilers with natural circulation arrangement require larger diameter tubes and also natural circulation system does not ensure safe and reliable operation. Boilers with forced circulation require smaller diameter tubes thereby reducing the weight of boiler. The system is helpful in giving the most rational shape to the boiler, providing the required rate of circulation and in such boilers, circulation ratio is small.

7.10.1 Once-through Forced Circulation Boilers

Modern boilers use forced circulation rather than the natural circulation. In these boilers, a pump is used below the down comer for circulating water through the risers. These are used at very high pressures (generally above 160 bar). These are often called pump-assisted circulation units. Both these natural and pump-assisted circulation boilers are designated as recirculating units, since only a fraction of water is evaporated in the riser tubes, and the steam separated in the drum proceeds to the superheater.

In an *once-through* (OT) boiler, the feedwater goes through the economiser, furnace walls, and superheater sections, changing sequentially to saturated water, saturated steam, and superheater steam in one continuous pass (Fig. 7.19).

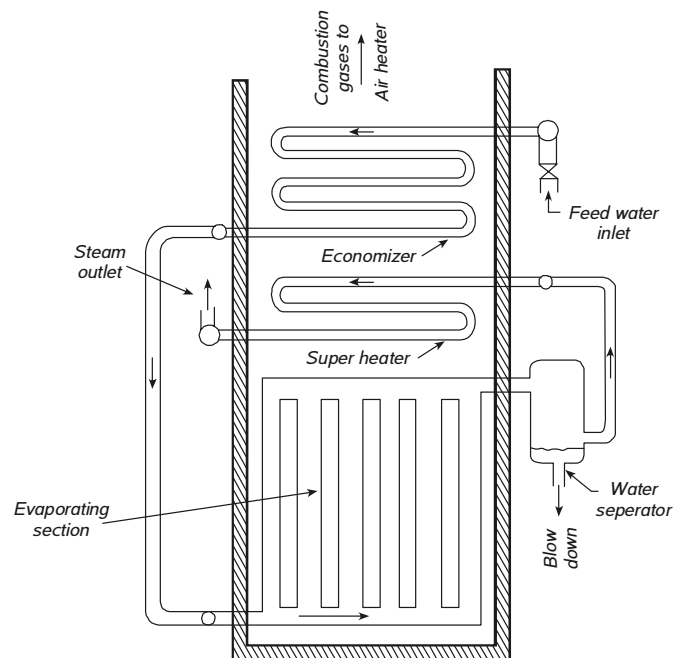


Fig.7.19. Once-through forced circulation boiler.

In these, no steam drum is required to separate saturated steam from boiling water and no water recirculation takes place. These boilers are the only type suited to *super-critical pressure operation* (above 221.2 bar) because there is no latent heat of vaporisation involved, and water on being heated at this pressure suddenly flashes into vapour, there being no saturation temperature as such, and the vapour is further heated to the desired temperature in the superheaters. Although there is no drum, but there should be a transition section where the water is likely to flash in order to accommodate the large increase in volume.

In controlled circulation boilers, it is possible to utilise high steam pressure between 150 to 200 bar, if water walls is used and by providing boiler circulating pump. Upto 150 bar the boilers can have natural circulation, as above this pressure the 'film boiling' may result, in which tube metal remains in contact with steam bubbles and therefore tube metal sharply deteriorates due to high metal temperature leading to boiler tube failures.

Beyond 200 bar when a drumless boiler is envisaged, a separator vessel is used to separate out salts from steam water mixture in sub-critical range (< 225 bar), but in super critical range (> 225 bar) the separator vessel cannot function and only 'once through (monotube) boiler is adopted.'

It is now a general practice that 800 MW units are designed on supercritical steam pressure with a drumless boiler on once through principle.

Major difficulties faced during the design of once through boilers are:

1. The importance of purity of feed water and make-up water increases many folds because of elimination of boiler drum and separator vessel in supercritical boilers. Volatile internal treatment with hydrazine and NH_3 is to be used for internal cleaning of boilers.
2. The main limitation in the design of high pressure, high temperature boiler is the availability of suitable materials. The temperature limit of ferritic materials is 580°C are as such authentic steels are to be used for parts where metal temperature exceeds this limit. Therefore, the adoption of once through boilers require easy availability of suitable materials within the country.

Advantages of once through boilers

Once through boilers used for large thermal unit have following main advantages:

1. Highest pressure can be used to achieve high thermal efficiency.
2. Full steam temperature can be maintained over a wider load range in once through boiler.
3. Once through circuit permits greater freedom in arrangement and location of heating surfaces.
4. Elimination of heavy drum decreases the metallurgical sensitivity of boiler against pressure changes.
5. Its size is smaller and requires about 60% space. Its weight is also less and requires about 65% foundation cost as compared to natural circulation boilers.
6. Faster start-up and cooling down of the boiler is possible.
7. Steam temperature can be easily controlled during start-up and shut-down in accordance with predetermined characteristics.
8. Variable pressure operation can be easily adopted for better performance at part load operation.

A flow diagram for natural circulation and also for assisted circulation boilers (designated as recirculating units) having steam drum for steam separation before proceeding to superheaters is shown in Fig. 7.20(a). Flow diagram for once-through (OT) boiler is shown in Fig. 7.20 (b).

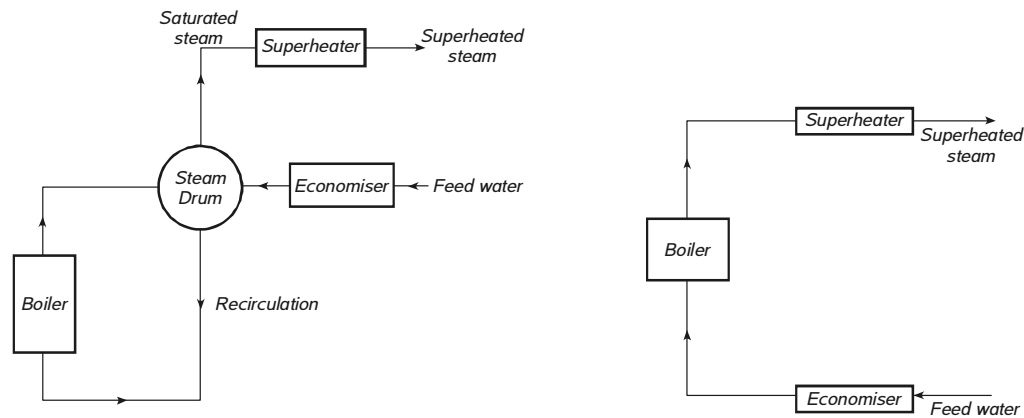


Fig. 7.20. Schematic flow diagram (a) drum type, and (b) Once-through boilers.

7.11 FEED WATER REGULATORS

Water, fuel and air are the three variables entering into the production of steam of boiler. It is necessary that feed water should flow into the boiler almost as rapidly as steam flows out. Maintaining a fixed water level is not advisable. Every change in load on boiler must be met with a change in feed water input. Feed water regulation should be automatic as it cannot be done successfully by hand.

The commonly used feed water regulators are of following types:

- (i) Float type.
- (ii) Vapour pressure type.
- (iii) Thermostatic expansion type.

Fig. 7.21 shows a float type feed water regulator. This regulator has a float chamber piped to boiler drum. It is installed opposite the normal water level of the boiler so that level can be duplicated in the float chamber. With the change in level the motion of float mechanically opens or closes a balanced regulating valve in feed water line.

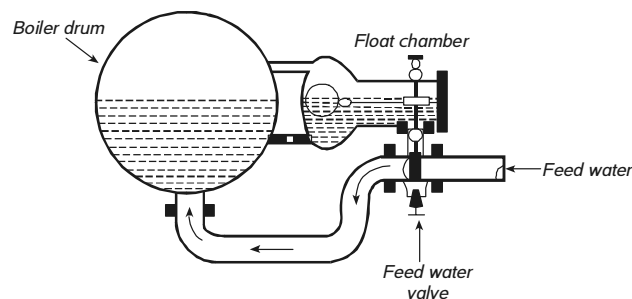


Fig. 7.21. Float type feed water regulator.

7.11.1 Feed Water Control

The supply of feed water depends upon the plant load. Some of the automatic method used to control feed water flow are as follows:

- (i) Single element pilot operated system
- (ii) Single element self operated system
- (iii) Two element pilot-operated system
- (iv) Three element pilot-operated system

Fig. 7.22 shows a single element pilot operated feed water control system. This allows manual correction of water level with flow variation. In this system the water level recorder (*R*) operates an air pilot valve which through a relay and indicator (*P*) controls the feed water regulating valve.

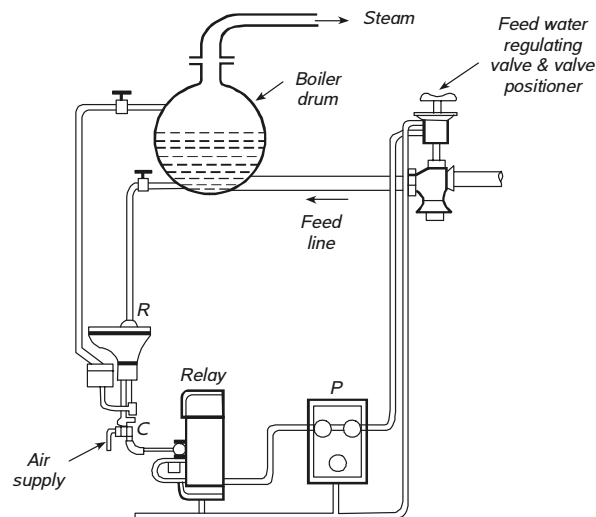


Fig. 7.22. Feed water control system.

The pilot valve (*C*) can be automatically reset and this makes it possible to hold boiler drum level constant at all rates of steam flow. Pilot valve's manual adjustment allows lowering the water level at low load and *vice versa*.

7.12 FEED WATER TREATMENT

Natural water supplies contain solid, liquid and gaseous impurities and as such this water cannot be used for the generation of steam in the boilers. This requires that the impurities present in the raw water should be removed before it can be used in the boiler.

Although in steam power plants the main condensate returns to the boiler as feed water but make up water is needed to replace the losses due to blow down, leakage, etc., in the cycle.

The various impurities present in the natural water (raw water) may be in the following forms:

- (i) Dissolve salts such as carbonate, sulphates chlorides of calcium, sodium and magnesium. Sometimes some iron, aluminium or silica salts are also present
- (ii) Dissolved gases such as carbon dioxide, oxygen and SO_2
- (iii) Mineral acids
- (iv) Suspended matter such as alumina and silica may be present as mud and salt.

In the water intended for steam boilers all the impurities are harmful especially if present in considerable quantities. The salts of calcium and magnesium are extremely harmful and when water containing them is heated and steam generated they precipitate as solid residue and form a hard scale on wall surface. The scale so formed is not desirable as it hampers the process of heat transfer.

Presence of mineral acids in water is always undesirable as it may result in chemical reaction with the boiler materials

These impurities may cause the following troubles:

- | | |
|---------------------------|---------------------|
| (i) Scale formation | (ii) Corrosion |
| (iii) Foaming and priming | (iv) Embrittlement. |

1. **Scale formation.** Impurities present in the water may cause scale formation in the boiler drums or header tubes and feed water piping system. This will reduce the heat transfer rate and will cause over-heating of tubes which may result in blistering and rupturing. The scale formation in the feed water pipes chokes the flow which requires higher pressure to maintain the water flow. Scale is due mainly to salts of calcium and magnesium. When scale has formed, the tubes should be cleaned with water or electric powered rotary brushes and cutters which are pushed through the tubes during boiler overhauls.
2. **Corrosion.** Corrosion may occur in the boiler shell, tubes, plates due to acidity present in water. This reduces the life of construction materials. Corrosion is the destructive conversion of metal into oxides or salts. Corrosion takes place due to the presence of oxygen, carbon dioxide or chlorides dissolved in water. Corrosion due to oxygen produces small pits.
3. **Foaming and priming.** A layer of foam is caused in the boiler drum by soluble and insoluble salts and other organic impurities which are carried in suspension. Foaming prevents the free escape of steam bubbles as they rise to the surface of water. Oil and other impurities which may be present in boiler water may cause foaming. Priming or carrying-over is the passing of small water particles with steam as it leaves the boiler. These water particles make the steam unfit for use in engines. This is caused by the impurities in water, a high water level and the method of operation of the boiler.
To prevent foaming and priming the following precautions should be observed:
 - (i) Oil, soap and other suspended impurities should not be present in boiler water.
 - (ii) Various valves should not be opened suddenly to maximum.
 - (iii) Water in the boiler should be at its minimum possible level.
4. **Embrittlement.** Caustic embrittlement is caused due to caustic impurities present in water. Presence of certain concentration of sodium hydroxide causes embrittlement. Due to this the boiler metal becomes brittle and inner cracks appear along the seams below the water level.

7.13 METHODS OF FEED WATER TREATMENT

It is desirable that the water to be used in the boiler should be free from various impurities. The impure water is chemically treated in different ways depending upon the nature and concentration of impurities. The different treatments adopted to remove the various impurities are as follows:

7.13.1 Mechanical treatment

It includes sedimentation, coagulation and filtration. Suspended matter can be removed easily by these processes. Sedimentation involves allowing the water to stand quietly for some time. In this way the solid matter settles down and is removed periodically. In case of coagulation some coagulants like aluminium sulphate, sodium aluminate or ferrous sulphate are added to the impure water. This removes the minute colloidal suspensions. Filtration consists in passing the water through filters. The suspended matter adheres to the filter material. The filters may be either gravity filter or pressure filter.

Fig. 7.23 shows a pressure filter. The raw water is passed through alum pot and then through tank containing fine sand, graded layer of gravel. These filters can be easily installed on the pipe line as much lesser space is required.

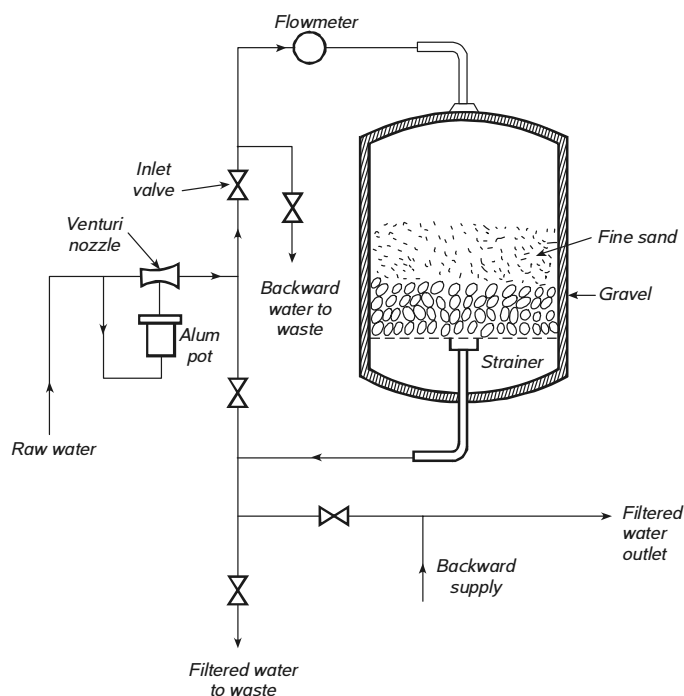


Fig. 7.23. Pressure filter—A mechanical treatment.

7.13.2. Thermal treatment

It includes distillation and deaerative heating of water. By these processes dissolved gases of water are removed.

Deaerating Heater. A tray type deaerating heater is shown in Fig. 7.24. In this heater feed water after passing through the vent condenser is sprayed upwards in the spray pipe. Water falls in the form of uniform showers over the heating trays and air separating trays and finally gets collected in the storage space. Steam enters the heater through a nozzle fitted in the side of heater shell. The entire space between the shell and tray compartment gets filled with steam. The steam makes its way downwards through the perforation in the top plate of tray compartment. While flowing downward the steam comes in contact with the falling water.

Most of the steam condenses in between the spray and heating trays. From the bottom of heating trays, the remaining steam and separated gases such as oxygen etc. flow to the vent condenser. The steam used for heating may be the main turbine bled steam or may be from other sources. Storage tank with controls helps to add make up water when needed to maintain the feed water flow.

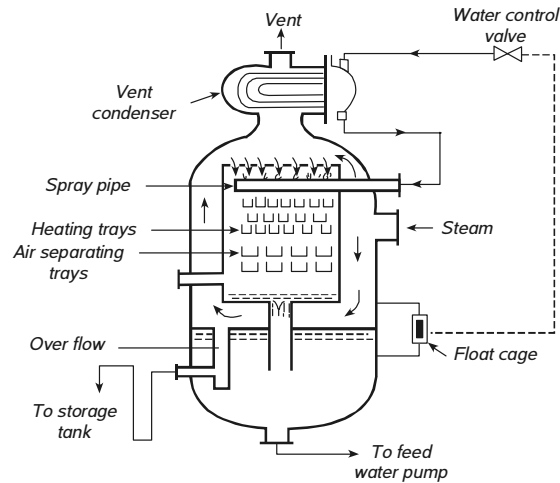


Fig. 7.24. Deaerating heater.

Closed feed water heater. Fig.7.25. shows a closed feed water heater in which steam bled from steam turbine is used for heating the feed water. This process is also called regenerative process and increases the efficiency of the generation cycle.

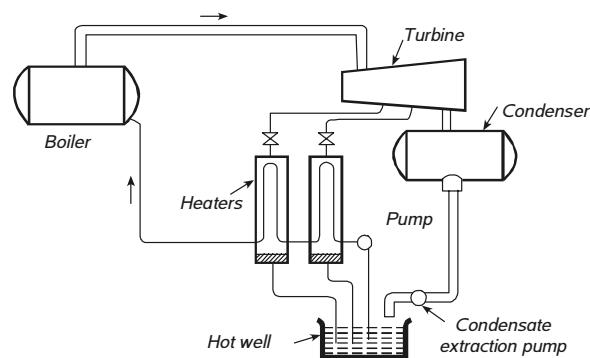
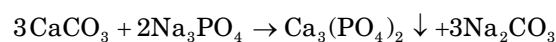


Fig. 7.25. Closed feed water heater

7.13.3. Chemical treatment

It includes addition of some chemicals to cause precipitation or impurities.

- (i) **Internal treatment.** In this process the reagents are added to water present in boiler. Various reagents used are sodium carbonate, sodium phosphate, sodium hydroxide etc. Sodium phosphate when added removes carbonate impurities.

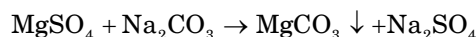
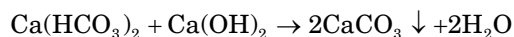


Internal treatment is suitable for low pressure (about 300 p.s.i) boilers.

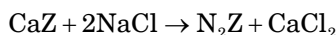
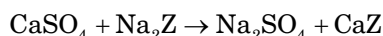
- (ii) **External treatment.** In this treatment raw water is received in a tank where the reagent are mixed. The reaction becomes rapid if the water is heated before the addition of reagents.

The various reagents used in external treatment are as follows:

- (a) *Lime soda treatment.* In this method lime $\text{Ca}(\text{OH})_2$ is added to remove the carbonate hardness of water and soda ash Na_2CO_3 is added to remove the sulphate hardness.



- (b) *Ion exchange process.* In this process sodium zeolite N_2Z ($\text{Na}_2 \text{Al}_2 \text{Si}_2 \text{O}_2$) is added. It reacts with calcium and magnesium salts to form zeolities. Zeolite is reproduced on the addition of brine (sodium chloride) solution.



A zeolite softener (Fig. 7.26) consists of a shell holding a bed of active sodium zeolite supported by layers of graded gravel lying over a water distribution and collection system. The raw water enters the zeolite softener at the top. It flows downwards through the bed although the system can be by-passed while back washing, if necessary. The figure also shows arrangements for the periodic back washing with brine needed by this system.

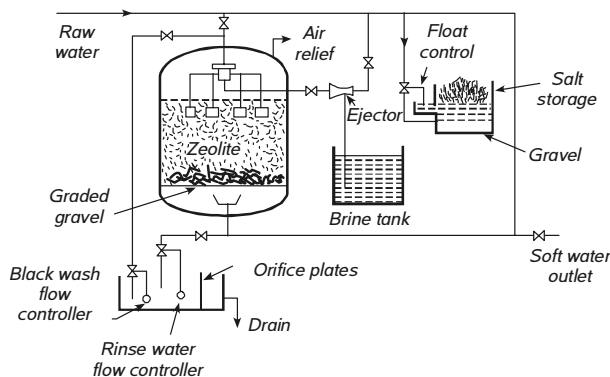


Fig. 7.26. Ion exchange system.

7.13.4. Demineralisation

It is used to remove mineral contents of water. In this system (Fig. 7.27) raw water enters the hydrogen zeolite exchanger (cation removal) at the top and then flows to the anion exchanger and degasified and finally passes through silicon absorber.

In spite of the best treatment some quantity of dissolved and suspended impurities enter the boiler. Table 7.1 indicates the recommended boiler water concentration of impurities permitted at different operating pressures (p).

Table 7.1

$P \text{ kg/cm}^2$	Silica PPM	Suspended solids PPM	Alkalinity PPM	Total solid PPM
0-25	100	300	650	3000
25-50	60	150	500	2500
50-100	8	40	200	1000
Above	6	4	70	400

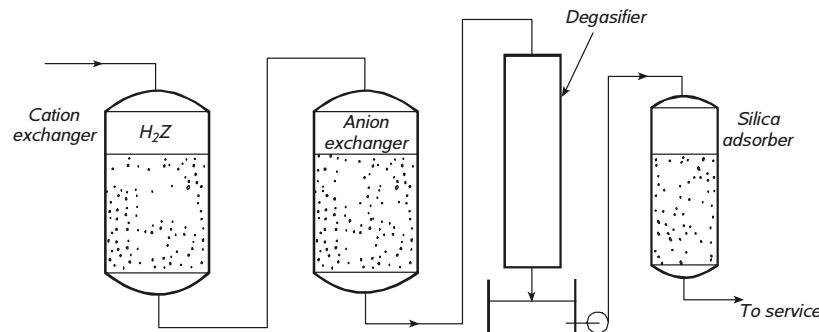


Fig. 7.27. Demineralisation of feed water

The above values shows that safe limit of concentration decreases with increase in boiler working pressure. Concentration of these solids can be reduced to a great extent by blowing down some quantity of water from the bottom of boiler drum. Periodical blow down helps in maintaining the concentration of impurities within the allowable limits.

7.13.5 Blow down

Water entering the boiler may contain some dissolved solids. The concentration of these solids goes on increasing as the water is vaporised. Beyond a certain limit of concentration, these solids may cause foaming and priming. The concentration of these solids can be reduced by drawing off some of the quantity of the boiler water from the bottom of boiler drum. This is called blowing down and discharged water is known as blow down.

The blow down mainly contains the undesirable impurities which concentrate at the bottom of drum. As a result of blow down the concentration of these impurities inside the boiler drum can be temporarily reduced. Therefore a boiler may have periodical blow down so that the concentration of impurities can be kept within permissible limits.

Determination of Blow Down

Draining of some of the boiler water carrying excessive concentration of solids and replacing it with fresh feed water keep the solid concentration within safe limits. This is called blowing down the boiler and the discharged water is called blow down.

The blow down is calculated as follows :

$$K = \text{Blow down \%} = W / W_f$$

where W = weight of water blow down from boiler in kg per hour

W_f = weight of feed water supplied to boiler for steam and blow down in kg per hour.

Blow down may be continuous or intermittent. Continuous blow down is about 1 to 10% of the incoming feed water.

The impurities in blow down are found by the following relation:

$$M = M_1 + M_2 + M_3$$

where

M = Impurities in blow down (weight of blow down) $\times$ (Allowable salt concentration) ppm.

M_1 = Feed water impurities = (weight of feed water) $\times$ (Impurity concentration) ppm.

M_2 = Internal treatment chemicals.

M_3 = Steam impurities = (weight of steam) $\times$ (Impurity concentration) ppm.

Fig. 7.28. shows mass balance of impurities and treatment chemicals in boiler.

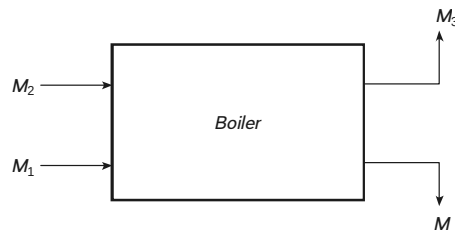


Fig. 7.28. Mass balance of impurities.

pH Value of Water

It is the logarithm of the reciprocal of hydrogen ion concentration in water. In water either hydroxyl ions (OH^-) or hydrogen ions (H^+) predominate causing either an alkaline or acidic condition. Acidity or alkalinity is measured in pH values ranging from 1 to 14. pH value 1 is strongly acidic. pH 14 is strongly basic and pH 7 indicates a neutral solution. pH value of a sample of water can be measured by a pH meter.

7.14 FEED WATER HEATERS

Feed water heaters are used to heat the feed water before it is supplied to the boiler. Heating of boiler feed water serves the following purposes:

1. It causes scale forming dissolved salts to precipitate outside that boiler.
2. It removes dissolved gases such as oxygen and carbon dioxide which corrode boiler metal.
3. By using pre-heated feed water, the steaming capacity of boiler is increased.
4. It avoids the thermal stresses which can be induced in the boiler surface by cold water entering a hot drum.

There are two types of feed water heaters:

1. **Contact or open heaters.** In these heaters feed water get mixed with the heating steam.

2. **Surface or closed heaters.** Surface heaters have shell and tube construction, and may be steam tube type or water tube type. Generally water tube type heaters are used. They ordinarily use bled steam for heating. Heating the feed water with steam at a lower pressure than boiler pressure raises the over-all efficiency of the plant. Fig. 7.29 shows a closed feed water heater in which water flow through tubes and steam flows all around the tubes. Steam while flowing transfers its heat to the feed water. Surface heater has construction similar to a surface condenser except that surface heater is designed for higher pressure and temperatures.

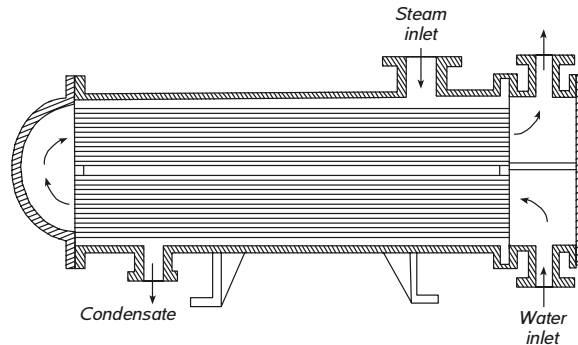


Fig. 7.29. Closed feed water heaters.

Chemical tests are performed at frequent intervals on samples of water to have information of some important items such as variation in mineral characteristics of raw water supply, condition of boiler water with respect to treatment adjustment and blow down adjustment percentage of condensate and make up in the feed water steam purity and correction of feed water and boiler water for corrosion control, etc.

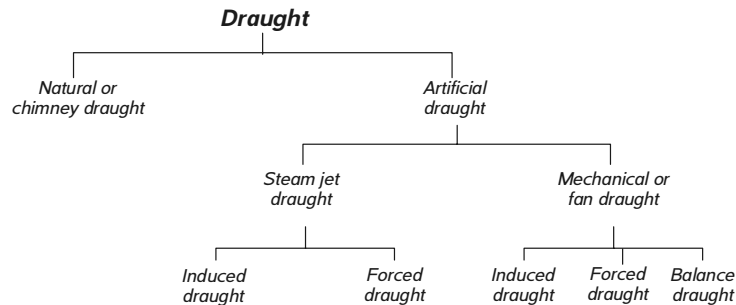
7.15 DRAUGHT

The rate of steam generation in a boiler depends upon the rate of fuel burned, which further depends upon the availability of fresh air supply to the combustion chamber. In order to maintain a continuous flow of fresh air into the combustion chamber, it is necessary to exhaust the products of combustion from the combustion chamber through the chimney into the atmosphere. The difference of pressure causing the flow of fresh air in and exhaust gases out is known as draught.

To move the air through the fuel bed and to produce a flow of hot gases through the boiler, economiser, preheater and chimney require a difference of pressure equal to that necessary to accelerate the burnt gases to their final velocity and to overcome the pressure losses. The draft can be obtained by use of chimney, fan, steam or air jet or combination of these.

The draft produced by the chimney is known as *natural draught*. The draft produced by mechanical equipment like fan is known as *mechanical draught*. The draught produced with the help of high pressure steam is known as *steam jet draught*.

Boiler draught is classified as given below:



The purpose of draught is as follows:

- (i) To supply required amount of air to the furnace for the combustion of fuel. The amount of fuel that can be burnt per square foot of grate area depends upon the quantity of air circulated through fuel bed.
- (ii) To remove the gaseous products of combustion.

Draught is defined as the difference between absolute gas pressure at any point in a gas flow passage and the ambient (same elevation) atmospheric pressure. Draught is plus if $P_{atm} < P_{gas}$ and it is minus $P_{atm} > P_{gas}$. Draught is achieved by small pressure difference which causes the flow of air or gas to take place. It is measured in millimetre (mm) of water.

If only a chimney is used to create the necessary draught, the system is called natural draught system and if in addition to chimney a forced draught (F.D.) fan or an induced draught (I.D.), fan or both are used, the system is called mechanical draught system. Fans or chimneys produce positive pressure and is called available draught whereas fuel bed resistance, turbulence and friction in air ducts, gas breechings, chimney, etc., create negative pressure and is called the required draught.

7.15.1 Natural or Chimney draught

Natural draught system is used in boilers of smaller capacities. Natural draught is created by the difference in weight of a column of cold external air and that of a similar column of hot gases in the chimney. This system is dependent upon the height of chimney and average temperature of the gases in the chimney.

Natural draught is obtained by the use of chimney. A chimney is a verticle tubular structure of brick, masonry or steel built for the purpose of enclosing a column of gases to produce draught.

Now a days the chimney is not used for creating draught in steam power-plants as it has no flexibility, the total draught produced is insufficient for high generating capacity. By using chimney draught can be increased by allowing the flue gases to leave the combustion chamber at higher temperature and this reduces the overall efficiency of the power plant. The chimney is, therefore, used only to discharge the flue gases.

Advantages of Chimney Draught

1. It does not require any external power for producing the draught.
2. The capital investment is less than that of artificial draught. Maintenance cost is nil as there is no mechanical part.

3. Chimney keeps the flue gases at a high place in the atmosphere which prevents the contamination of atmosphere.
4. It has long life.

Limitations:

1. For producing sufficient draught, the flue gases have to be discharged at comparatively high temperatures resulting in the loss of overall efficiency of the plant.
2. Due to low velocity of air, there is no thorough mixing of air and fuel in the combustion chamber resulting into poor consumption.
3. Nearly 20% heat released by the fuel is lost to the flue gases.

7.15.2. Mechanical Draught (Artificial Draught)

In boilers of larger capacities, fans are employed to create the necessary draught in order to reduce the height of chimney, to obtain draught that is independent of weather conditions and to control the draught easily.

Mechanical draft may be induced, forced or balanced draft.

Induced draught (I.D.) system shown in Fig. 7.30(a) is created by chimney and fan located in the gas passage on the chimney side of the boiler. In this system gas movement is achieved as result of a vacuum.

The various pressures indicated are as follows:

- P_1 = Inlet pressure of forced draft fan.
- P_2 = Outlet pressure of forced draft fan.
- P_3 = Pressure below grate.
- P_4 = Pressure above the grate.
- P_5 = Inlet pressure of induced draft fan.
- P_6 = Outlet pressure of induced draft fan.

Induced draught is not as simple and direct as forced because fans used in induced draft system operate in gases of much higher temperature (nearly 500° — 900°F). This becomes more expensive.

The fan sucks in gas from the boiler side and discharges it to the chimney (stack).

The draught produced is independent of the temperature of the hot gases and, therefore, the gases may be discharged as cold as possible after recovering as much heat possible in air preheater and economiser.

In **forced draught (F.D.)** system Fig. 7.30(b) the fan installed near the boiler base supplies the air at a pressure above that of atmosphere and delivers it through air duct to the furnace. Most high rating combustion equipment employs forced draught fans for supplying air to the furnace. Forced draught is used in under fed stokers carrying a thick fuel bed.

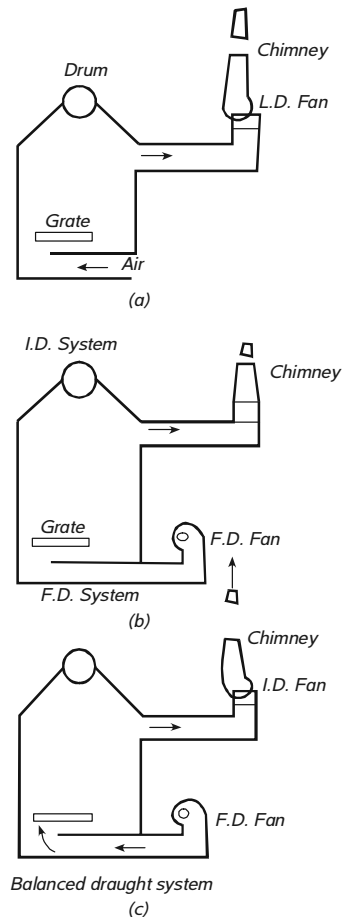


Fig. 7.30. Artificial draughts.

Balanced draught system is a combination of induced and forced draught systems. The forced draught fan forces the air through the fuel bed on to the top of grate and the induced draught fan sucks in gases from the boiler side and discharges them to the chimney. This system is used where pressure above fire is slightly below atmospheric, Fig.7.30(c) shows this system.

Fig. 7.31 shows the pressure distribution for the balanced system.

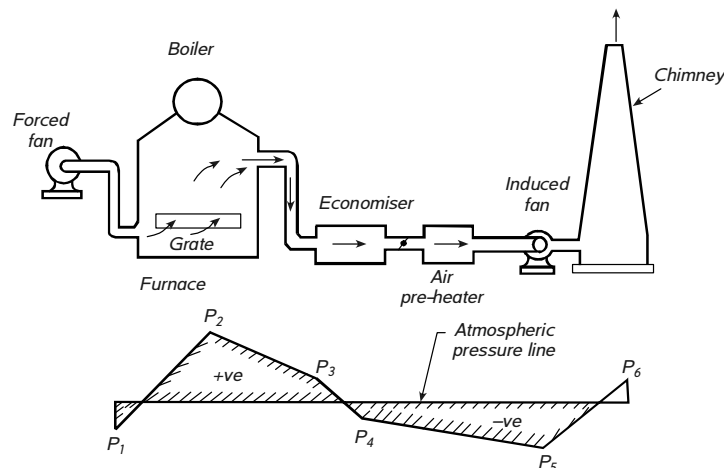


Fig. 7.31. Balanced draft.

Construction such as shielding or water cooling or water protect the bearings of fans. The I.D. fans handle gases laden with dust which causes wear of blades. In forced draught system the fans handle cool and clean air and the fan can be located where convenient. Balanced draft system is more efficient. In balanced draft system about 0.1 inch water vacuum is maintained over the fuel bed.

Multivane centrifugal fans are generally used for moving large volume of air and gases. The performance of a fan depends on the shapes of blades which are, in general, of three types:

1. Backward curved blades.
2. Forward curved blades.
3. Straight radial blades.

Various types of blade forms are shown in Fig. 7.32.

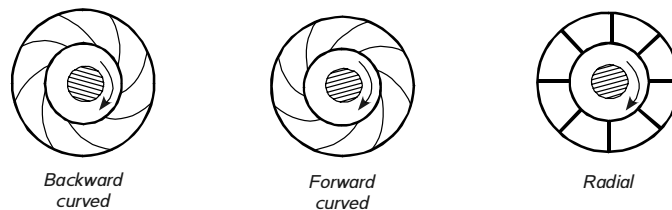


Fig.7.32. Types of fan blades.

The air leaving the tips of backward curved blades possesses low velocity. This makes them suitable for high rotor speed. Fans with backward curved blades are used in forced draught system. Forward and radial blades are used in induced draught fans.

Comparisons of forced and induced draughts

The forced draught system has the following advantages as compared to induced draught:

- (i) The induced draught handles more volume of gases and at elevated temperature. Therefore, the size and power required for induced draught fan is more than forced fan.
- (ii) Improved route of burning of fuel is achieved by using forced draught because there is more uniform flow of air through the grate and furnace and also the air penetrates better into the fire bed.
- (iii) In case of induced draught when doors are opened for firing there will be rush of cold air into the furnace and this reduces the heat transmission.
- (iv) The induced draught fan handles flue gases at high temperature and thus water cooled bearing are needed for induced draught fan.

Advantages of mechanical draught over natural draught

1. The rate of combustion is high due to the greater draught available.
2. Due to better mixing of air and fuel with artificial draught, the quantity of air supplied per kg of fuel burned is reduced. This also increase the efficiency of combustion.
3. The air flow can be regulated by changing the draught pressure as per requirement, *i.e.*, as per the load on the boiler.
4. Maximum heat can be recovered from the exhaust gases in the economiser and air-pre-heater as artificial draught is independent of gas temperature.
5. The height of the chimney can be reduced as the function of the chimney is now only to exhaust the gases high in the atmosphere.
6. The efficiency of the artificial draught is much higher as compared to natural draught.
7. It prevents the formation of soot and smoke as complete combustion is possible even with less percentage of excess air.

7.15.3. Steam Jet Draught

Steam jet draught may be induced or forced draught depending upon the location of steam jet producing the draught.

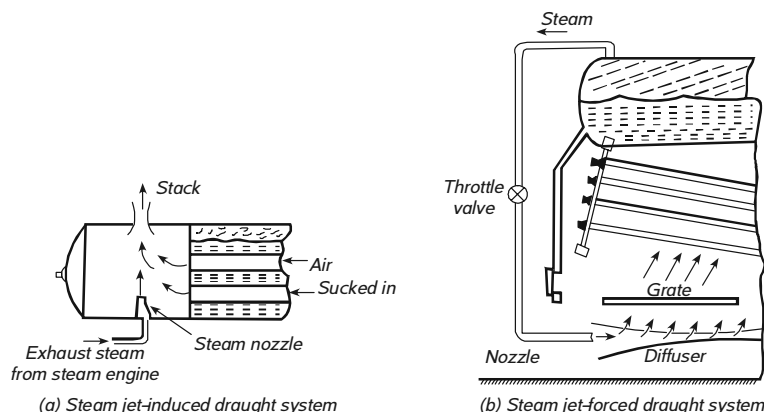


Fig. 7.33. Steam jet draught.

Induced draught produced by steam jet is shown in Fig. 7.33(a). This system is used in locomotive boilers. Exhaust steam from the engine enters the smoke box through a nozzle to create draught. The air is induced through the flues, the grate and ash pit to the smoke box.

Fig. 7.33 (b) shows a forced draught developed by steam jet. Steam from the boiler is passed through a throttle valve, throttle pressure being 1.5 to 2 kg/cm² gauge. Then the steam passes through a nozzle projecting in diffuser pipe. The steam comes out of nozzle with great velocity and drags a column of air along with it thus allowing the fresh air to enter. The mixture of steam and air possesses high kinetic energy and passes through the diffuser pipe. The kinetic energy gets converted into pressure energy and thus air is forced through the coal bed, furnace and flows to the chimney. Steam jet system is simple, requires less space and is economical. But it can be used only if steam at high pressure is available.

Draught Measurement

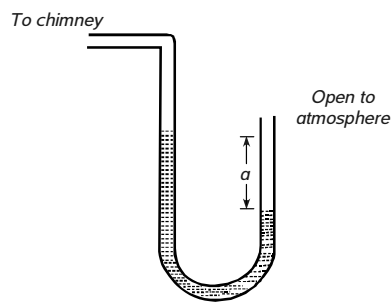


Fig. 7.34. U-tube manometer.

A U-tube manometer shown in Fig. 7.34 is employed to measure the draught. One end of manometer is open to atmosphere and the other end is connected to the boiler furnace, chimney or any other point in the gas flow passage where draught is to be measured. Due to pressure difference on the two sides of the tube the level of water in two sides is altered to balance the pressure. The difference in level a in the two sides measures the draught.

Draft-loss. It is the pressure loss caused by friction between two points in the gas flow path.

Draft-system flow resistance. The following relation may be used to evaluate the flow pressure pattern in a draft system.

$$P_F + P_S = P_A + P_G + P_V$$

where P_F = Total fan effect pressure

P_S = Net stack effect (Chimney $\pm$ vertical passage)

P_A = Draft pressure loss on air side.

= Sum of friction losses in following :

- | | |
|----------------------|---|
| (i) Air ducts, bends | (ii) Air heaters |
| (iii) Stoker bed | (iv) Secondary air pressure at fuel burner. |

P_G = Draft pressure loss on gas side.

This is sum of friction losses in the following :

- | | |
|----------------------|-------------------|
| (i) Gas ducts, bends | (ii) Economiser |
| (iii) Air heater | (iv) Super heater |
| (v) Boiler setting | (vi) Chimney |

P_V = Gas exit velocity pressure.

7.16 CHIMNEY (STACK)

Chimneys are made up of steel or bricks and concrete. Concrete chimneys are more popular. The average life of concrete chimneys is 50 years and that of steel chimneys is about 15 years depending upon the care taken to prevent corrosion. The net area of chimney depends on the following factors:

- (i) Volume of gases to be discharged when the boilers operate at maximum rating.
- (ii) Draught to be produced.

Chimney generally denotes brick concrete construction whereas stack means steel construction.

Fig. 7.35(a) shows a brick chimney and Fig. 7.35(b) shows reinforced chimney. Chimney is provided with a lightning conductor, aircraft warning lights, and various means of access and inspection.

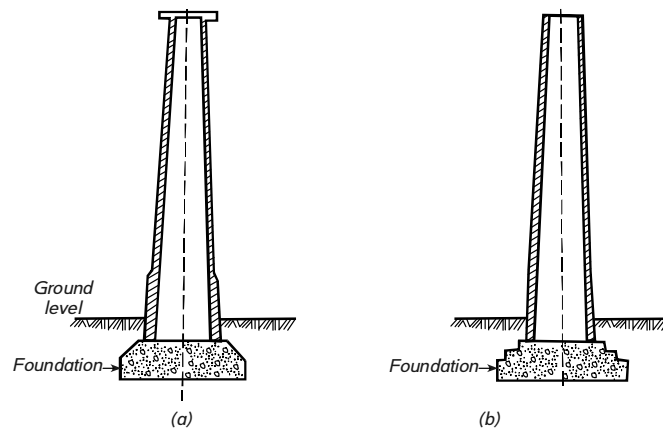


Fig. 7.35. (a) Brick and (b) Reinforced chimney.

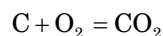
The various advantages of steel chimneys over masonry chimneys are as follows :

- (i) Lower overall cost,
- (ii) Slightly high efficiency,
- (iii) Easier construction,
- (iv) Requires lesser space,
- (iv) Lighter in weight.

R.C.C. chimneys are also becoming popular.

7.16.1 Calculation of Chimney Height

Air is required to burn coal or fuel in the furnace. We know



This shows that volume of chimney gases produced by the complete combustion of 1 kg of coal or oil is same as that of air required to support combustion if the temperature of flue gases and air is same.

Let H = Height of chimney above grate level in metres.

W = Weight of air required in kg per kg of fuel.

T = Average absolute temperature of chimney gases in °K

T_1 = Absolute temperature of air outside the chimney in °K

Weight of chimney gas produced

$$= (W + 1) \text{ kg of fuel burnt}$$

Volume of chimney gases at 273°K

$$= \text{volume of 1 kg air at 273°K}$$

$$= \frac{RT}{P} = \frac{29.27 \times 273}{1.03 \times 10^4} \times 0.7734 \text{ m}^3$$

Therefore, volume of one kg of air T_1 °K

$$= 0.7734 \times \frac{T_1}{273}$$

Thus volume of W kg of air at T_1 °K

$$= \frac{0.7734 \times T_1 \times W}{273}$$

$$\text{Density of air } T_1 \text{°K} = \frac{\text{Mass}}{\text{Volume}} = W \times \frac{273}{0.7734 \times T_1 \times W}$$

$$= \frac{273}{0.7734 \times T_1}$$

Therefore, pressure at the grate due to a column of cool air of

H metres height

$$= \frac{273}{0.7734 \times T_1} \times H = 1.293 \times \frac{273}{T_1} \times H \text{ kg/m}^2 \quad \dots(i)$$

Volume of chimney gases at T °K per kg of air

$$= 0.7734 \times \frac{T}{273}$$

Volume of $(W + 1)$ kg of chimney gases at T °K

$$= 0.7734 \times \frac{T}{273} \times W$$

Density of chimney gases at $T^\circ\text{K}$

$$\begin{aligned} &= \frac{\text{Mass}}{\text{Volume}} = (W+1) \times \frac{273}{0.7734 \times T \times W} \\ &= 1.293 \left(\frac{W+1}{W} \right) \times \frac{273}{T} = 353 \times \left[\frac{W+1}{W} \right] \times \frac{1}{T} \end{aligned}$$

Therefore, pressure at the grate by a column of hot gases of height H metre

$$\begin{aligned} &= \text{Density} \times H \text{ kg/m}^2 \\ &= 1.293 \left(\frac{W+1}{W} \right) \times \frac{273}{T} \times H \text{ kg/m}^2 \end{aligned} \quad \dots(ii)$$

Therefore, pressure P causing the draught

$$\begin{aligned} P &= \left[\left(1.293 \times \frac{273 \times H}{T_1} \right) - \left(1.293 \times \frac{W+1}{W} \times \frac{273}{T} \times H \right) \right] \\ &= 353 H \left[\frac{1}{T_1} - \left(\frac{W+1}{W} \right) \times \frac{1}{T} \right] \text{ kg/m}^2. \end{aligned}$$

If this draught is h mm of water as measured by a U-tube water manometer.

$$h = 353 H \left[\frac{1}{T_1} - \left(\frac{W+1}{W} \right) \times \frac{1}{T} \right] \text{ mm of water}$$

(We know $1 \text{ kg/m}^2 = 1 \text{ mm of water column}$)

Let h' = height of column of hot gases.

$$\begin{aligned} h' \times 1.293 \times \left(\frac{W+1}{W} \right) \times \frac{273}{T} &= P \\ &= 353 H \left[\frac{1}{T_1} - \left(\frac{W+1}{W} \right) \times \frac{1}{T} \right] \\ h' &= \frac{353 H \left[\frac{1}{T_1} - \left(\frac{W+1}{W} \right) \times \frac{1}{T} \right]}{1.293 \times 273 \times \left(\frac{W+1}{W} \right) \times \frac{1}{T}} \\ h' &= H \left[\left(\frac{W}{W+1} \right) \times \frac{T}{T_1} - 1 \right] \end{aligned}$$

Example 7.1 A chimney is 28 metres high and the temperature of hot gases inside the chimney is 320°C. The temperature of outside air is 27°C and furnace is supplied with 15 kg of air per kg of coal burnt. Calculate:

- (a) Draught in mm of water.
 (b) Draught height in metres of hot gases.

Solution. We have

$$\begin{aligned}
 (a) \quad h &= 353 \times H \left[\frac{1}{T_1} - \left(\frac{W+1}{W} \right) \times \frac{1}{T} \right] \\
 &= 353 \times 28 \left[\frac{1}{27 + 273} - \frac{15+1}{15} \times \frac{1}{(320 + 273)} \right] \\
 &= 353 \times 28 \left[\frac{1}{300} - \frac{16}{15} \times \frac{1}{593} \right] \\
 &= 353 \times 28 \left[3.33 \times 10^{-3} - 1.8 \times 10^{-3} \right] \\
 &= 353 \times 28 \left[1.53 \times 10^{-3} \right] = 15.1 \text{ mm of water.} \quad \text{Ans}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad h' &= H \left[\frac{W}{W+1} \times \frac{T}{T_1} - 1 \right] = 28 \left[\frac{15}{16} \times \frac{593}{300} - 1 \right] \\
 &= 24.88 \text{ metres.} \quad \text{Ans.}
 \end{aligned}$$

Example 7.2 Determine the height of chimney to produce a static draught of 22 mm of water if the mean flue gas temperature in a chimney is 290°C and ambient temperature in boiler house is 20°C. The gas constant (R) for air is 29.26 kgfm / kg °K and for chimney flue gas is 26.2 kgfm / kg °K. Assume barometer reading as 760 mm of mercury.

Solution. Let

P = Absolute pressure of gas (kg/m²)

V = Volume of gas (m³)

R = Gas constant

T = Absolute temperature of gas (°K)

Now $PV = RT$

$$V = \frac{RT}{P}$$

Difference of pressure (ΔP) = Height of chimney (H) $\times$ ($\rho_{air} - \rho_{flue}$) kg/mm²

$$\Delta P = 22 \text{ mm of water} = 22 \text{ kg/m}^2$$

(As $1 \text{ kg/m}^2 = 1 \text{ mm of water}$)

$$\begin{aligned}\rho_{\text{air}} &= \frac{1}{V} = \frac{P}{RT} \\ &= \frac{1.033 \times 10^4}{29.26 \times (273 + 20)} = 1.2 \text{ kg/m}^3\end{aligned}$$

(P = Pressure = 760 mm of Hg = 1.033 kg/cm²)

$$\rho_{\text{flue}} = \frac{1}{V} = \frac{P}{RT} = \frac{1.033 \times 10^4}{26.2 \times (273 + 290)} = 0.69 \text{ kg/m}^3.$$

$$\begin{aligned}\text{Height of chimney (H)} &= \frac{\Delta P}{(\rho_{\text{air}} - \rho_{\text{flue}})} = \frac{22}{1.2 - 0.69} \\ &= 43.1 \text{ metres} \quad \quad \quad \mathbf{Ans}\end{aligned}$$

Example 7.3 Determine the height of chimney to get net draught of 12 mm if the total draft losses are 4 mm. The temperature of air is 25°C and the temperature of chimney gases is 300°C. The mass of air used per kg of fuel used is 18 kg. One kg of air occupies a volume of 0.7734 m³ at N.T.P.

Solution. Let

H = Height of chimney in metre

A = Gross sectional area of chimney in m²

$$\text{Density of air} = \frac{1}{0.7734} \times \frac{273}{273 + 25} = 1.24 \text{ kg/m}^3$$

Density of chimney gases at 0°C

$$= \frac{18 + 1}{18 \times 0.7734} = \frac{19}{18 \times 0.7734}$$

Density of chimney gases at 300°C

$$= \frac{18}{18 \times 0.7734} \times \frac{273}{273 + 300} = 0.65 \text{ kg/m}^3$$

Mass of chimney gases = $A \times H \times 0.65 \text{ kg}$.

Mass of equal column of external air = $A \times H \times 1.24 \text{ kg}$.

Difference in masses = $1.24 AH - 0.65 AH = 0.59 A \times H \text{ kg}$.

$$P = \text{Draft in kg/m}^2 = \frac{0.59AH}{A} = 0.59 H \text{ kg/m}^2 = 0.59 H \text{ mm of water.}$$

(As $1 \text{ kg/m}^2 = 1 \text{ mm of water}$)

$$\therefore 0.59 H = 16 \quad (\text{as given draft} = 12 + 4 \text{ mm})$$

$$H = 27.1 \text{ metres.} \quad \text{Ans.}$$

Example 7.4. A chimney produces a draught of 1.8 cm of water when temperature of flue gases is 280°C and ambient temperature is 21°C . The flue gases formed per kg of fuel burnt are 24 kg. Taking diameter of chimney as 1.7 m, determine the mass of flue gases flowing through the chimney.

Solution.

$$h = \text{Draught} = 1.8 \text{ cm} = 18 \text{ mm of water}$$

$$T = \text{Temperature of flue gases} = 280^\circ + 273^\circ = 553^\circ\text{K}$$

$$T_1 = \text{Ambient temperature} = 21 + 273 = 294^\circ\text{K}$$

$$W = \text{Mass of air reqd. per kg of fuel} = 24 - 1 = 23 \text{ kg}$$

$$h = 353H \left[\frac{1}{T_1} - \frac{W+1}{W} \times \frac{1}{T} \right]$$

$$18 = 353H \left[\frac{1}{294} - \frac{23+1}{23} \times \frac{1}{553} \right]$$

$$H = 34.1 \text{ m of air}$$

$$h' = \text{Height of column of hot gases}$$

$$= H \left[\left(\frac{W}{W+1} \right) \times \frac{T}{T_1} - 1 \right] = 34.1 \left[\frac{23}{24} \times \frac{553}{294} - 1 \right] = 28.9 \text{ m of air}$$

$$C = \text{Velocity of gases} = \sqrt{2g \times h'} = \sqrt{2 \times 9.81 \times 28.9} = 23.5 \text{ m/s}$$

$$W_g = \text{mass of flue gases} = A \times C \times \rho_g$$

$$\text{where } A = \text{Area of chimney} = \frac{\pi}{4} \times (1.7)^2 \text{ m}^2$$

$$\text{and} \quad \rho_g = 353 \left[\frac{W+1}{W} \right] \times \frac{1}{T} = 353 \left[\frac{23+1}{23} \right] \times \frac{1}{553} = 0.67 \text{ kg/m}^3$$

$$\therefore W_g = \frac{\pi}{4} (1.7)^2 \times 23.5 \times 0.67$$

$$= 35.7 \text{ kg/s} \quad \text{Ans.}$$

7.17 BOILER ACCESSORIES

Boiler accessories are auxiliary parts required for the smooth function of the boiler and to increase the overall efficiency of the boiler. As huge quantity of fuel is used in thermal power plant and very large quantity of heat is carried by the exhaust gases. The heat carried by the exhaust gases is about 25%. In the present age of costly fuel, it is necessary to conserve the fuel by utilising the wasted energy to the atmosphere. This is done in all modern power plants by incorporating economiser and air preheater.

The common equipments used in thermal power plants to increase the thermal efficiency are economisers, and air preheaters. The heat carried with the flue gases is partly recovered in air preheaters and economisers and reduces the fuel supplied to the boiler. The preheating of air with the gases increases the combustion efficiency and reduces the fuel consumption. The other important accessories are superheaters, and reheaters.

7.18 ECONOMISERS

The economiser is a feed water heater deriving heat from the flue gases discharged from the boiler.

There are steaming economisers in which the water is raised to the boiling point and partially (10 – 20%) evaporates and non-steaming economisers in which the temperature of water is below the boiling point by 20 – 30°C. The advantage of an economiser are as follows:

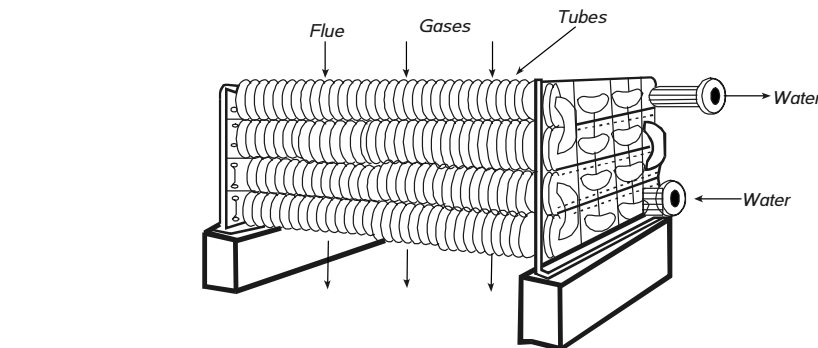


Fig. 7.36. Iron tube economiser.

- (i) It reduces the losses of heat with the flue gases.
- (ii) It reduces the consumption of fuel.
- (iii) It improves the efficiency of the boiler installation by increasing the steaming capacity.
- (iv) It enhances the life of boiler.

Economiser may have iron or steel tubes. Smooth or ribbed. Fig. 7.36 shows a general view of an iron economiser.

Flue gases flow over the tubes.

Fig. 3.37 shows an economiser, consisting of series of steel tubes through which the feed water flow. The combustion gases pass over the tubes and transfer some of their

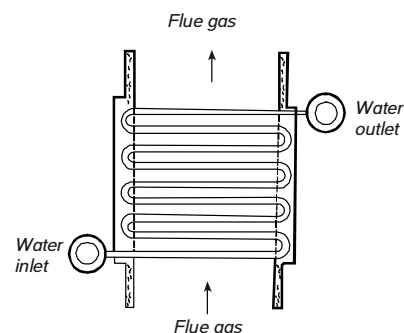


Fig. 7.37. Steel tube economiser

heat to the feed water. The boiler efficiency rises by about 1% for each 10°F rise in feed water temperature. Economisers may be parallel-flow or counter-flow, when the gas flow and water-flow are in the same direction economisers are called parallel-flow whereas in counter-flow economiser the gas flow and water-flow are in the opposite direction. Installation of an economiser depends on its initial cost, type of boiler and nature of feed water used.

Fig. 7.38 shows by-pass arrangement which enables to isolate or include the economiser in the path of the flue gases.

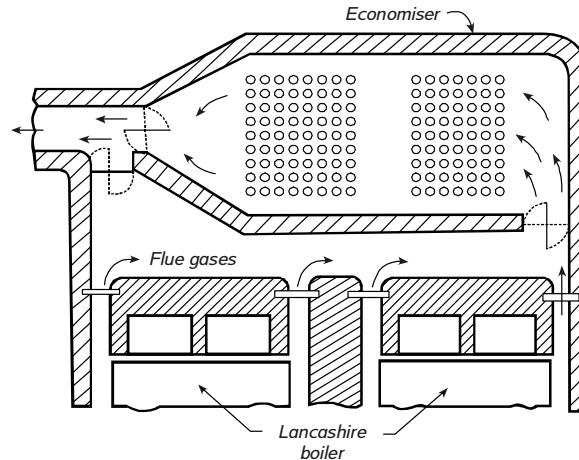


Fig. 7.38. By-pass arrangement for economiser.

From the above it is seen that an economiser is a heat exchanger which raises the temperature of the feed water leaving the highest feed water heater to about the saturation temperature corresponding to the boiler pressure. This is done by the hot flue gases leaving the last superheater or reheater at a temperature varying from 370°C to 540°C. Since by utilising the exhaust flue gases in heating feed water, higher efficiency and better economy were achieved, hence, the heat exchanger is known as “economiser”. Modern economisers are often designed to allow some boiling of the feedwater in the outlet sections, and are therefore, termed as “steaming economisers”.

For an economiser, following *design requirements* should be satisfied:

1. It must be able to extract maximum possible heat from exhaust gases, with minimum heat transfer surface.
2. The pressure loss should be minimum to reduce the running expenses of ID fans.
3. There must be free circulation of water around the economiser to prevent the overheating and boiling during the period when there is no feed-flow during early pressure rising stages.

7.19 AIR PREHEATERS

The heat carried with the flue gases coming out of economiser is further utilised for preheating the air before supplying to the combustion chamber. The air heater is not only considered in terms of boiler efficiency in modern power plant, but also as necessary equipment for supply of hot air for drying the coal in pulverised fuel system to facilitate grinding and satisfactory combustion of fuel in the furnace.

Thus, with the application of air preheater, it is possible to cool the exhaust gases in the heating of air supplied for the combustion of fuel. The use of hot air makes the combustion process more efficient by making it more stable and reducing the energy losses due to incomplete combustion and unburnt carbon.

It consists of plates or tubes with hot gases on one side and air on the other. It preheats the air to be supplied to the furnace. Preheated air accelerates the combustion and facilitates the burning of coal. Degree of preheating depends on the type of fuel, type of fuel burning equipment and the rating at which the boiler and furnace operated. The principal benefits of preheating the air are increased thermal efficiency and increased steam capacity per square metre of boiler surface. There are two types of air preheater.

1. Tabular type

2. Plate type

A **tabular type air preheater** as shown in Fig.7.39. After leaving the boiler or economiser the gaseous products of combustion travel through the inside of the tubes of air preheater in a direction opposite to that of air travel and transfer some of their heat to the air to be supplied to the furnace. Thus, the air gets initially heated before being supplied to the furnace. The horizontal baffles are provided as shown in the figure to increase time of contact which will help for higher heat transfer.

The gases reverse their direction near the bottom of the air heater, and a soot hopper is fitted to the bottom of air heater casing to collect soot.

In **plate type air preheater** the air absorbs heat from the hot gases being swept through the heater at high velocity on the opposite side of a plate. It consists of rectangular flat plates spaced from 1.5 to 2.5 cm apart leaving alternate air and gas passages. This type of air-preheater is not used in modern installations, as it is more expensive both as to initial cost and maintenance cost and also in efficiency as compared with tubular air preheater.

Finally the products of combustion leave the stack (chimney) to make their passage to the atmosphere. It is desirable that the temperature of the gases leaving the stack should be kept as low as possible to keep the heat loss to the stack at minimum.

By installing economiser and air preheater less fuel is required per unit mass of steam raised and boiler efficiency is increased. The justifiable cost of economiser and air preheater depends upon the gain in boiler efficiency.

Air preheater should be used where a study of costs indicates that some money can be saved or some beneficial action on combustion can be obtained by its use. Some factors that need to be taken into account in examining a case for justification of air preheat system are as follows:

- (i) Improvement in combustion efficiency.
- (ii) Cost of the equipment and estimated maintenance cost.
- (iii) Cost of extra draft.

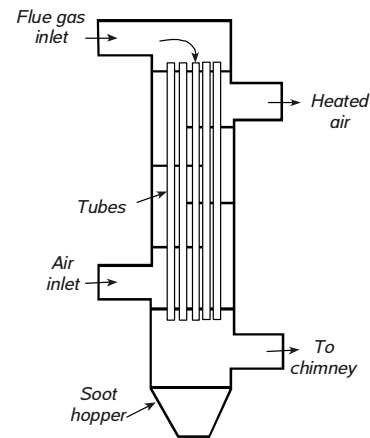


Fig. 7.39. Tabular type air preheater.

- (iv) The extent to which air can be preheated.
- (v) Saving in-heat discharged to the chimney.

7.19.1 Heat Transfer in Economiser and Air Preheaters

In economiser water is passed through the tubes and hot flue gases pass over the tubes whereas in air preheater air passes through the tubes and flue gases pass over the tubes and transfer the heat to the air. The most effective use of given heat transfer surface is obtained when two fluids (water or air and flue gases) travel in the opposite direction (counter flow). Fig 7.40 shows the variation of temperature along the flow paths of the fluids.

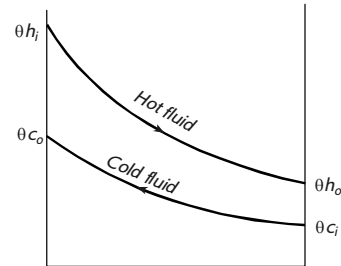


Fig. 7.40. Temperature variation in fluid flow paths.

The heat transfer from the hot flue gases to water in case of economiser and to air in case of air preheater is found as follows:

$$Q = U.A. \theta_m$$

where

Q = Quantity of heat transferred

U = Overall heat transfer coefficient

A = Area of heat transfer surface

θ_m = Log mean temperature difference

$$= \frac{\theta_i - \theta_o}{\log \frac{\theta_i}{\theta_o}}$$

θ_{co} = Outlet temperature of cold fluid

θ_{ei} = Inlet temperature of cold fluid

θ_{ho} = Outlet temperature of hot fluid

θ_{hi} = Inlet temperature of hot fluid

θ_i = Temperature difference at inlet = $\theta_{hi} - \theta_{ci}$

θ_o = Temperature difference at outlet = $\theta_{ho} - \theta_{co}$

7.20 SOOT BLOWERS

The fuels used in steam power plants create soot and this is deposited on the boiler tubes, economiser tubes, and air preheater. The soot deposited on these exchangers drastically reduces heat transfer and significantly increase the amount of fuel that must be burned to produce heat equivalent to that generated by clean tube surfaces.

In a steam power plant the heating surfaces of boilers especially coal fired water tube boiler have a tendency to become coated with a layer of soot, cinder and flyash. This loosely adhering layer does not allow the heat transfer to take place properly and should be removed after it has built up sufficiently to result in a significant increase in flue gas temperature. Soot blower is used to remove this layer of soot, cinder and flyash.

Soot blowers should be operated frequently enough to keep the tubes clean.

Soot blower is a device which blows off the soot from the economiser tubes. Soot which gets deposited on the economiser tubes due to the flow of flue-gases over the tubes, obstructs the transfer of heat from flue gases of the feed water.

A soot blower consists of a tube having a number of nozzles. High pressure steam is blown through the nozzles which strikes against the tubes and removes the soot.

7.21 SUPER HEATERS

The steam produced in the boiler is nearly saturated. This steam as such should not be used in the turbine because the dryness fraction of the steam leaving boiler will be low. This results in the presence of moisture which causes corrosion of turbine blades, etc. To raise the temperature of steam super heater is used. It consists of several tube circuits in parallel with one or more return bends connected between headers. Super heater supplies steam at constant temperature at different loads. The use of super heated steam increases turbine efficiency.

The function of the superheater in the thermal power plant is to remove the last traces of moisture from the saturated steam coming out of boiler and to increase its temperature sufficiently above the saturation temperature. The superheating raises overall cycle efficiency as well as avoids too much condensation in the last stages of the turbine, which avoids the blade erosion. The heat of combustion gases from furnace is utilised for the removal of moisture from steam and to superheat the steam. In modern utility high pressure boilers, more than 40% of the total heat absorbed in the generation of steam takes place in the superheaters, therefore, large surface area is required to be provided for superheating of steam. Superheaters and reheaters are made of 20 to 50 mm outer diameter.

There are three types of super-heaters:

1. Convective super heater
2. Radiant super heater
3. Combination of the two

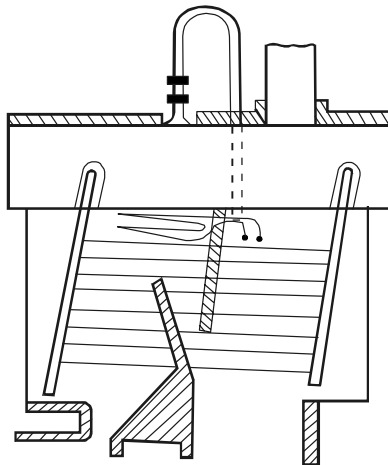


Fig. 7.41. Over deck superheater.

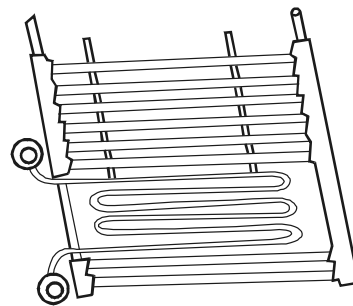


Fig. 7.42. Inter deck superheater.

Convective super heater makes use of heat in the gases entirely by convective by locating them in the convective zone of the furnace, usually ahead of the economiser, whereas a **radiant super heater** is placed in the radiant zone of the furnace near the water wall and receive heat from the burning fuel through radiation process.

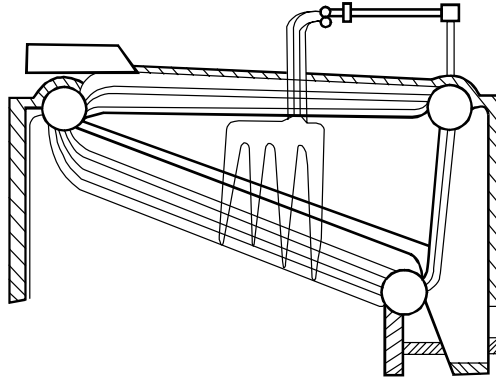


Fig. 7.43. Inter tube superheater.

The final temperature of steam depends upon the gas flow rate, quantity of gas flow and the temperature of the gases leaving the super heated section. The flue gas temperature should be nearly 175°C higher than the temperature of super heated steam. Material used for super-heater tubes should have high temperature strength and high resistance to oxidation. Special steel alloys such a chromium molybdenum alloy is used for tubes of super-heater for modern high pressure boilers.

According to location the super-heaters are classified as follow:

- | | |
|------------------|------------------|
| (i) Over deck | (ii) Inter deck |
| (iii) Inter tube | (iv) Inter bank. |

Fig. 7.41 shows over deck location of a super heater and inter deck location is shown in Fig. 7.42. Inter tubes and inter bank location of super heater is shown in Fig. 7.43 and Fig. 7.44 respectively.

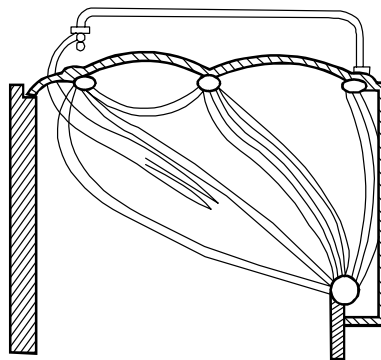


Fig. 7.44. Inter bank superheater.

Over deck and inter deck super heaters are essentially convective super heater. The heat transfer conditions in a super heater vary with load. When load is decreased the gas mass flow decreases proportionately and in a convective superheater fewer degrees of super heat are

obtained whereas in a radiant superheat steam receives more heat than at higher loads. A radiant superheater has falling characteristic with increased steam output of boiler. Modern boilers using high pressures use combination of convective and radiant superheater. Fig. 7.45 shows the locating of both convective and radiant super-heaters in a bent tube boiler.

If superheater is located in the spaces over the water tubes of the boiler, it is called over deck. If it is placed in the space within the tube deck, this is called inter deck superheater, and when located between the banks of water tubes, it is known as inter bank.

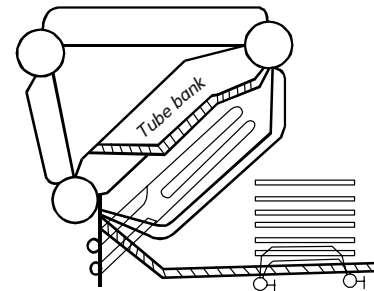


Fig. 7.45.

Superheaters may be horizontal or pendant, the former have the advantage of being drained easily, while in the latter the condensed steam has to be blown off.

Depending on the direction of flows of steam and combustion products, superheaters may be partial flow, counter-flow or mixed flow type.

Radiant superheater. If the heating surface of a radiant superheater in the drum-type boiler is not large, it can be located on the furnace roof. Otherwise, it can occupy part of the vertical walls of the furnace. In once-through boilers, the radiant superheater usually occupies the furnace roof, the upper and medium radiation sections, the walls of the horizontal flue duct.

7.21.1. Sugden Superheater

Fig. 7.46 shows sugden's super heater used in a Lancashire boiler. This super heater uses two steel headers to which are attached solid drawn 'U' tubes of steel. These tubes are arranged

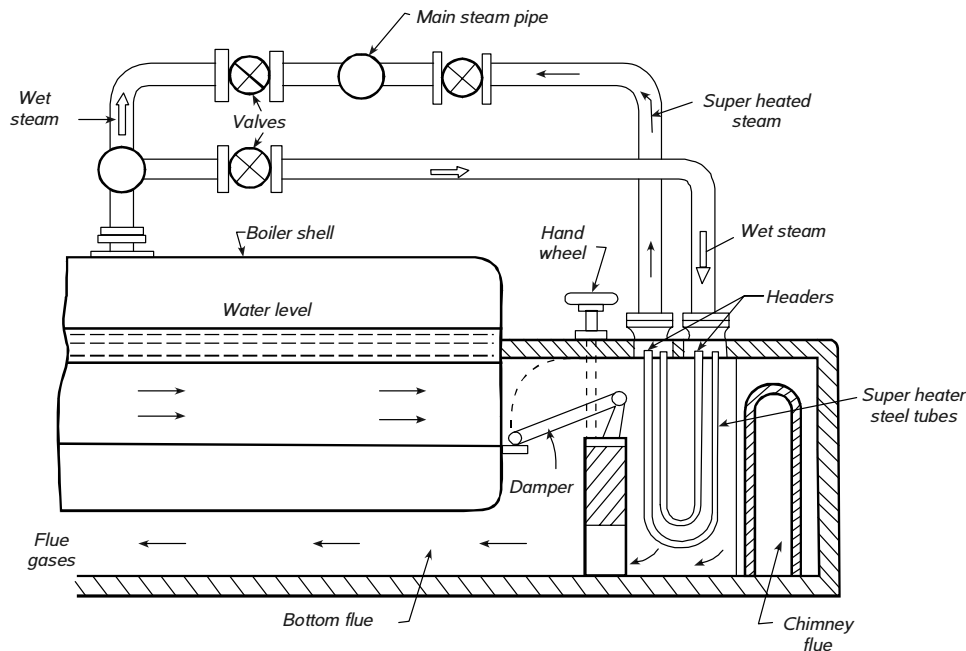


Fig. 7.46. Sugden superheater.

in groups of four and one pair of the headers generally carries ten of these groups or total of forty tubes. The steam from the boiler enters and leaves the headers as shown by the arrows. It shows how the steam pipes may be arranged so as to pass the steam through the superheater or direct to the main steam pipe.

Advantages of Super-heated Steam

Super-heated steam is vapour whose temperature has been increased above that of its boiling point at that pressure.

The various advantages of using super-heated steam are as follows:

- (i) Super-heated steam has an increased capacity to work due to a higher heat contact. Therefore, an economy in steam consumption in steam turbines and steam engine achieved.
- (ii) Super-heating raises the over all efficiency of the plant. The temperature of the super-heated steam being higher it gives a higher thermal efficiency when used for working a prime mover.
- (iii) Super-heating of steam avoids the erosion of turbine blades in the last stages of expansion of steam. In order to avoid blade erosion it is desirable to limit the moisture content 10 to 12% in the exhaust of the steam turbines.

7.12.2. Super-heat Control

It is desirable that there should be a close control over the final temperature of steam over a reasonably wide range of load.

The various methods employed to achieve this are as follow:

1. **Use of Desuperheater.** To control the temperature of steam a desuperheater (attenuator) is used. In the desuperheater (Fig. 7.47) some quantity of cold water is injected into or around the pipe carrying the steam. This causes the evaporation of water so injected and thus the temperature of steam is lowered.
2. **Use of Tilting Burners.** Tilting burners in the furnace are used to regulate the temperature of gases leaving the furnace (Fig. 7.48)

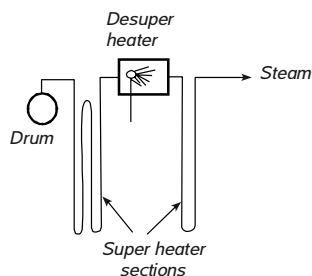


Fig. 7.47. Desuperheater.

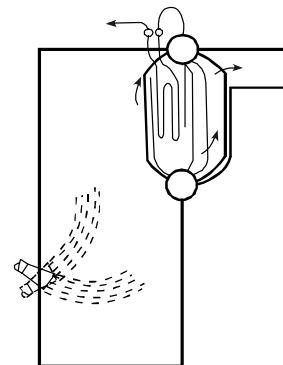


Fig. 7.48. Tilting Burners.

3. **Use of Dampers.** Dampers are provided to control the direction and flow of hot combustion gases in order to vary the quantity of gas passing through the superheater.
4. **Use of Auxiliary Burners.** Auxiliary burners (Fig. 7.49) can be used to control degree of super-heat.
5. **Twin Furnace.** Arrangement (Fig. 7.50) may be used for the control of super-heating temperature.

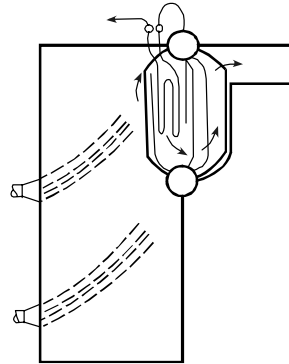


Fig. 7.49. Auxiliary Burners.

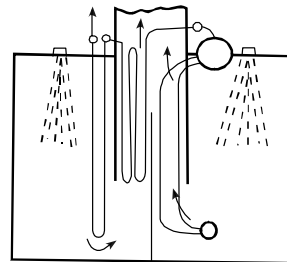


Fig. 7.50. Twin furnace.

7.22 REHEATERS

The superheaters discussed above which superheat the saturated steam just produced in the boiler are known as *primary superheaters*. The *reheaters*, also known as *intermediate (or reheat) superheaters*, in which the worked-off steam returned from the turbine is again superheated.

As discussed in earlier chapters, reheat cycles need two or more stages of superheating, one for turbine throttle steam and one or more for steam taken from the turbine for re-superheating. This is done for taking full advantage of the improved cycle efficiency offered by the high steam pressure and also avoid excessive wetness at the low pressure end of the turbine.

In the reheaters, the steam is withdrawn from the turbine before it becomes saturated during expansion and is reheated before passing back to the turbine to complete its expansion. In the reheater, steam comes from the high-pressure turbine and after being reheated it goes to the intermediate-pressure turbine for further expansion.

The type of reheaters installed are very similar to those of the superheater, the only difference being in the use, generally, of a slightly larger tube size. The modern units have reheaters in the same gas passages of the boiler as the superheaters.

The reheaters should be designed to avoid large pressure drops. Use of higher reheat pressure results in reduced steam volume and economical tube size, although theoretically lower reheat pressures are conducive to greater cycle efficiency.

Since the reheaters have lower pressure as compared to the superheaters, the stresses are lower and hence a lower grade steel alloy can be used. The reheater is usually located above the primary or convective superheater in the convective zone of utility boilers.

7.23 HEAT FLOW

A power plant consists mainly of equipment for energy flow and transformation. The basic modes of heat transfer are as follows.

- (a) Conduction
- (b) Convection
- (c) Radiation.

All the three forms of heat transfer are found in power plant engineering usually in combination with each other. The principal heat transfer calculations are associated with the following:

- (i) Radiation from fuel bed and luminous gases to boiler tubes and water walls.
- (ii) Radiation and conduction in heat insulators like refractories and pipe coverings.
- (iii) Conduction of heat through boiler, economiser and air preheater surfaces.
- (iv) Conduction of heat through condenser or heater tubes.
- (v) Heat convection from combustion region to more remote boiler tubes, economiser tubes and air preheater surfaces.

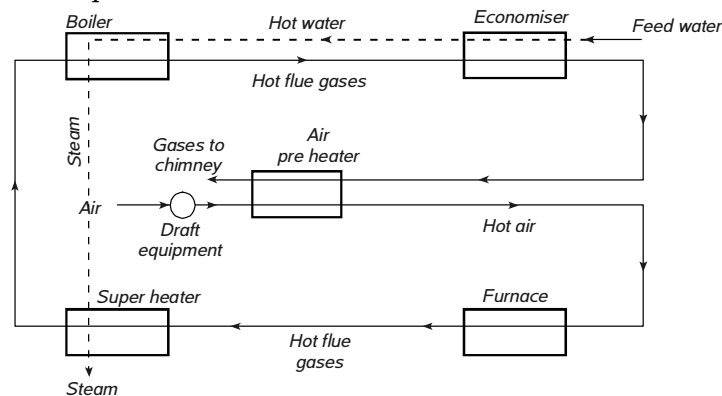


Fig. 7.51. Heat flow from flue gases.

Fig. 7.51 shows the flow of flue gas and feedwater and steam flow in the boiler plant. The heat of flue gases is utilised in heating the feedwater in economiser, in heating the air to be supplied to boiler furnace in air pre-heater and also in superheater.

7.24 BOILER PERFORMANCE

The performance of a steam boiler may be expressed in terms of the following:

- (i) Heat release per cubic metre of furnace volume.
- (ii) Efficiency.
- (iii) Heat transferred per square metre of the heating surface area per hour.
- (iv) Rate of combustion in kcal per square metre of the grate area per hour for solid fuel.
- (v) Amount (kg) of steam produced per hour.

Equivalent Evaporation

To compare evaporative capacity or performance of different boilers working under different conditions it is desirable to provide a common base so that water be supposed to be evaporated under standard conditions.

The standard conditions adopted are temperature of feed water 100°C and converted into dry and saturated steam at 100°C. As per these standard conditions 1 kg of water at 100°C requires 2257 kJ to get converted to steam at 100°C.

Therefore equivalent evaporation is defined as the amount of water evaporated from water at 100°C to dry and saturated steam at 100°C. However (i) the evaporation rate of boiler may also be expressed in terms of kg of steam per kg of fuel.

(ii) The capacity of a boiler may also be expressed in terms of total heat added per hour.

Factor of evaporation (Fe)

It is defined as follows

$$Fe = \frac{H_1}{H_2}$$

where H_1 = Heat received by 1 kg of water under working conditions.

H_2 = Heat received by one kg of water evaporated from and at 100°C

Heat losses in a boiler plant

The heat losses which may take place in a boiler plant are as follows:

- (i) heat loss due to incomplete combustion of fuel
- (ii) heat loss due to unburnt fuel
- (iii) convection and radiation heat loss
- (iv) heat lost in flue gases.

Boiler Trial

Steam is generated in boilers under certain conditions of inlet water and exist steam while a certain rate of fuel is being burnt. In order to study the performance of boiler the experiments are conducted by operating the boiler for a certain length of time and recording the data. This procedure is known as boiler trial. A heat balance sheet is prepared which indicates the heat supplied, heat utilised through various sources and heat wasted.

The main objects of boiler trial are as follow:

- (i) To determine the thermal efficiency of the boiler when working at a definite pressure.
- (ii) To draw up heat balance sheet for the boiler to check the performance of the boiler.

Heat Loss in Boilers

The various causes of heat loss in a boiler are as follows:

- (i) Loss due to heat carried away in chimney gases.
- (ii) Loss due to incomplete combustion of fuel. This can be due to insufficient air supply or fuel bed being in poor conditions.

- (iii) Loss due to moisture formed by combustion of hydrogen in fuel.
- (iv) Loss due to moisture in coal.
- (v) Loss due to radiation and convection from boiler.
- (vi) Heat loss due unburnt fuel which is passed on to the ash pit.

Thermal Efficiency of Boiler

Thermal efficiency of a boiler is defined as the ratio of heat actually utilised by water in generation of steam to the heat supplied during combustion of fuel.

$$\eta_{\text{Boiler}} = \frac{W(H_2 - H_1)}{W_1 \times CV}$$

where W = Weight of steam produced, kg

H_1 = Enthalpy of feed water, kcal/kg

H_2 = Total heat of steam, kcal/kg at the generation pressure

W_1 = Weight of fuel burnt, kg

$C.V.$ = Calorific value of fuel, kcal/kg.

Although the major portion of heat produced during combustion of fuel is utilised to heat the feed water for conversion into steam but some portion of heat produced is lost in flue gases leaving the chimney and some amount of heat is lost due to radiation from furnace to the outside. The boiler efficiency is nearly 75% without heat recovery equipment and it varies from 85 to 90% with heat recovery equipment.

Boiler performance can be improved by taking into account the following factors:

- (i) Proper supervision should be carried out.
- (ii) Too much excessive air should not be supplied.
- (iii) Boiler, superheater, economiser and air heater surfaces should be cleaned at proper intervals.
- (iv) Type of fuel used also controls boiler efficiency. Fig. 7.52 compares the performance in burning bituminous coal, oil or natural gas in the boiler.

Efficiency (η) indicates steam generating unit efficiency and C represents per cent of rated capacity.

- (v) Proper supply of excess air.
- (vi) Minimising combustible in refuse.
- (vii) Minimising combustible in flue gas.
- (viii) Maintaining required cleanliness of heat transfer surfaces.

Efficiency of a boiler depends on the following factors:

- (i) Steam pressure
- (ii) Calorific value
- (iii) Effectiveness in burning of fuel.

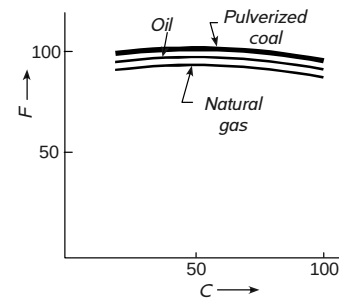


Fig. 7.52. Efficiency with different fuels

Indication of the effective burning of fuel is given by the percentage of the carbon dioxide (CO_2) in the flue gases. The percentage of CO_2 to give maximum efficiency and satisfactory operation will depend on:

- (a) type of fuel such as oil, gas, coal.
- (b) method of firing.
- (iv) Capacity of boilers
- (v) Use of heat recovery apparatus such as air preheater, economiser etc.

Ways of Increasing Thermal Efficiency.

The thermal efficiency of a steam power plant can be increased by the following ways :

- (i) An increase in the initial pressure of steam raises the thermal efficiency.
- (ii) The thermal efficiency can be increased by raising the initial temperature of the steam without changing the pressure.
- (iii) Intermediate reheating of steam improves the thermal efficiency of the plant. As already mentioned an increase in the initial pressure of steam improves the thermal efficiency steam power plant but the wetness fraction of such steam also grows at the end of expansion. The wetness is so high that it causes wear of the blades of the last stages of a steam turbine. This short-coming is eliminated by intermediate reheating of the steam. The steam from boiler after partial expansion in the first stage turbine is fed into super heater, where it is reheated at a constant pressure either to the temperature of live steam or to a temperature slightly below that one. The re-heated steam then flows to the last stages of the turbine where the process of expansion is completed (Fig. 7.53).
- (iv) Thermal efficiency of steam power plant can be improved by carrying out regenerative heating of the feed water. Such heating of the water is carried out by using the

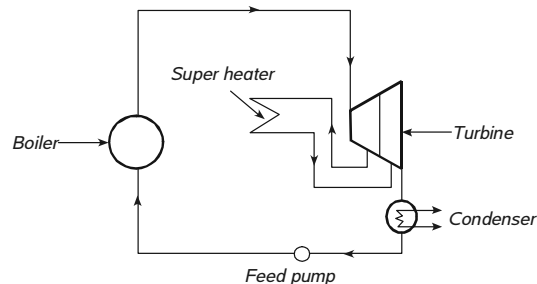


Fig. 7.53. Reheating of the steam

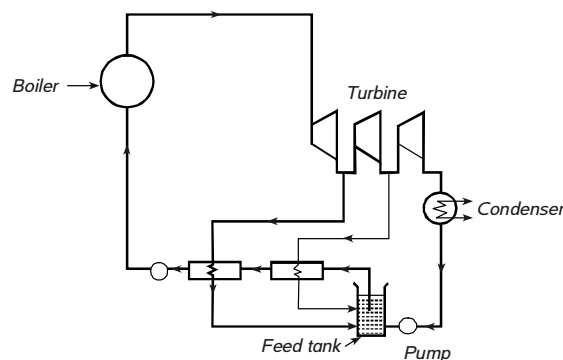


Fig. 7.54. Regenerative heating of feedwater.

heat of steam partly tapped from the turbine. In this method (Fig. 7.54) the steam from boiler flows into steam turbine. After partial expansion some of the steam is tapped from the first stage of the turbine and directed to feed water heater and then to feed tank. The remaining steam enters the second stage of the turbine where it continues to expand. At the outlet from the second stage some of the steam is directed into water heater and then to feed tank. The other part of steam goes to the third stage of the turbine and expands there to the final pressure and enters the condenser. The condensate is delivered by pump to feed tank.

Example 7.5 Determine the boiler efficiency if the boiler generate 23,000 kg of steam at 10.2 kg/cm² dry and saturated. The amount of coal used is 2100 kg and the calorific value of coal is 8600 kcal/kg. The feed water temperature is 105°C.

Solution. The heat of steam at 10.2 kg/cm² = 664 kcal/kg

$$\eta_{\text{Boiler}} = \frac{23,000(664 - 105)}{2100 \times 8600} = 0.71 = 71\% \text{ Ans}$$

7.25 BOILER INSPECTION AND MAINTENANCE

1. Boiler Inspection. The boiler are required to be tested regularly as per Indian Boilers Act. Before inspection, boiler should be made empty and all manhole doors, caps, mud drums, fire bars, fire arches, front end bridge plates, oil burners should be removed and boiler should be cleaned. All tubes, smoke boxes and external flues should be cleaned and checked. All the steam and water connections should be disconnected and then it should be made ready for conducting the following tests.

- (a) **Hydraulic Test.** For this test, the mountings should be fitted tightly except the safety valves. The openings of the safety valves should be closed. All the joints should be made tight. Boiler is filled with water and inspector's gauges should be fitted to the boiler. Generally the boilers is tested at twice the working pressure of the boiler. During this test no part of boiler should deflect.
- (b) **Steam Test.** This test is conducted to check whether the safety valves operate properly to relieve the steam when the steam pressure inside the boiler exceeds the rated pressure. This operation is tested keeping the feed valve and steam stop valve closed.

2. Boiler Maintenance. Regular maintenance and careful supervision of working of various components of boiler result in the efficient operation of the boiler. The various factors which should be carefully observed for proper maintenance of the boiler are as follow:

1. The combustion equipment should be so adjusted that the temperature in the furnace does not exceed the designed value. The air supply to the furnace should be in correct proportion.
2. The water level in the boiler should not be allowed to fall beyond the minimum level.
3. The temperature should change slowly and uniformly in the various parts of the boiler. Rapid changes in temperature cause unequal expansion.
4. The water used for steam generation should be free from scale forming impurities because the scale if deposited on tube and boiler shell does not allow the heat trans-

fer to take place effectively and sometimes causes over heating of tubes and boiler shell which may result in tubes failure and boiler explosion.

5. Soot and ash deposited in tubes on gas side should be removed regularly.
6. Bearings of pumps, stokers, pulverisers and fans, etc., should be lubricated regularly.
7. Oil or scale in the tubes is undesirable, hence such tubes should be located and problem is removed.
8. All stays should be in even tension and they should be properly adjusted.
9. The tube leakage is caused due to blistering of tubes as it is exposed to high temperature. The boiler should be stopped when such leakage is noted and it must be immediately repaired.

7.25.1 Boiler Tube Failure and Prevention

With the introduction of 200 and 500 MW capacity boiler, the outages of boilers due to tube leakage vary from 10 to 15% of the total outages. Therefore, it is necessary to combat this problem in the initial stages. The basic causes of the tube failures are corrosion, stress cracking, thermal fracture, stress rupture and creep distortion.

When the water and steam chemistry are not maintained within the limits recommended by the boiler manufacturer, corrosion or corrosion related damage may occur in feedwater heaters, economiser, and water wall. Over-heating damage may also occur in these tubes as well as in superheater and reheater tubes, if deposits are build up in them due to poor water treatment and improper boiler operation. Overheating of the boiler tubes is the main cause of their failure.

The problem of corrosion have already been discussed while dealing with feedwater treatment in earlier sections. The outer surface erosion of the water tubes is caused by an erosive action of ash particles in the gases. Erosion is enhanced by high flue gas velocities. Erosion creates spots, wall thinning and finally tube failure. These can be reduced upto some extent by controlling the excess air supplied to the combustion chamber, and by providing baffles, and tube shields can be welded in the maximum affected zones.

7.26 MODERN TRENDS IN BOILERS

1. In modern boilers, water wall (water-cooled furnace walls) eventually led to the integration of furnace, economiser, boiler, superheater, reheater and air preheater.
2. One boiler per turbine is provided to simplify the piping system and the use of steam reheating.
3. No drum is provided in boilers operating at pressures above the critical pressure (221 bar) since no separation of the water and steam occurs. The water enters the bottom of the tubes and is completely transformed to steam by the time it reaches the top, passing through the tubes only once. Therefore, these are known as a 'once-through boiler' (also referred as drumless boiler).
4. A large number of steam generation units are designed to work in the ranges of 125 bar and 510°C to 300 bar and 600°C which are basically characterised as sub-critical and super-critical. Usually a sub-critical boiler consists of three distinct section as economiser, evaporator and superheater and in case of super critical boiler the only economiser (preheater) and superheater are required.

5. The most recent method to produce economical thermal power is by the use of super-critical cycle. It is a general practice to use super-critical boilers above 300 MW capacity units, as super-critical boilers have higher thermal efficiency, least problems of corrosion and erosion, and are easy to operate. Presently, about 250 bar and 540°C are used for 500 MW capacity units.
6. Use of supercharged boilers are also becoming common, in which the combustion is carried out under pressure in the combustion chamber by supplying the compressed air. In this, in addition to generating steam in the boiler for use in turbine like other boilers, the exhaust gases from the combustion chamber are used to run gas turbine as they are exhaust at high pressure. The part of the gas turbine output can be used to drive other auxiliaries.
7. Modern boilers have sufficiently large furnace to complete combustion of fuel and contain enough heat transfer surface in suitable arrangement, to prevent excessive slagging on the water walls and tubes.

SOLVED PROBLEMS

Example 7.6. *The overall thermal efficiency of a 40 MW turbo alternator is 30%. If the load factor of the power station is 50% and coal burnt has a calorific value of 6800 kcal, determine the following:*

- (i) Coal consumption per kWh.
- (ii) Coal consumption per day.

Solution.

$$\text{Load factor} = \frac{\text{Average load}}{\text{Maximum demand}}$$

$$0.50 = \frac{\text{Average load}}{40 \times 1000}$$

assuming maximum demand equal to capacity of the power station.

$$\text{Average load} = 0.5 \times 40 \times 1000 = 20,000 \text{ kW}$$

$$\begin{aligned} \text{Energy produced per day} &= \text{Average load} \times 24 \\ &= 20,000 \times 24 = 48 \times 10^4 \text{ kWh.} \end{aligned}$$

$$\text{Overall thermal efficiency} = \frac{860}{W \times 6800}$$

$$W = \text{Coal consumption per kWh.}$$

$$\therefore (1 \text{ kWh} = 860 \text{ kcal})$$

$$0.3 = \frac{860}{W \times 6800}$$

$$\therefore W = \frac{860}{0.3 \times 6800} = 0.42 \text{ kg. Ans}$$

Coal required per day = Energy produced $\times W$

$$= 48 \times 10^4 \times 0.42 \text{ kg.} = \frac{48 \times 10^4 \times 0.42}{1000} \text{ tonnes.}$$

= 2011 tonnes. Ans.

Example 7.7 In a steam power station the coal consumption is 0.4 kg per kWh output. If the calorific value of coal is 6800 kcal per kg, boiler efficiency is 70% and mechanical and electrical efficiency of alternator is 90%. Prepare roughly the heat balance sheet for the power station.

Solution.

Let output = 1 kWh

$$\therefore \text{Heat input} = 0.4 \times 6800 = 2720 \text{ kcal.}$$

$$\text{Electrical energy input} = \frac{2720}{860} = 3.16 \text{ kWh}$$

Losses. (i) Boiler house loss

$$= 3.16 \times 0.3 = 0.948 \text{ kWh}$$

(as boiler efficiency is 70%).

$$\text{Heat output of steam} = 3.16 - 0.948 = 2.212 \text{ kWh.}$$

$$\text{Input to alternator} = \frac{1}{0.90} = 1.11 \text{ kWh}$$

$$(ii) \text{ Loss in alternator} = 1.11 - 1 = 0.11 \text{ kWh.}$$

$$(iii) \text{ Loss in turbine} = 2.212 - 1.11 = 1.102 \text{ kWh.}$$

This loss may be considered as rejected to cooling water.

$$\text{Hence, output} = 1 \text{ kWh.}$$

$$\text{Loss in boiler} = 0.948 \text{ kWh.}$$

$$\text{Loss in alternator} = 0.11 \text{ kWh}$$

$$\text{Loss in alternator} = 1.102 \text{ kWh.}$$

$$\text{Total} = 1 + 0.948 + 0.11 + 1.102 = 3.16 \text{ kWh.}$$

Example 7.8 The daily output of a steam power station is 18×10^5 kWh. If the coal consumption is 700 tonnes per day, calculate the thermal efficiency of power station.

Assume calorific value of coal 8500 kcal.kg.

$$\text{Solution. Coal used} = 700 \times 1000 \text{ kg.}$$

$$\text{Energy output per day} = 700 \times 1000 \times 8500 \times 595 \times 10^7 \text{ kcal}$$

$$\text{Energy output per day} = 18 \times 10^5 \text{ kWh}$$

$$= 18 \times 10^5 \times 860 \text{ kcal.}$$

(as 1 kWh = 860 kcal).

$$\text{Efficiency} = \frac{\text{Output}}{\text{Input}} = \frac{18 \times 10^5 \times 860}{595 \times 10^7} = 26\%. \text{ Ans}$$

Example 7.9. Prepare the heat balance sheet for a boiler having the following data:

Steam pressure	= 13 kg/cm ²
Water evaporated	= 300,000 kg
Coal fired	= 30,000 kg
Superheat temperature	= 300°C

Air temperature

(a) To air preheater	= 45°C
(b) From air preheater	= 90°C

Feed water temperature

(a) To economiser	= 50°C
(b) From economiser	= 120°C
Flue gas exit temperature	= 115°C

Boiler house temperature	= 25°C
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Mean specific heat of the flue gases = 0.24

Coal analysis (as fired)

$$C = 64\%; H_2 = 4.5\%$$

$$O_2 = 5.5\%; H_2 = 1.5\%$$

$$S = 0.7\%;$$

Ash	= 9.8 %
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Total moisture	= 14%
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Calorific value of coal	= 6800 kcal/kg
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Duration of trial	= 8 hours
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Flue gas (dry) analysis

$$CO_2 = 12.5\% ; CO = 0.4\%$$

$$O_2 = 6\% ; N_2 = 81.1\%.$$

Solution. (i) To find weight of water evaporated.

Let W = Actual weight of dry flue gas per kg of coal

$$W = \frac{11CO_2 + 8O_2 + 7(CO + N_2)}{3(CO_2 + CO)} \times \left(C + \frac{S}{1.83} \right)$$

$$= \frac{11 \times 12.5 + 8 \times 6 + 7(0.4 + 81.1)}{3(12.5 + 0.4)} \times \left(0.64 + \frac{0.007}{1.83} \right)$$

$$= 12.6 \text{ kg.}$$

W_1 = Total weight of flue gas per kg of coal

= Actual weight of dry flue gas + Moisture evaporated from coal + Water produced from H

$$= 12.6 + 0.14 + 9(0.045) = 13.14 \text{ kg}$$

Now ash content

$$= 9.8\%$$

W_2 = Weight of fuel burnt

$$= 1 - 0.098 = 0.902$$

W_3 = Weight of air supplied per kg of coal fired

$$W_3 = W_1 - W_2 = 13.14 - 0.902 = 12.238 \text{ kg}$$

W_4 = Weight of water evaporated per kg of fuel fired

$$= \frac{300,000}{30,000} = 10 \text{ kg.}$$

(ii) To Calculate Heat Absorbed:

H_1 = Heat absorbed by air preheater per kg of coal fired.

= Weight of air $\times$ Temperature rise $\times$ Specific heat

$$= W_2 \times (90 - 45) \times 0.24$$

$$= 12.238 \times 45 \times 0.24 = 132 \text{ kg cal.}$$

H_2 = Heat absorbed by boiler

= Weight of water evaporated per kg of coal $\times$ (Total heat in 1 kg of steam – Heat in feed water – Heat in air)

$$= 10 [666 - (120 - 0) - 132] = 4140 \text{ kg cal.}$$

H_3 = Heat absorbed by economiser

= Weight of water evaporated $\times$ Temperature rise

$$= 10 [120 - 50] = 700 \text{ kg cal.}$$

H_4 = Heat absorbed by superheater

= Weight of water evaporated $\times$ [Super heat temperature – Saturated temperature] $\times$ Specific heat

$$= 10 [300 - 190] \times 0.5 = 10 \times 110 \times 0.5 = 550 \text{ kcal.}$$

(iii) To Calculate Heat Lost:

$$\begin{aligned}
 H_5 &= \text{Heat lost in dry flue gases} \\
 &= \text{Weight of dry flue gases per kg of coal} \times \text{Temperature rise} \\
 &\quad \times \text{Specific heat} \\
 &= W \times (115 - 25) \times 0.25 \\
 &= 12.6 \times 90 \times 0.25 = 283.6 \text{ kg cal}
 \end{aligned}$$

$$\begin{aligned}
 H_6 &= \text{Heat loss due to moisture in fuel} \\
 &= w (100 - t) + L + k (T - 100)
 \end{aligned}$$

where w is the weight of moisture evaporated from coal

$$= 0.14 [(100 - 25) + 539 + (115 - 100) \times 0.47] = 16 \text{ kg cal}$$

(Assuming specific heat as 0.47)

$$\begin{aligned}
 H_7 &= \text{Heat lost due to water formed by combustion of hydrogen} \\
 &= 9 H [(100 - t) + L + k (T - 100)]
 \end{aligned}$$

where H weight of hydrogen per kg of fuel, t is the atmospheric temperature, T is exist gas temperature and k is the specific heat of steam and L is latent heat of evaporation

$$\begin{aligned}
 &= 9 \times 0.045 [(100 - 25) + 539 + (115 - 100) \times 0.47] \\
 &= 0.4 \times 621 = 248.4 \text{ kg cal}
 \end{aligned}$$

$$\begin{aligned}
 H_8 &= \text{Heat lost due to incomplete combustion of carbon} \\
 &= C \times \text{C.V.} \cdot \frac{\text{CO}}{\text{CO} + \text{CO}_2}
 \end{aligned}$$

where C is the weight of carbon per kg of fuel, and CO and CO_2 represent the per cent of volume of carbon monoxide and carbon dioxide in flue gases respectively and C.V. is the calorific value of coal

$$= 0.64 \times 6800 \frac{0.4}{12.6 \times 0.4} = 130 \text{ kg cal}$$

$$\begin{aligned}
 H_9 &= \text{Heat unaccounted for loses} \\
 &= 6800 - [H_1 + H_2 + H_3 + H_4 + H_5 + H_6 + H_7 + H_8] \\
 &= 6800 - 6170 = 630 \text{ kcal.}
 \end{aligned}$$

Example 7.10. A boiler with maximum continuous steam rating of 1.2×10^5 kg per hour is fired with bituminous coal having 10% moisture and 10% ash and a higher calorific value of 7500 kcal per kg. If it takes 1000 kcal to evaporate 1 kg of feed water entering the boiler and superheat it to final temperature and if overall efficiency of boiler is 80% determine the following:

(i) Hourly coal supply.

(ii) Grate length, if building dimensions limit grate width to 6 metre. Given that the maxi-

Heat produced is 3×10^6 kcal per square metre per hour.

Solution. H = Heat required to convert water into steam
 $= 1.2 \times 10^5 \times 1000 = 1.2 \times 10^8$ kcal per hour
 Efficiency (η) of boiler $= 0.8$

Heat to be produced by the fuel $= \frac{H}{0.8}$

$$H = \frac{1.2}{0.8} \times 10^8$$

$$H = 1.5 \times 10^8 \text{ kcal per hour}$$

Coal required $= \frac{1.5 \times 10^8}{7500} \text{ kg} = \frac{1.5 \times 10^8}{7500 \times 10^3} \text{ tonnes}$
 $= 20 \text{ tonnes per hour}$

Grate area $= \frac{H}{3 \times 10^6} = \frac{1.5 \times 10^8}{3 \times 10^6} = 50 \text{ square metres}$

Grate length $= \frac{50}{6} = 8.33 \text{ metres. Ans.}$

Example 7.11. The following data is supplied for a boiler plant:

- (i) *Boiler*
 Mass of coal supplied $= 220 \text{ kg/hr.}$
 Calorific value of coal $= 7200 \text{ kcal/kg}$
 Mass of feed water $= 2000 \text{ kg/hr}$
 Enthalpy of steam produced $= 670 \text{ kcal/kg}$
- (ii) *Economiser*
 Inlet temperature of feed water $= 18^\circ\text{C}$
 Outlet temperature of feed water $= 88^\circ\text{C}$
 Atmospheric temperature $= 19^\circ\text{C}$
 Temperature of flue gas entering $= 350^\circ\text{C}$
 Mass of flue gases $= 4150 \text{ kg/hr.}$

Determine the following:

- (a) Efficiency of boiler
 (b) Efficiency of economiser
 (c) Efficiency of whole boiler plant.

Solution. (a) W = Mass of coal supplied $= 220 \text{ kg/hr}$
 C_v = Calorific value of coal $= 7200 \text{ kcal/kg}$

H_1 = Heat obtained by combustion of coal

$$= W \times C_v = 220 \times 7200$$

$$= 1584 \times 10^3 \text{ k. cal/hr.}$$

W_1 = Mass of feed water = 2000 kg/hr.

H_2 = Heat utilised in evaporation of steam

$$= W_1 (h - h_1)$$

where h = Enthalpy steam = 670 k. cal/kg

h_1 = Enthalpy of feed water = 88.k cal/kg.

$$H_2 = 2000 (670 - 88)$$

$$= 2000 \times 582 = 1164 \times 10^3 \text{ kcal./hr}$$

$$= 1164 \times 10^3 \text{ kcal.}$$

$$\therefore \text{Boiler Efficiency} = \frac{H_2}{H_1} \times 100 = \frac{1164 \times 10^3}{1584 \times 10^3} \times 100$$

$$= 73.5\%. \text{ Ans.}$$

(b)

T_1 = Inlet temperature of feed water = 18°C

T_2 = Outlet temperature of feed water = 88°C

H_3 = Heat utilised by economiser

$$= W_1 (T_2 - T_1) = 2000 (88 - 18)$$

$$= 14 \times 10^4 \text{ k. cal/hr.}$$

W_2 = Mass of flue gases = 4150 kg/hr.

T_3 = Temperature of flue gases = 350°C

T_4 = Atmospheric temperature = 19°C

K_p = Specific heat of flue gases = 0.24

H_4 = Heat supplied to economiser

$$= W_2 (T_3 - T_4) \times K_p = 4150 (350 - 19) \times 0.24$$

$$= 4150 \times 331 \times 0.24 = 329 \times 10^3 \text{ k cal/hr.}$$

$\therefore$ Economiser efficiency

$$= \frac{H_3}{H_4} \times 100 = \frac{14 \times 10^4}{329 \times 10^3} \times 100 = 42.5\%. \text{ Ans.}$$

(c)

H = Total heat utilised

$$= H_2 + H_3 = 1164 \times 10^3 + 14 \times 10^4 = 1304 \times 10^3 \text{ kcal/hr}$$

η = Over all efficiency

$$\begin{aligned}
 &= \frac{H}{H_1} \times 100 \\
 &= \frac{1304 \times 10^3}{1584 \times 10^3} \times 100 = 82.3\%. \quad \text{Ans.}
 \end{aligned}$$

This shows that by installing economiser the efficiency of the boiler plant is increased.

Example 7.12 Determine the quantity of air required per kg of coal burnt in a steam power plant furnace fitted with a 62 m high stack. The draft produced is 38 mm of water and temperature of flue gases is 410°C. Boiler house temperature is 29°C.

Solution.

h = Draught = 38 mm of water

W = Weight of air required per kg of coal burnt

T_1 = Absolute temperature of air outside the chimney

$$= 29 + 273 = 302^\circ\text{K}$$

T = Average absolute temperature of chimney (stack) gases.

$$= 410 + 273 = 683^\circ\text{K}$$

H = Height of chimney = 62 m.

$$h = 353H \left[\frac{1}{T_1} - \left(\frac{W+1}{W} \right) \times \frac{1}{T} \right]$$

$$38 = 353 \times 62 \left[\frac{1}{302} - \left(\frac{W+1}{W} \right) \times \frac{1}{683} \right]$$

$$W = 16 \text{ kg.} \quad \text{Ans.}$$

Example 7.13 In a steam power plant an endless rope haulage raises 300 tonnes of coal through a vertical distance of 80 metres in one hour. If the efficiency of the haulage system is 40% and the system runs for 8 hours per day for 5 days, calculate the following:

(a) H.P. output of system

(b) Power input

(c) Weekly cost of coal raising. The charge for electricity is Rs. 2.00 paise per kWh.

Solution.

$$t = \text{Time} = 1 \text{ hour} = 3600 \text{ sec.}$$

P = Power output

$$= mgh = 300 \times 1000 \times 9.8 \times 80$$

$$= 2352 \times 10^5 \text{ J}$$

$$= \frac{2352 \times 10^5}{t} = \frac{2352 \times 10^5}{3600} = 65 \times 10^3 \text{ W}$$

(As 1 W = 1 J per sec)

$$\text{Power output (H.P.)} = \frac{65 \times 10^3}{735.5} = 88 \quad \text{Ans.}$$

$$\eta = \text{Efficiency} = 0.4$$

$$\text{Power input} = \frac{88}{0.4} = 220 \text{ H.P.} \quad \text{Ans.}$$

$$\text{Hours per week} = 8 \times 5 = 40$$

$$\text{Power input} = \frac{\text{Power output}}{\eta} = \frac{65 \times 10^3}{0.4} = 162 \times 10^3 \text{ W} \quad \text{Ans.}$$

$$E = \text{Energy consumed}$$

$$= \frac{65 \times 10^3}{1000} \times 40 = 6480 \text{ kWh.}$$

$$\text{Rate per kWh} = \text{Rs. 2.00}$$

$$\text{Cost} = 6480 \times \text{Rs. 2.00}$$

$$= \text{Rs. 12960} \quad \text{Ans.}$$

Example 7.14 A steam power plant uses coal 5000 kg / hr. The heat conversion efficiency is 30% and calorific value of coal is 7000 k.cal / kg, calculate the electric energy produced per day.

Solution.

$$W_1 = \text{Amount of coal used} = 5000 \text{ kg/h}$$

$$C = \text{Calorific value of coal} = 7000 \text{ kcal/kg}$$

$$\text{and conversion efficiency} = 0.3$$

$$H_1 = \text{Net heat utilised}$$

$$= \text{W.C. } \eta = 5000 \times 7000 \times 0.3 = 10.5 \times 10^6 \text{ kcal/hr}$$

$$E = \text{Energy produced per hour} = \frac{H}{860} \text{ kWh}$$

$$(\text{As } 1 \text{ kWh} = 860 \text{ kcal}) \quad E = \frac{10.5 \times 10^6}{860} = 12,210 \text{ kWh}$$

$$\begin{aligned} \text{Energy produced per day} &= E \times 24 = 12,210 \times 24 = 293 \times 10^3 \text{ kWh.} \\ &= 293 \text{ MWh} \quad \text{Ans.} \end{aligned}$$

Example 7.15. A steam power plant of 150 kW capacity used gas of calorific value of 1200 kcal/m³.

Determine the volume of gas required per hour when the plant is running at full load conditions.

Take the overall efficiency of the plant as 30%.

Solution.

$$E = \text{Energy generated per hour} = 150 \times 1 = 150 \text{ kWh}$$

$$H_1 = \text{Heat required for } E = E \times 860 = 150 \times 860 \text{ kcal}$$

(As 1 kWh = 860 kcal)

Let

$$V = \text{volume of gas required per hour}$$

$$\text{C.V.} = \text{Calorific value of gas} = 1200 \text{ kcal/m}^3$$

$$\eta = \text{Efficiency of plant} = 0.3$$

$$H_2 = \text{Heat produced by burning the gas}$$

$$H_2 = V \times \text{C.V.} \times \eta = V \times 1200 \times 0.3$$

$$H_1 = H_2$$

$$150 \times 860 = V \times 1200 \times 0.3$$

$$V = 358 \text{ m}^3/\text{hr} \quad \text{Ans.}$$

Example. 7.16 Two boilers one with a super heater and the other without a super heater are supplying equal quantities of steam into a common main. The temperature of steam from the boiler with super heater is 330°C and that of steam in the main is 260°C. If the pressure in the boiler and the main is 15 kg/cm² and specific heat of superheated steam is 0.54 determine the quality of steam supplied by boiler without super heater.

Solution.

Let

$$q = \text{Dryness fraction of steam supplied by boiler without super heater}$$

At 15 kg/cm² (From steam tables)

$$t = \text{Saturation temperature of steam} = 197.4^\circ\text{C}$$

$$L = \text{Latent heat} = 466.7 \text{ kcal}$$

$$H_w = 200.7 \text{ kcal}$$

Let us assume that each boiler supplies one kg of steam into the common main.

$$H' = \text{Heat of steam} = H_w + qL = 200.7 + q \times 466.78 \text{ kcal.}$$

Now t_s = Temperature of super heated steam = 330°C

$$C_p = 0.54$$

$$H_s = \text{Heat of super heated steam} = H + C_p (t_s - t)$$

$$= 667.4 + 0.54 (330 - 197.4) \text{ kcal} = 739 \text{ kcal}$$

$$H_1 = H' + H_s = 200.7 + q \times 466.7 + 739 \text{ kcal.}$$

where H_1 is the heat of steam supplied by both the boilers

Since the temperature of steam in the common main is 260°C and saturation temperature is 197.4°C it is obvious that steam is still in super heated condition.

$$\begin{aligned} H_2 &= \text{Heat of 2 kg of steam in the main} \\ &= [H + C_p (t_s - t)] \times 2 \\ &= [667.4 + 0.54 (260 - 197.40)] \times 2 = 1402.4 \text{ kcal.} \end{aligned}$$

$$\begin{aligned} \text{Now} \quad H_1 &= H_2 \\ 200.7 + q \times 466.7 + 739 &= 1402.4 \end{aligned}$$

$$\therefore q = 0.91. \quad \text{Ans.}$$

Example 7.17. Calculate the efficiency of a boiler in which coal consumption is 65 kg per hour and which generates 370 kg of steam per hour at 0.93 dryness fraction and at a pressure of 8 bar absolute. Coal used has the following composition per kg.

Carbon	= 0.71
Hydrogen	= 0.05
Oxygen	= 0.11
Sulphur	= 0.02
Ash	= 0.11
Feed water temperature	= 24°C

Solution.

$$P = \text{Pressure of steam} = 8 \text{ bar}$$

$$\text{From steam tables} \quad hf = \text{Sensible heat} = 721 \text{ kJ/kg}$$

$$hfg = \text{Latent heat of steam} = 2046.5 \text{ kJ/kg}$$

$$x = \text{Dryness fraction}$$

$$H = \text{Heat of 1 kg of steam}$$

$$= hf + x \cdot hfg = 721 + 0.93 \times 2046.5 = 2624.3 \text{ kJ/kg}$$

$$H_1 = \text{Heat supplied to steam per kg}$$

$$= H - \text{heat contained in 1 kg of feed water}$$

$$= 2624.3 - 1 \times 4.18 \times (24 - 0) = 2524 \text{ kJ/kg}$$

$$W = \text{Weight of steam produced per hour} = 370 \text{ kg}$$

$$H_2 = \text{Heat of steam} = W \times H_1 = 370 \times 2524 = 933880$$

$$C_v = \text{Calorific value of coal}$$

$$= 33800 \text{ C} + 144000 \left(H - \frac{0}{8} \right) + 9270 \text{ S}$$

where

C = Amount of hydrogen per kg of coal = 0.71

H = Amount of hydrogen per kg of coal = 0.05

O = Amount of oxygen per kg of coal = 0.11

S = Amount of sulphur per kg of coal = 0.02

$$C_v = 33800 \times 0.71 + 144000 \left(0.05 - \frac{0.11}{8} \right) + 9270 + 0.02$$

$$= 29367 \text{ kJ/kg}$$

W_1 = Amount of coal burnt per hour = 65 kg

H_3 = Total heat input

$$= C_v \times W_1 = 29367.4 \times 65 = 1908855 \text{ kJ}$$

$$\eta = \text{Boiler efficiency} \frac{H_2}{H_3} \times 100 = \frac{933880}{1908855} \times 100 = 49\%. \quad \text{Ans.}$$

Example 7.18. A boiler working at a pressure of 10 bar generates 2100 kg of dry and saturated steam per hour. The feed water is heated by an economiser to a temperature of 105°C. Coal consumed is 208 kg and calorific value of coal is 30200 kJ/kg. If 12% of coal remains unburnt determine:

(a) thermal efficiency of boiler

(b) thermal efficiency of boiler and grate combined.

Solution.

x = Dryness fraction = 1

m = Rate of production of steam = 2100 kg/h

ΔT = Feed water temp. rise = 105°C

P = Pressure = 10 Bar

h = Heat of steam at 10 bar pressure = h_g = 2776 kJ/kg

h_{f1} = Heat contained in feed water

$$= 1 \times 4.18 \times \Delta T = 1 \times 4.18 \times 105 = 439 \text{ kJ/kg}$$

h_1 = Heat used to produce 1 kg of steam in boiler.

$$= h_1 - h_{f1} = 2776 - 439 = 2337 \text{ kJ/kg}$$

Unburnt coal

$$= 10\%$$

W = Coal consumed = 208 kg

$$m_1 = \text{Mass of coal actually burnt} = 208 \times \frac{90}{100} = 187.2 \text{ kg}$$

m_2 = Mass of steam produced per kg of coal actually burnt.

$$= \frac{m}{m_1} = \frac{2100}{187.2} = 11.2 \text{ kg}$$

$$\begin{aligned} H_1 &= \text{Total heat of steam} = m_2 \times h_1 \\ &= 11.2 \times 2337 = 26174.4 \text{ kJ} \end{aligned}$$

$$C_v = \text{Calorific value of coal} = 30200$$

$$\eta_b = \text{Boiler efficiency} = \frac{H_1}{C_v} \times 100$$

$$= \frac{26174.4}{30200} \times 100 = 86.6\% \quad \text{Ans.}$$

$$\eta_g = \text{Efficiency of boiler and grate}$$

$$= \frac{\frac{m}{w} \times (h_1)}{C_v} = \frac{2100}{208} \times \frac{2337}{30200} = 0.78 = 78\% \quad \text{Ans.}$$

Example 7.19. Sketch and describe in brief a steam power plant indicating various parts of the plant.

Solution. In a steam power plant, fuel and air enter the power plant and products of combustion leave the unit. There is a transfer of heat to the cooling water, and work is done in the form of the electrical energy leaving the power plant. The overall objective of a power plant is to convert the availability (to do work) of the fuel into work (in the form of electrical energy) in the most efficient manner, taking into consideration cost, space, safety, and environmental concerns.

A schematic diagram of a steam power plant is shown in Fig. 7.55. High-pressure superheated steam leaves the boiler, which is also referred to as a steam generator, and enters the turbine. The steam expands in the turbine and, in doing so, does work, which enables the turbine to drive the electric generator. The low-pressure steam leaves the turbine and enters the condenser, where heat is transferred from the steam (causing it to condense) to the cooling.

Example 7.20. Determine the boiler efficiency if the boiler generates 23,000 kg of steam at 10.2 kg/cm² dry and saturated. The amount of coal used is 2100 kg and the calorific value of coal is 8600 kcal/kg. The feed water temperature is 105°C.

Solution. Total heat of steam at 10.2 kg/cm².

$$= 664 \text{ kcal/kg}$$

$$\begin{aligned} \therefore \eta_{\text{boiler}} &= \frac{23000(664 - 105)}{2100 \times 8600} = 0.71 \\ &= 71\% \quad \text{Ans.} \end{aligned}$$

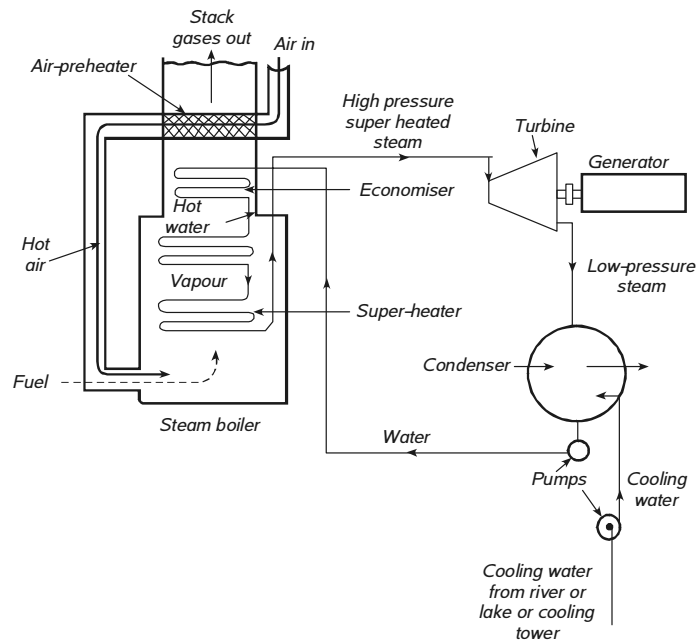


Fig.7.55. Steam power plant.

QUESTIONS

- How will you classify various types of boilers?
- What is the difference between water-tube and fire-tube boilers? Describe the working principle of Cochran or Lancashire boilers.
- Write short notes on the following:
 - Boiler mountings.
 - Boiler accessories.
 - Efficiency of a boiler.
- What is a superheater? Describe different types of super-heaters.
- State the advantages of superheated steam. Explain the working of a superheater.
- Write short notes on the following.
 - Maintenance of a boiler.
 - Feed water treatment.
 - Feed water control.
- Explain the methods used to increase thermal efficiency of a steam power plant.
- Write short notes on the following:
 - pH value of water
 - Power plant pumps.
 - Comparison of forced and induced draft system for boilers.
- Describe with a neat sketch,
 - water tube boiler,
 - fire tube boiler.
- Describe a feed water regulator.
- For a boiler, draw a flow diagram showing flue gas flow and water steam flow.
- State the advantages of high pressure boilers.
- Write short notes on the following:
 - Selection of boiler.
 - Gas fired boilers.

14. Discuss modern trends in generating steam.
15. Discuss modern trends in boilers.
16. Write short notes on the following:
 - (a) Economiser
 - (b) Hydraulic test of a boiler.
17. Describe the systematic arrangement in a modern steam power station and detail the various heat saving devices used.
18. What are the advantages of using water tube boilers and what are the special characteristics which enable water tube boilers best suited for modern power plant?
19. What are the high pressure boilers and how they differ in construction and working principle from ordinary boilers? What are the advantages gained by forced water circulation?
20. Describe construction and working of La Mont, Loeffler, Benson and Velox boilers giving suitable sketches.
21. What is the function of a superheater in a power plant?
22. Describe advantages gained by using economizers in a modern power plant.
23. What are the main types of air heaters employed in a steam power plant? Describe the regenerative air heater.
24. (a) Why is the feed water treatment necessary before it is supplied to the boiler?
(b) What are the effects of impurities in feed water on the boiler?
25. Describe different methods used for feed water treatment.
26. Write short notes on the following:
 - (a) Internal water treatment.
 - (b) External treatment of feed water.
 - (c) Chemical treatment of feed water.
 - (d) Steam purity.
27. What is the function of a steam generator? Write down its main components.
28. How does an industrial boiler differ from a power plant boiler?
29. Explain the characteristic features of a water tube boiler. How is it different from a fire tube boiler?
30. Explain the following boiler mountings in brief:
 - (a) Fusible plug.
 - (b) Spring loaded safety valve.
 - (c) Blow off cock.
 - (d) Water level indicator.
31. What is circulation in a boiler ? What is the difference between natural circulation and forced circulation?
32. (a) Why is too much steaming not desired in a riser tube?
(b) Why should no moisture flow along with steam from the drum to the superheater?
(c) What is blow-down? Why is it needed?
33. (a) What are the function of the steam drum in a water tube boiler?
(b) What do you mean by primary and foaming?
34. (a) What is a water wall?
(b) What is an assisted circulation steam generator?
35. What are once-through boilers? How do they differ from drum boilers?
36. Explain the functions of primary air and secondary air?
37. What is super-critical boilers? What are its merits and demerits?
38. Write short notes on:
 - (a) Pendant superheater.
 - (b) Desuperheater.
 - (c) Tilting burners.
 - (d) Feed water softening
39. (a) How is the erosion rate of fly ash related to gas velocity.
(b) Where is reheater located?
(c) How is the water level in steam drum controlled?
40. What are the major difference between mountings and accessories? Give five examples of each?

41. Define draught and name different methods of producing draught.
42. What do you understand by natural draught? What are its limitations?
43. Explain the principles used in forced and induced draughts. Discuss the merits and demerits of each over the other.
44. List out the merits of mechanical draught over natural draught.
45. What do you understand by 'steam jet draught'? Explain its working principle. Where is it used?
46. Name various draught systems? Describes the operation of a balanced draught system.
47. Name the different types of chimneys used. State the advantages of steel chimney.
48. What are the harmful effects caused by using impure water in boilers? Describe the various methods of purifying feed water.
49. What is meant by make up water of boiler and how is this water fed into a boiler?
50. (a) What are the advantages gained by using superheated steam?
(b) Describe the various factors which determine the location of a steam power station.
51. State the advantages and disadvantages of a steam power station as compared to hydro-electric power station and nuclear power station.
52. Write short notes on the following:
(a) Hydraulic test of a boiler. (b) Draught fans.
(c) Steam separator. (d) Heat flow in steam plant.
53. What are natural draft and mechanical draft? How is the latter more advantageous? What is the utility of balanced draft? How is the balanced draft produced?
54. Describe the main types of fans used for production of draft in the boilers. What is the fields of application of each type?
55. What do you understand by draught ? How the draughts are classified?
56. (a) What are the limitations of chimney draught?
(b) Discuss the merits and demerits of forced draught over induced draught.
(c) Why balanced draught is preferred over forced or induced draught?
57. Draw a neat diagram of a volve boiler and discuss its merits and demerits with Benson boiler.
58. (a) What do you understand by supercharged boiler?
(b) What are the considerations in locating the superheater and economiser?
59. (a) What is the necessity of a soot blower?
(b) How is the slag removed from a boiler?
60. Write short notes on:
(a) Boiler trial (b) Boiler balance sheet
(c) Boiler maintenance (d) Boiler inspection.